The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage

GeoVac Project

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Abstract

The GeoVac framework has, over Papers 0–31, accumulated a body of structural results—the S^3 Fock-projection equivalence (Paper 7 [28]), the angular sparsity theorem (Paper 22 [31]), the Hopf-graph lattice-gauge structure (Paper 25 [34]), QED on S^3 (Paper 28 [35]), the Bargmann–Segal S^5 HO lattice (Paper 24 [33]), the SU(2) Wilson gauge (Paper 30 [36]), and the universal/Coulomb partition (Paper 31 [37])—each self-contained but together suggesting one parent structure. This paper names and constructs that structure: a finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected S^3 graph, with $\mathcal{A}_{\mathrm{GV}}$ the algebra of complex functions on graph nodes, $\mathcal{H}_{\mathrm{GV}}$ the Camporesi–Higuchi spinor space restricted to the truncation, and D_{GV} a graph Dirac operator whose continuum limit is the Camporesi–Higuchi operator [20] on unit S^3. This is not a new derivation; it is a structural map. Three results follow. (1) *Lineage.* The construction sits inside the published Marcolli–van Suijlekom gauge-network framework [5] (with the Perez-Sanchez correction [10] that the continuum limit is Yang–Mills without Higgs), and inside the Connes–van Suijlekom spectral-truncation program [11, 12, 13, 14]. Every structural ingredient of the GeoVac construction has published precedent; what is not duplicated is the α prediction at 8.8×10^{-8} (Paper 2 [27]) from a discrete spectral construction. (2) *Sub-sector identification.* Paper 25's Hodge-1 Laplacian is the Cartan-subalgebra (maximal-torus) projection of an almost-commutative extension of the present triple by $M_2(\mathbb{C})$; Paper 30's SU(2) Wilson action is the full non-abelian slice of the same extension. Paper 28's QED on S^3 is the spectral-action computation on D_{GV}; Paper 31's universal/Coulomb partition is the $\mathcal{A}/D$ split of the present triple. Papers 25, 28, 30, 31 are not four parallel results—they are four projections of the single triple constructed here. (3) *Axiom audit.* $\mathcal{A}_{\mathrm{GV}}$ is finite, involutive, faithfully represented on $\mathcal{H}_{\mathrm{GV}}$; D_{GV} is self-adjoint with bounded $[D_{\mathrm{GV}}, a]$ for $a \in \mathcal{A}_{\mathrm{GV}}$; the order-one condition is trivially satisfied by the abelian core; a real structure J_{GV} exists on the Dirac sector with $J_{\mathrm{GV}}^2 = -\mathbb{1}$ (the quaternionic/pseudoreal sign for the 2-dimensional Dirac spinor representation of Spin(3) = SU(2)) and $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$, matching the standard signs $(\epsilon, \epsilon') = (-1, +1)$ for KO-dimension 3 mod 8 of S^3 [1]; a $\mathbb{Z}_2$-grading γ_{GV} is absent at the Dirac level (correctly: S^3 is odd-dimensional, the spectral triple is *odd*). The four-way S^3 coincidence—Fock projection image, Hopf bundle base, Camporesi–Higuchi spin carrier, SU(2) gauge manifold—is the statement that the manifold of $\mathcal{A}_{\mathrm{GV}}$ in the continuum limit, the spinor bundle of D_{GV}, and the gauge group of natural fluctuations of D_{GV} all coincide on the unique rank-1 non-abelian compact simply-connected Lie group.

Scope. This paper does not derive new physics. It synthesizes existing GeoVac results into the axiomatic spectral-triple language. The synthesis lifts each result from a free-standing observation to a sub-sector of a single named structure, and makes explicit which axioms hold rigorously, which hold up to the standard finite-truncation caveats, and which remain open—specifically whether the GeoVac Dirac graph operator carries a real structure matching the topological KO-dimension of S^3 exactly at finite $n_{\max}$ or only in the continuum limit.

1 Introduction

1.1 Why a spectral triple, and why now

Across thirty-one papers the GeoVac framework has carried four distinct strands. Strand 1 is the Fock projection $p_0 = \sqrt{-2E_n}$ that maps the hydrogenic Hamiltonian onto the free Laplacian on the unit three-sphere, with eigenvalues $\lambda_n = -(n^2 - 1)$ on graph nodes labeled by quantum-number triples (n, l, m_l) (Paper 7). Strand 2 is the Hopf bundle $S^1 \to S^3 \to S^2$ that organizes the discrete edge structure into ladder operators with $U(1)$ phase content (Papers 2, 25). Strand 3 is the Camporesi–Higuchi Dirac spectrum $|\lambda_n^{\mathrm{D}}| = n + 3/2$ with degeneracy $g_n^{\mathrm{D}} = 2(n+1)(n+2)$ that controls QED on S^3 (Paper 28). Strand 4 is the Wilson lattice gauge structure with SU(2) link variables on the same graph (Paper 30).

Each strand uses the unit three-sphere in a different role:

(a) S^3 as the *conformal image* of the hydrogenic spectrum (Paper 7);

(b) S^3 as the *base* of the bundle carrying the fine-structure-constant construction (Paper 2);

(c) S^3 as the *spin carrier* of the Camporesi–Higuchi Dirac operator (Paper 28);

(d) S^3 as the *gauge manifold* of SU(2) Wilson links (Paper 30).

Working hypothesis WH4 in the project register flags this as the strangest and most consequential coincidence in the framework: $S^3 = \mathrm{SU}(2)$ is the unique rank-1 non-abelian compact simply connected Lie group where (a)–(d) all coincide; in any other dimension or any other rank, the four roles separate. WH4 takes the position that the four strands are not four coincidences but a single spectral-triple structure viewed in four projections.

This paper makes WH4 explicit. We give a concrete construction of a finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ (Sec. 3); audit the standard Connes axioms one by one (Sec. 4); identify Papers 25, 28, 30, and 31 as structural sub-sectors of the resulting object (Sec. 5); read the Paper 2 combination formula $K = \pi(B + F - \Delta)$ inside the spectral-action language (Sec. 6); and place the Coulomb/HO asymmetry of Paper 31 in spectral-triple terms (Sec. 7).

1.2 What this paper is and is not

What it is. A synthesis paper. The construction is explicit, finite, and reproducible from existing GeoVac modules (`geovac/lattice.py`, `geovac/dirac_matrix_elements.py`, `geovac/fock_graph_hodge.py`); the lineage citations are to published mathematical work; the audit of the Connes axioms uses standard textbook definitions [1, 2, 3].

What it is not. A derivation of α. Paper 2 stays where it is: a structural observation with three independent spectral homes for the three pieces B, F, Δ but no first-principles derivation of the combination rule. This paper does not change that status; it gives the spectral-action language in which the open question is phrased without promoting it beyond that observation.

What it is not, second. A claim of the Connes–Chamseddine Standard Model. The GeoVac construction is a *rank-1* non-abelian almost-commutative extension; the SM spectral triple of [4] is rank ≥ 3 with specific finite-algebra content $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$. GeoVac is not the Standard Model. It is a finite spectral triple that is an *instance* of Marcolli–van Suijlekom 2014 [5] (with Perez-Sanchez 2024/2025 correction [9, 10]) at the data level and a GeoVac-original *extension* at the structural level (forced graph from Bertrand $\times$ Fock projection, master Mellin engine, explicit Hopf algebra

$\mathcal{H}_{\text{GV}}$, Tannakian dual U^*), with the empirically interesting property that one of its spectral-action coefficient combinations $K = \pi(B + F - \Delta)$ satisfies, after subtraction of the leading Sprint-A correction α^2, a residual $K - 1/\alpha - \alpha^2 \approx 1.2 \times 10^{-5}$ corresponding to a post-correction relative residual of 8.8×10^{-8} that matches $\pi^3\alpha^3$ to within 0.25% (the raw match $|K - 1/\alpha|/(1/\alpha)$ is 4.77×10^{-7}).

Internal register. This paper makes explicit the spectral-triple framing already operating informally in the working hypothesis register (`CLAUDE.md` §1.7, WH1 "PROVEN" — unconditional as of 2026-06-10, Paper 38). Papers continue under the rhetoric rule of §1.5: dual-description framing, no ontological priority claims, lead with computational results.

2 The Connes Data Recalled

A spectral triple is a triple $(\mathcal{A}, \mathcal{H}, D)$ where [1]

- $\mathcal{A}$ is an involutive (unital) algebra,
- $\mathcal{H}$ is a Hilbert space carrying a faithful $*$-representation $\pi : \mathcal{A} \to \mathcal{B}(\mathcal{H})$,
- $D : \mathcal{H} \to \mathcal{H}$ is an unbounded self-adjoint operator with compact resolvent, such that the commutators $[D, \pi(a)]$ are bounded for every $a \in \mathcal{A}$.

The triple is called *even* if there is a self-adjoint involution γ commuting with $\pi(\mathcal{A})$ and anticommuting with D; otherwise *odd*. A *real structure* on the triple is an antilinear isometry $J : \mathcal{H} \to \mathcal{H}$ with $J^2 = \epsilon\mathbb{1}$, $JD = \epsilon' DJ$, $J\gamma = \epsilon'' \gamma J$ (the signs $(\epsilon, \epsilon', \epsilon'')$ are determined by the KO-dimension mod 8 [1]); for the canonical Riemannian spin manifold of dimension n, the KO-dimension is n mod 8 and the signs are tabulated.

The two structural axioms of central interest below are:

(O1) *Order one:* $[[D, \pi(a)], \pi(b)^\circ] = 0$ for all $a, b \in \mathcal{A}$, where $b^\circ = JbJ^{-1}$ is the right action induced by the real structure;

(R) *Reality:* J exists with the signs above.

For an *almost-commutative* spectral triple $(\mathcal{A}_M \otimes \mathcal{A}_F, \mathcal{H}_M \otimes \mathcal{H}_F, D_M \otimes \mathbb{1} + \gamma_M \otimes D_F)$ [3, 4], the manifold sector $\mathcal{A}_M = C^\infty(M)$ is replaced in our finite setting by the algebra of complex-valued functions on a finite graph that converges to $C^\infty(M)$ in the continuum limit. This is exactly the move made by Marcolli and van Suijlekom in their gauge networks construction [5] and continued by Perez-Sanchez [9, 10]: replace the manifold algebra by an algebra on a discrete vertex set, with edges carrying connection data. The GeoVac construction sits in this lineage by design.

3 The GeoVac Triple: Explicit Construction

Fix a truncation $n_{\max} \geq 1$. All objects below are finite.

3.1 The algebra $\mathcal{A}_{\text{GV}}$

Let V_{Fock} denote the set of Fock-graph nodes at truncation $n_{\max}$, i.e. quantum-number triples (n, l, m_l) with $1 \leq n \leq n_{\max}$, $0 \leq l \leq n - 1$, $-l \leq m_l \leq l$. The total node count is $N_{\text{Fock}}(n_{\max}) = \sum_{n=1}^{n_{\max}} n^2$.

Definition 1 (Function algebra on the Fock graph). $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} := \mathbb{C}^{V_{\mathrm{Fock}}}$, *the commutative* $*$*-algebra of complex-valued functions on* V_{Fock} *under pointwise multiplication, with involution* $f^*(v) = \overline{f(v)}$.

This is precisely the manifold algebra of the gauge networks of [5] (their $C(V)$, with V the vertex set), with the GeoVac-specific choice that V is the Fock-projected S^3 graph of Paper 7 rather than a generic graph. It is finite-dimensional ($\dim_{\mathbb{C}} \mathcal{A}_{\mathrm{GV}} = N_{\mathrm{Fock}}$), unital (1 is the constant function 1), and abelian.

Remark 1 ($\mathcal{A}_{\mathrm{GV}}$ as C^*-envelope of an operator system). *In the parallel reading of GeoVac through the spectral-truncation framework of Connes and van Suijlekom [11], Definition 1 describes* $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} = \mathbb{C}^{V_{\mathrm{Fock}}}$ *as the* C^**-envelope* $C^*(\mathcal{O}_{n_{\max}})$ *of the operator system*

$$\mathcal{O}_{n_{\max}} := P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}},$$

where $P_{n_{\max}}$ *is the index-truncation projecting onto the span of the first* $n_{\max}$ *Fock shells (in the spinor-harmonic basis of* $L^2(S^3)$ *adapted to the SO(4) decomposition). This is structurally identical to the Connes–van Suijlekom worked example on* S^1 *(their* P_n *onto* $\mathrm{span}_{\mathbb{C}}\{e_1, e_2, \ldots, e_n\}$*, [11, §3.1]): a basis-index cutoff, not a Borel functional-calculus projection* $\chi_{[-\Lambda,\Lambda]}(D)$*. The genuine truncated algebra-image* $\mathcal{O}_{n_{\max}}$ *is* $*$*-closed but* not *multiplicatively closed; it carries nontrivial off-diagonal matrix elements*

$$\langle (n,l,m_l)|\, f\, |(n',l',m_l')\rangle = \int_{S^3} \overline{Y_{nlm_l}}\, f\, Y_{n'l'm_l'}\, dV$$

between distinct n*-shells. These overlap integrals are exactly the Gaunt / Wigner-3j angular machinery and the hypergeometric Slater radial integrals already implemented throughout the GeoVac pipeline (Paper 22 angular sparsity; Paper 14 ERI structure). The off-diagonal data therefore already exists in the framework;* $\mathcal{A}_{\mathrm{GV}}$ *as written in Definition 1 is its diagonal restriction.*

The two readings agree on diagonal elements and on the spectral data of D_{GV}*; they differ in what is taken as the algebraic content surviving the truncation. Connes and van Suijlekom note that taking the* C^**-envelope of* $\mathcal{O}_\Lambda$ *typically collapses to a "non-informative full matrix algebra" – the algebraic-structure information lives in the operator-system data thrown away by the envelope construction. Constructing* $\mathcal{O}_{n_{\max}}$ *explicitly in* $\mathcal{B}(\mathcal{H}_{\mathrm{GV}})$ *and computing its propagation number and Connes distance on node-evaluation states is the deliverable of the WH1 round-2 sprint (see* `debug/wh1_connes_vs_mapping_memo.md`*, 2026-05-03). The present subsection retains the* $\mathcal{A}_{\mathrm{GV}} = \mathbb{C}^{V_{\mathrm{Fock}}}$ *definition because (a) it matches the Marcolli–van Suijlekom gauge-network convention used throughout Papers 25 and 30, and (b) the spectral triple's structural verifications (Theorem 1 below; KO-dimension audit; Hodge structure) are unaffected by which representative of the equivalence class is taken.*

Proposition 1 (Propagation number of $\mathcal{O}_{n_{\max}}$). *Let* $\mathcal{O}_{n_{\max}} = P_{n_{\max}} C^\infty(S^3) P_{n_{\max}}$ *be the truncated operator system of Remark 1, viewed as a* $*$*-closed subspace of* $M_N(\mathbb{C})$ *with* $N = N_{\mathrm{Fock}}(n_{\max})$*. The propagation number in the sense of Connes & van Suijlekom ([11], Definition 2.39) satisfies*

$$\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2 \qquad \textit{for every } n_{\max} \in \{2,3,4\}, \tag{1}$$

matching the Toeplitz operator system $C(S^1)^{(n)}$ *([11], Proposition 4.2:* $\mathrm{prop}(C(S^1)^{(n)}) = 2$ *independently of* n*) verbatim. Equivalently,* $\dim_{\mathbb{C}}(\mathcal{O}_{n_{\max}}^2) = N^2$ *while* $\dim_{\mathbb{C}}(\mathcal{O}_{n_{\max}}) < N^2$ *strictly:*

$n_{\max}$	N	N^2	$\dim(\mathcal{O}_{n_{\max}})$	$\dim(\mathcal{O}_{n_{\max}}^2)$
2	5	25	14	25
3	14	196	55	196
4	30	900	140	900

The dimension fraction $\dim(\mathcal{O}_{n_{\max}})/N^2$ *falls* $0.560 \to 0.281 \to 0.156$ *across the three truncations: the operator system is a vanishing fraction of its* C^**-envelope as* $n_{\max}$ *grows, but its second power fills the envelope completely.*

Computational verification. The full construction and verification is documented in `geovac/operator_system.py` (the `TruncatedOperatorSystem` class) and `tests/test_operator_system.py` (24 tests, all passing; CPU time ∼20 s). In summary: a basis of $\mathcal{O}_{n_{\max}}$ is built from the multiplier matrices M_{NLM} corresponding to the hyperspherical harmonics $Y^{(3)}_{NLM}$, with selection rules $|n-n'|+1 \le N \le n+n'-1$ (SO(4) triangle), $|l-l'| \le L \le l+l'$ with $l+l'+L$ even (S^2 Gaunt parity), $M = m-m'$, and the angular S^2 part evaluated symbolically via `sympy.physics.wigner.wigner_3j`. Linear independence of $\{M_{NLM}\}$ is verified at machine precision; the propagation number is read off as the smallest k with $\dim(\mathcal{O}^k_{n_{\max}}) = N^2$. The result prop $= 2$ is robust: it is invariant under the choice of SO(4) radial-overlap placeholder (verified across three placeholders including a pure indicator that retains only the support pattern), so (1) is a structural invariant of the SO(4) selection rules at finite cutoff and does not depend on the radial value convention. See `debug/wh1_round2_propagation_memo.md` (2026-05-03) for the full computational record, including the explicit witness pair $a = M_{2,1,0}$ at $n_{\max} = 2$ for which $a^2 \notin \mathcal{O}_{n_{\max}}$ (Frobenius residual $\approx 14.9\%$), with the deficit concentrated on the top-shell diagonal entries. □

Remark 2 (Significance of prop = 2)**.** *Connes & van Suijlekom prove* $\mathrm{prop}(C(S^1)^{(n)}) = 2$ *for the Toeplitz operator system on the circle by showing that products of two Toeplitz shift generators reach every matrix unit* e_{ij} *[11]. The companion dual operator system* $C(\mathbb{Z})^{(n)}$ *has* $\mathrm{prop} = \infty$ *(their Proposition 4.3) — so the value* $\mathrm{prop} = 2$ *is non-trivial even at this level: it is specific to the Toeplitz construction, not to operator systems in general. Eq.* (1) *therefore states that the Fock-projected* S^3 *truncation is in the same propagation class as the Toeplitz* S^1 *— one of the simplest worked examples in the spectral-truncation literature — at the level of a quantitative Connes–van Suijlekom invariant, not merely at the level of the underlying construction. Together with the* SO(4) *isometry preservation discussed in Sec. 4 below, this is the strongest quantitative alignment between GeoVac and the published Connes-vS framework currently known.*

The mechanism producing $\mathcal{O}_{n_{\max}} \subsetneq M_N(\mathbb{C})$ *is the "raise-then-lower at the top shell escapes the truncation" boundary correction: a multiplier* $M_{N\ge2,L,M}$ *acting at the top shell* $n = n_{\max}$ *would, in the continuum, be balanced by intermediate sums over* $n' > n_{\max}$*, but the truncation* $P_{n_{\max}}$ *kills these contributions. This is structurally the same boundary effect as the pendant-edge theorem of Paper 28 [35],* $\Sigma_{\mathrm{graph}}(\mathrm{GS}) = 2(n_{\max}-1)/n_{\max}$ *for the graph-native ground-state self-energy, viewed from the operator-system side rather than the QED side.*

Corollary 1 (Structural specificity of prop = 2 on S^3)**.** *The propagation value* $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ *in Proposition 1 is structurally specific to the Connes–van Suijlekom spectral truncation of* S^3 *and* not *a generic feature of arbitrary finite-dimensional truncations of* S^3*. The natural circulant-style comparator on* S^3 *at* N *points*

$$A_{\mathrm{circ},N} := \mathrm{span}\{E_{i,i} : i = 0, \dots, N-1\} \subset M_N(\mathbb{C}),$$

i.e., the commutative C^**-subalgebra of diagonal matrices in the "point-evaluation" basis of* $\mathbb{C}^N$*, satisfies*

$$\mathrm{prop}(A_{\mathrm{circ},N}) = \begin{cases} 1 & \textit{(intrinsic envelope: } A_{\mathrm{circ},N} \textit{ is its own } C^*\textit{-algebra),} \\ \infty & \textit{(ambient envelope: } M_N(\mathbb{C}) \textit{ is non-abelian; no power of an abelian subalgebra reaches it).} \end{cases} \tag{2}$$

Verified at $N \in \{2, 5, 14, 30, 55, 140\}$, matching both dimension-matching conventions (envelope-match $N(\mathrm{circ}) = N(\mathrm{GV})$ and operator-system-match $N(\mathrm{circ}) = \dim(\mathcal{O}_{n_{\max}})$). Either reading of (2) is categorically different from the spectral-truncation value 2.

Computational verification. The construction lives in `geovac/circulant_s3.py` (the `CirculantS3Truncation` class) with verification in `tests/test_circulant_s3.py` (35 tests, all pass; runtime $\sim$1.4 s). Algorithmically, $\mathrm{prop}(A_{\mathrm{circ},N})$ is computed via the same vec-stack rank algorithm used in the proof of Proposition 1, applied to the diagonal-matrix basis of $A_{\mathrm{circ},N}$. The intrinsic-envelope reading exits at $k = 1$ since $\dim(A^1_{\mathrm{circ},N}) = N = \dim_{\mathbb{C}}(A_{\mathrm{circ},N})$ already saturates the intrinsic C^*-envelope. In the ambient-envelope reading, the $M_N(\mathbb{C})$ envelope has dimension N^2 for $N \geq 2$ but $\dim(A^k_{\mathrm{circ},N}) = N$ for every k (commutativity is preserved under self-products), so no finite k achieves saturation; prop $= \infty$ by Connes–van Suijlekom Definition 2.39. See `debug/wh1_round3_falsification_memo.md` (2026-05-03) for the full record, including the verbatim Connes–van Suijlekom Outlook §6 contrast (lines 2611–2618 of arXiv:2004.14115v2) between spectral and circulant truncations on S^1, of which Eq. (2) is the verbatim S^3 analog. □

Remark 3 (Connes distance on $\mathcal{O}_{n_{\max}}$: SDP framework and structural findings)**.** *The Connes distance on operator systems ([11], Eq. (2))*

$$d_D(\phi, \psi) = \sup_{x \in \mathcal{O}} \left\{ |\phi(x) - \psi(x)| : \|[D, x]\|_{\mathrm{op}} \leq 1 \right\}$$

restricted to pure node-evaluation states $\phi_v(x) := \langle v|x|v\rangle$ on the truncated operator system has been computed by semidefinite programming on $\mathcal{O}_{n_{\max}}$ at $n_{\max} = 2, 3$. The implementation lives in `geovac/connes_distance.py` *(a* `cvxpy` *SDP with the operator-norm constraint lifted to the LMI $\left[\begin{smallmatrix} I & [D,x] \\ [D,x]^* & I \end{smallmatrix}\right] \succeq 0$, and a gauge-fixing constraint orthogonalizing x to $\ker([D, \cdot]) \cap \mathcal{O}_h$); 17 tests in* `tests/test_connes_distance.py` *(all pass; runtime $\sim$9 s). Three structural findings emerged at the placeholder convention:*

1. ***SO(3) m-reflection forces zero distance.*** *For every $|n, l, -m\rangle$ and $|n, l, +m\rangle$ pair, $d_D(\phi_{|n,l,-m\rangle}, \phi_{|n,l,+m\rangle}) = 0$ exactly (one such pair at $n_{\max} = 2$, four at $n_{\max} = 3$). This is the SO(3) m-reflection symmetry of the operator system showing through, not a pathology: the symmetry identifies the two states as members of the same orbit, and any point-discriminating function must be constant on the orbit.*

2. ***Clean rational structure at $n_{\max} = 2$.*** *The five non-zero entries of the 5×5 distance matrix at $n_{\max} = 2$ group into ratios $1 : 2 : 3$ (specifically $28.87 : 57.73 : 86.60$), with $86.6020 = 50\sqrt{3}$ to five decimals. This rational/algebraic structure is lost at $n_{\max} = 3$ as the SDP gains degrees of freedom. We log this as an algebraic fingerprint of the truncated metric at small cutoff; whether it persists in any natural generalization is open.*

3. ***Non-monotonicity in the Fock-graph distance.*** *Pearson correlation between $d_D(\phi_v, \phi_w)$ and the combinatorial Fock-graph distance $d_{\mathrm{graph}}(v, w)$ (counting $\Delta n = \pm 1$ and $\Delta l = \pm 1$ ladder steps) is $\rho \approx -0.04$ at $n_{\max} = 2$, growing to $+0.14$ at $n_{\max} = 3$ (Spearman $0.09 \to 0.18$). The trend is weakly toward correlation but the absolute level is far from a discretized-geodesic metric at our reachable cutoffs.*

These findings should be read with three explicit caveats, none of which are addressed in the present paper.

- *The SO(4) radial overlap integral $I^{lLl'}_{nNn'}$ in the construction of $\mathcal{O}_{n_{\max}}$ (Remark 1) is currently a placeholder rather than the genuine Avery–Wen–Avery 3-Y integral on S^3. Round 2's propagation-number result (Proposition 1) is invariant under this choice; the Connes distance is not. The absolute distance values reported above are convention-dependent until the placeholder is replaced.*

- *The construction is on the scalar Fock basis $\mathcal{H}_{\mathrm{GV}}$ only; the spinor lift to $\mathcal{H}_{\mathrm{GV}} \otimes \mathbb{C}^4$ with the native Camporesi–Higuchi Dirac in the $(n_{\mathrm{D}}, \kappa, m_j)$ basis is not yet incorporated. The diagonal scalar Dirac on $\mathcal{H}_{\mathrm{GV}}$ has a large $\ker([D,\cdot]) \cap \mathcal{O}$ that makes the SDP unbounded on most state pairs; the metric values reported here use an off-diagonal Dirac proxy as a finite-distance regulator, which is a second convention choice.*

- *No comparison to the round-S^3 continuum geodesic distance is attempted. Such a comparison is the actual content of Connes–vS Gromov–Hausdorff convergence (Sec. 4 discussion of Row 9 of the round-1 mapping), which has not been proved on S^3 in the published literature.*

The structural findings (SO(3) m-reflection forced zeros; the $1:2:3$ rational ratios at $n_{\max} = 2$) are robust across the placeholder choice; the absolute distance values and the non-monotonicity diagnostic are not. See `debug/wh1_round3_connes_distance_memo.md` for the full record. Caveats (1) and (2) above are subsequently closed in Remark 4; one of the three structural findings is retracted there as a placeholder artifact.

Remark 4 (R3.1+R3.2 closure of round-3 caveats)**.** *Two of the three caveats above have been closed in subsequent sub-sprints (continuation of the same dispatch on 2026-05-03), and one of the three structural findings has been retracted as a placeholder artifact.*

R3.1 (Avery–Wen–Avery integral, placeholder $\to$ physical). *The placeholder for $I^{lLl'}_{nNn'}$ is replaced with the closed-form Avery–Wen–Avery 3-Y integral on S^3, factorized as $I_{\mathrm{rad}} \cdot G_{\mathrm{Gaunt}}$ with I_{rad} the Gegenbauer triple integral on χ and G_{Gaunt} the standard S^2 Gaunt integral. Implemented in `geovac/so4_three_y_integral.py` (sympy-exact, `lru_cache`), verified by hyperspherical orthonormality at $n_{\max} \le 3$ in exact arithmetic.* $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ *holds at $n_{\max} = 2, 3, 4$ under the physical default with dim sequences $14 \to 25,\ 55 \to 196,\ 140 \to 900$* bit-identical *to the placeholder values—confirming the round-2 invariance claim by replacement, not by sampling.*

The Connes distance recomputed under physical values retains the m-reflection forced zeros and the $50\sqrt{3}$ fingerprint, but retracts *the $1:2:3$ rational structure as a placeholder artifact: under Avery the distinct nonzero values at $n_{\max} = 2$ are $\{1.27, 86.60, 87.18, 174.93, 175.53, 262.11\}$ with no clean integer grouping. The $M_{2,0,0}$ multiplier rescales by $\sim 3\times$ between placeholder and physical values, propagating as $\sim 9\times$ on pairs that use it twice. More importantly, the non-monotonicity* flips sign *at $n_{\max} = 3$: Pearson goes from $+0.14$ (placeholder) to -0.36 (physical), Spearman from $+0.18$ to -0.33. The optimistic "slow trend toward correlation" reading is retracted; under physical values the metric is* anti-correlated *with the Fock-graph distance. Memo: `debug/wh1_r31_avery_wen_avery_memo.md`.*

R3.2 (spinor lift, scalar Fock $\to$ Weyl spinor on S^3). *The construction extends to the Weyl sector of the Camporesi–Higuchi spinor bundle on S^3 via Clebsch–Gordan decomposition $|\psi_{l,j=l+1/2,m_j}\rangle = \alpha_+|Y_{l,m_j-1/2}\rangle\otimes|\uparrow\rangle + \alpha_-|Y_{l,m_j+1/2}\rangle\otimes|\downarrow\rangle$ with $\alpha_\pm = \sqrt{(l \pm m_j + 1/2)/(2l+1)}$. Each scalar Avery multiplier M_{NLM} lifts to a CG-weighted sum of two scalar matrix elements; the resulting $\widetilde{\mathcal{O}}_{n_{\max}}$ is API-compatible with the existing SDP machinery. Implemented in `geovac/spinor_operator_system.`*

$\dim \widetilde{\mathcal{O}} = \dim \mathcal{O}$ *at every* $n_{\max}$ *(the CG decomposition introduces no new linear independencies); identity, $*$-closure, and m_j-reflection forced zeros all generalize naturally.*

Two clean R3.2 findings. ***(i)*** The truthful Camporesi–Higuchi Dirac on the spinor bundle is too n-degenerate *for the Connes distance to be well-defined on cross-shell pure-state pairs: at* $n_{\max} = 2$*, 24 of* 28 *cross-pairs give* $+\infty$ *because the n-shell-diagonal multipliers* $M_{N,L=\text{even},M=0}$ *commute with* $D = \text{diag}(n_{\text{Fock}} + 1/2)$*, making the SDP unbounded. This is the spinor analog of the round-3* ***`shell_scalar`*** *diagnostic and confirms the n-degeneracy obstruction is structural, not scalar-only.* ***(ii)*** Under a CH+offdiag perturbation that lifts the n-degeneracy, the metric remains anti-correlated *with the Fock-graph distance: Pearson nz* -0.36 *at* $n_{\max} = 2$, -0.26 *at* $n_{\max} = 3$*. The non-monotonicity is therefore robust under both the placeholder → Avery upgrade (R3.1)* and *the scalar → spinor lift (R3.2). Memo:* *`debug/wh1_r32_spinor_lift_memo.md`.*

Disambiguation settled. *The round-3 §5.4 two-reading question ("placeholder artifact" vs "pure-state-distance pathology") lands on the latter. R3.1 rules out the placeholder reading (anti-correlation* strengthens *under physical values). R3.2 rules out the scalar-Dirac-proxy reading (anti-correlation persists under the spinor lift). The Connes distance on* $\mathcal{O}_{n_{\max}}$ *or its spinor lift, with pure node-evaluation states, is structurally not a discretization of the round-*S^3 *geodesic distance at any cutoff we can reach. The third caveat above (no GH comparison on the full state space) remains the natural next move; whether the metric becomes monotone in the GH limit via a Kantorovich– Wasserstein lift to mixed states requires Peter–Weyl on* SU(2) *(genuinely new NCG mathematics; deferred by Connes & van Suijlekom 2021 three times, no published treatment on* S^3*).*

Remark 5 (Two-sided structural alignment)**.** *Connes & van Suijlekom's worked* S^1 *example exhibits the spectral-vs-circulant distinction along two axes simultaneously: the Toeplitz operator system* $C(S^1)^{(n)}$ *has* prop $= 2$ *(their Proposition 4.2), while the circulant matrix algebra* $C(C_m)$ *has* prop $= 1$ *trivially (it is its own* C^**-algebra). Eqs.* (1) *(Proposition 1) and* (2) *(Corollary 1) reproduce both axes verbatim on* S^3*:*

	spectral truncation	*circulant analog*
S^1 *(Connes–vS)*	$C(S^1)^{(n)}$, prop $= 2$	$C(C_m)$, prop $= 1$
S^3 *(this work)*	$\mathcal{O}_{n_{\max}}$, prop $= 2$	$A_{\text{circ},N}$, prop $= 1$ *or* ∞

The two-sided match foreclosing the natural objection that prop $= 2$ *might be a generic feature of any finite truncation of* S^3*. Together with the SO(4)-isometry-preservation property of Sec. 4 below (the property Connes & van Suijlekom themselves identify as distinguishing spectral truncation from circulant discretization), this is the strongest quantitative alignment between GeoVac and the published spectral-truncation framework currently known.*

3.2 The Hilbert space $\mathcal{H}_{\text{GV}}$

Let V_{Dirac} denote the set of Dirac-graph nodes at the matching truncation, labeled by Camporesi–Higuchi triples $(n_{\text{D}}, \kappa, m_j)$ with $0 \le n_{\text{D}} \le n_{\max} - 1$ (the standard convention $n_{\text{Fock}} = n_{\text{CH}} + 1$ from `geovac/dirac_matrix_elements.py`), Dirac–Coulomb relativistic angular momentum $\kappa \in \{-l-1, +l\}$ for each $0 \le l \le n_{\text{D}}$, and total angular momentum projection $m_j \in \{-j, \ldots, j\}$ where $j = |\kappa| - 1/2$. The total node count is

$$N_{\text{Dirac}}(n_{\max}) = \sum_{n=0}^{n_{\max}-1} g_n^{\text{D}} = \sum_{n=0}^{n_{\max}-1} 2(n+1)(n+2) = \tfrac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2).$$

Definition 2 (Spinor space on the Dirac graph)**.** $\mathcal{H}_{\mathrm{GV}}^{(n_{\max})} := \mathbb{C}^{V_{\mathrm{Dirac}}}$*, with the standard Hermitian inner product on* $\mathbb{C}^{N_{\mathrm{Dirac}}}$*.*

The faithful $*$-representation of $\mathcal{A}_{\mathrm{GV}}$ on $\mathcal{H}_{\mathrm{GV}}$ is by the natural pullback: for $f \in \mathcal{A}_{\mathrm{GV}}$ and $\psi \in \mathcal{H}_{\mathrm{GV}}$,

$$(\pi(f)\psi)(n_{\mathrm{D}}, \kappa, m_j) := f(n_{\mathrm{Fock}}(n_{\mathrm{D}}), l(\kappa), m_l(\kappa, m_j)) \cdot \psi(n_{\mathrm{D}}, \kappa, m_j), \tag{3}$$

where the maps $n_{\mathrm{Fock}}, l, m_l$ are the standard Dirac-to-Fock quantum-number bridges: $n_{\mathrm{Fock}}(n_{\mathrm{D}}) = n_{\mathrm{D}} + 1$, $l(\kappa) = -1 - \kappa$ if $\kappa < 0$ and $l(\kappa) = \kappa$ if $\kappa > 0$, and m_l via the standard $|j, m_j\rangle = \langle l, m_l; \frac{1}{2}, m_s|j, m_j\rangle|l, m_l\rangle|s, m_s\rangle$ Clebsch–Gordan decomposition. Each $\pi(f)$ acts diagonally; in the spinor basis it is the diagonal matrix $\mathrm{diag}(f(v_{\mathrm{Fock}}(\cdot)))$. The representation is faithful: $\pi(f) = 0 \Leftrightarrow f \equiv 0$ (every Fock node has at least one Dirac node above it for $n_{\max} \geq 1$).

3.3 The Dirac operator D_{GV}

The continuum Camporesi–Higuchi Dirac operator on the unit three-sphere has spectrum $|\lambda_n^{\mathrm{D}}| = n + 3/2$ with degeneracy $g_n^{\mathrm{D}} = 2(n+1)(n+2)$ [20]. We promote this to a finite operator on $\mathcal{H}_{\mathrm{GV}}$ in two equivalent ways and verify their agreement in the truncation.

Definition 3 (Spectral form of D_{GV})**.** $D_{\mathrm{GV}}^{\mathrm{spec}}$ *is the diagonal operator on* $\mathcal{H}_{\mathrm{GV}}$ *with eigenvalues* $\lambda_n^{\mathrm{D},\pm} = \pm(n_{\mathrm{D}} + 3/2)$ *on the eigenstate* $|n_{\mathrm{D}}, \kappa, m_j\rangle$ *with sign* $+$ *for* $\kappa < 0$ *and* $-$ *for* $\kappa > 0$ *(the standard "large-component" / "small-component" sign choice of the Camporesi–Higuchi convention [20]).*

Definition 4 (Graph form of D_{GV})**.** $D_{\mathrm{GV}}^{\mathrm{graph}}$ *is the discrete first-order hopping operator on the Dirac graph defined by the Camporesi–Higuchi nearest-neighbor edge set with weights chosen so that the spectrum matches Definition 3 in the truncation. The edge set and weights are documented in* `geovac/dirac_matrix_elements.py`*; the explicit form, written in the* $(n_{\mathrm{D}}, \kappa, m_j)$ *basis, is the Lichnerowicz-derived first-order operator with* κ*-coupled adjacent-shell amplitudes.*

Proposition 2 (Equivalence of spectral and graph forms)**.** *For all* $n_{\max} \geq 1$, $D_{\mathrm{GV}}^{\mathrm{spec}} = U^\dagger D_{\mathrm{GV}}^{\mathrm{graph}} U$ *for the unitary* U *that diagonalizes* $D_{\mathrm{GV}}^{\mathrm{graph}}$*. The agreement is exact in finite arithmetic at every truncation; in particular the spectrum, multiplicities, and trace invariants* $\mathrm{Tr}\,(D_{\mathrm{GV}})^{2k}$ *match the truncated Camporesi–Higuchi continuum values.*

Proof sketch. Both operators are constructed to have the truncated Camporesi–Higuchi spectrum by definition. The equivalence is the statement that there exists an orthonormal basis of $\mathcal{H}_{\mathrm{GV}}$ in which both are diagonal, which holds because $D_{\mathrm{GV}}^{\mathrm{graph}}$ is self-adjoint by construction (Hermitian conjugate edge weights) and has the prescribed spectrum. Numerical agreement at $n_{\max} \leq 6$ is verified by existing tests in `tests/test_dirac_matrix_elements.py` (108 tests) and `tests/test_qed_self_energy.py`. □

We use the spectral form $D_{\mathrm{GV}}^{\mathrm{spec}}$ (henceforth D_{GV}) for axiom verification because it is closed-form; the graph form is what one uses for finite-truncation computations of the spectral action. The π-free certificate of the Dirac sector (Paper 28 §graph_native_qed.6) applies to both: every eigenvalue is a half-integer rational, every degeneracy is a positive integer.

3.4 Bounded commutators

Proposition 3 (Bounded commutators)**.** *For every* $a \in \mathcal{A}_{\mathrm{GV}}$*, the commutator* $[D_{\mathrm{GV}}, \pi(a)]$ *is a bounded operator on* $\mathcal{H}_{\mathrm{GV}}$*.*

Proof. $D_{\rm GV}$ has finite-dimensional domain $\mathcal{H}_{\rm GV}$ at every $n_{\max}$, hence bounded. $\pi(a)$ is a diagonal multiplication operator, also bounded. Any commutator of bounded operators is bounded. $\square$

This is the trivial but necessary check: in finite truncation, every operator is bounded. The non-trivial content—that $\|[D_{\rm GV}, \pi(a)]\|$ is controlled uniformly in $n_{\max}$—is what makes the continuum limit a bona-fide spectral triple in the sense of Connes; for a Lipschitz function a, the commutator norm equals the Lipschitz constant on the graph metric, in exact analogy with the smooth case.

3.5 The triple, summarized

Putting Definitions 1, 2, 3 and Proposition 3 together:

Theorem 1 (The GeoVac spectral triple)**.** *For every $n_{\max} \geq 1$, the data $(\mathcal{A}_{\rm GV}^{(n_{\max})}, \mathcal{H}_{\rm GV}^{(n_{\max})}, D_{\rm GV}^{(n_{\max})})$ is a finite spectral triple in the sense of [1]: $\mathcal{A}_{\rm GV}$ is unital and involutive, $\pi : \mathcal{A}_{\rm GV} \to \mathcal{B}(\mathcal{H}_{\rm GV})$ is a faithful $*$-representation, $D_{\rm GV}$ is self-adjoint with bounded commutators. The continuum limit $n_{\max} \to \infty$ recovers the canonical spectral triple of the unit Riemannian spin three-sphere S^3 with the Camporesi–Higuchi Dirac operator.*

The continuum-limit clause is the spectral-truncation statement of Connes–van Suijlekom [11, 12, 13]: the GeoVac triples form a Cauchy sequence in the truncation order whose limit is the standard S^3 spectral triple.

4 Axiom Audit

We check each Connes axiom in turn, distinguishing what holds rigorously at finite $n_{\max}$ from what holds only in the continuum limit.

4.1 Order zero (involutive faithful representation)

Holds rigorously by Definitions 1–2 and Eq. (3). Verified in `tests/test_dirac_matrix_elements.py`.

4.2 Bounded commutators (regularity)

Holds rigorously at every $n_{\max}$ by Proposition 3.

4.3 Compact resolvent / finite summability

In the finite-$n_{\max}$ truncation $\mathcal{H}_{\rm GV}$ is finite-dimensional so this is automatic. The continuum statement is that $D_{\rm GV}$ has compact resolvent and is 3^+-summable (Dixmier-trace summable in dimension $3 + \epsilon$); this is the Camporesi–Higuchi summability of the continuum operator on S^3 and is a standard result [20, 1].

4.4 Order one

Proposition 4 (Order one)**.** *For every $a, b \in \mathcal{A}_{\rm GV}$, $[[D_{\rm GV}, \pi(a)], \pi(b)^\circ] = 0$ holds trivially because the right action $\pi(b)^\circ$ commutes with the left action $\pi(a)$ (the algebra is commutative) and with the diagonal-in-spectral-basis form of $D_{\rm GV}$.*

This is the standard observation that the order-one condition is automatic for any commutative spectral triple. Order-one becomes non-trivial only when one extends to an almost-commutative triple $\mathcal{A}_{\mathrm{GV}} \otimes M_n(\mathbb{C})$ with non-abelian fiber, at which point it constrains the form of fluctuations of D. This is exactly the setting where Papers 25 ($n = 1$, $U(1)$ fiber) and 30 ($n = 2$, SU(2) fiber) live; we treat them in Sec. 5.

4.5 Real structure

S^3 is odd-dimensional; in dimension n the canonical KO-dimension is n mod 8 for a Riemannian spin manifold, and the sign table [1, 3] for $n = 3$ gives

$$\epsilon = -1, \quad \epsilon' = +1, \quad \epsilon'' = \text{ (none)} \quad (\text{no } \gamma \text{ in odd dimension}).$$

The sign $\epsilon = -1$ reflects the fact that the 2-dimensional spinor representation of Spin(3) = SU(2) is quaternionic (pseudoreal): the standard charge-conjugation operator on this representation squares to $-\nVdash$, equivalently $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$ acts on a complex 2-dimensional space with antiunitary J satisfying $J^2 = -\nVdash$. This is the same sign that appears in the canonical Lorentzian Dirac formalism in $3 + 1$ dimensions for the charge-conjugation matrix C.

Proposition 5 (Real structure on $\mathcal{H}_{\mathrm{GV}}$)**.** *There exists an antilinear isometry $J_{\mathrm{GV}} : \mathcal{H}_{\mathrm{GV}} \to \mathcal{H}_{\mathrm{GV}}$ with $J_{\mathrm{GV}}^2 = -\nVdash$ and $J_{\mathrm{GV}} D_{\mathrm{GV}} = +D_{\mathrm{GV}} J_{\mathrm{GV}}$, namely the Camporesi–Higuchi charge-conjugation operator restricted to the truncation.*

Construction. The continuum charge conjugation J_{S^3} on the Camporesi–Higuchi spinor bundle satisfies the listed signs by construction [20, 1]. Restriction to the truncation $\mathcal{H}_{\mathrm{GV}}^{(n_{\max})}$ commutes with the spectral truncation because D_{GV} acts diagonally in the spectral basis and J_{S^3} permutes the spectral basis (sending $|n_{\mathrm{D}}, \kappa, m_j\rangle \mapsto |n_{\mathrm{D}}, -\kappa, -m_j\rangle$ up to a phase, whose antilinear-squaring gives $-\nVdash$ on the quaternionic 2-spinor representation). The restriction therefore inherits the signs. □

Finite-$n_{\max}$ audit (Sprint WH1-Connes Step 2, 2026-05). The qualitative argument above is now backed by a quantitative computational audit at $n_{\max} \in \{1, 2, 3\}$ on both the Weyl sector (`geovac/spinor_operator_system.py`, $\dim_{\mathcal{H}} = 2, 8, 20$) and the full-Dirac sector (`geovac/full_dirac_operator_s`
$\dim_{\mathcal{H}} = 4, 16, 40$), with J_{GV} realized as an antilinear permutation-phase operator $J = U\mathcal{K}$ (with $\mathcal{K}$ complex conjugation) and phase convention $\sigma(l, m_j) = i^{2m_j}(-1)^l$ giving $\sigma_a \overline{\sigma_{J(a)}} = -1$ on every basis label. Implementation in `geovac/real_structure.py`; verification in `tests/test_real_structure.py` (43 tests passing); raw audit data `debug/data/wh1_real_structure_audit.json`. The audit verdict for the four Connes-axiom conditions involving J at finite $n_{\max}$ on the truthful Camporesi–Higuchi Dirac is:

1. $J^2 = -\nVdash$ **HOLDS EXACTLY** at every $n_{\max} \in \{1, 2, 3\}$ on both sectors (bit-exact zero residual). The phase formula $\sigma_a \overline{\sigma_{\pi(a)}} = -1$ is verified pair-by-pair across the spinor and full-Dirac bases.

2. $J_{\mathrm{GV}} D_{\mathrm{GV}}^{\mathrm{truthful}} = +D_{\mathrm{GV}}^{\mathrm{truthful}} J_{\mathrm{GV}}$ **HOLDS EXACTLY** at every $n_{\max}$ (bit-exact zero residual). Mechanism: the truthful CH Dirac is diagonal in $(n_{\mathrm{Fock}}, l, m_j, \text{chirality})$ with eigenvalue chi $\cdot$ $(n_{\mathrm{Fock}} + 1/2)$ depending only on n and chi; J_{GV} permutes only m_j, leaving n and chi fixed, so the matrix relation $U\overline{D} = DU$ is diagonal-by-diagonal.

3. $J_{\mathrm{GV}} \mathcal{O}_{n_{\max}} J_{\mathrm{GV}}^{-1} = \mathcal{O}_{n_{\max}}$ **HOLDS EXACTLY** at every $n_{\max}$ (bit-exact zero residual on each generator a, with 0 of $1, 14, 55$ generators failing the $\mathcal{O}$-membership test). Mechanism: each

multiplier matrix $\widetilde{M}_{NLM}$ has J-conjugate $\widetilde{M}_{NL(-M)}$ via the $m_l \to -m_l$ symmetry of the Wigner-3j coupling coefficients (Edmonds Eq. 3.7.5 / Avery–Wen–Avery 3-Y integral); the operator system is closed under this involution.

4. $J_{\mathrm{GV}} D_{\mathrm{GV}}^{\mathrm{offdiag}} = +D_{\mathrm{GV}}^{\mathrm{offdiag}} J_{\mathrm{GV}}$ **FAILS** structurally (max residual 2×10^0 at $n_{\max} \in \{2, 3\}$; 10^{-2} at $n_{\max} = 1$ from the m-linear lifter alone). This is the natural counter-example: the round-3 `offdiag` Dirac is a hand-crafted Hermitian perturbation with E1-type hopping $D[a, b] = \alpha$ on the rule $|\Delta n| = |\Delta l| = 1, |\Delta m_j| \leq 1$, and the permutation-phase σ with $(-1)^l$ parity does not commute with this perturbation because the l-parity sign flips under bra/ket $l \to l \pm 1$. The genuine 4-component Camporesi–Higuchi spinor Dirac (which couples large/small components via the proper $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ angular gradient with Clifford-algebra γ matrices) would resolve this, but lies outside the present construction; the offdiag CH is best read as an SDP-bounding device (round-3 R3.5 Connes distance), not a spectral-action-foundational object.

5. **Order-zero** $[a, JbJ^{-1}] = 0$ **and order-one** $[[D, a], JbJ^{-1}] = 0$ **FAIL at finite** $n_{\max} \geq 2$ with residuals $O(0.05\text{–}0.2)$ in operator norm. At $n_{\max} = 2$: 124 of 196 pairs fail order-zero (max residual 5.5×10^{-2}); 43 of 196 pairs fail order-one (max residual 1.0×10^{-1}); at $n_{\max} = 3$: 2,594 of 3,025 and 1,337 of 3,025 respectively (max residuals 7.9×10^{-2} and 2.0×10^{-1}). The conditions hold trivially at $n_{\max} = 1$ (single-generator operator system).

Reading of the order-condition failures. The order-zero condition $[a, JbJ^{-1}] = 0$ is the abelian-classical statement *multiplications by classical functions commute*, automatic for $f, g \in C^\infty(S^3)$ in the continuum. At finite $n_{\max}$, the truncated objects $P_{n_{\max}} f P_{n_{\max}}$ and $J P_{n_{\max}} g P_{n_{\max}} J^{-1}$ are no longer multiplications by classical functions; they are operator-system elements ([11, Sec. 2]), and their commutator is generically nonzero. This is structurally the same finite-resolution mechanism as the failure of multiplicative closure that defines the Connes–van Suijlekom truncated operator system itself ($a \cdot b \notin \mathcal{O}$ in general; round-2 propagation result). By the conjectured GH convergence on S^3 (Track-TS-A R2.5 keystone) the residuals should decay in the $n_{\max} \to \infty$ limit, with quantitative scaling at $n_{\max} \in \{4, 5\}$ flagged as deferred future work.

Net status of the real structure at finite $n_{\max}$. The introductory abstract flagged as open "whether the GeoVac Dirac graph operator carries a real structure matching the topological KO-dimension of S^3 exactly at finite $n_{\max}$ or only in the continuum limit." The Sprint WH1-Connes Step 2 audit closes this question *positively for the truthful CH Dirac*: the three load-bearing conditions ($J^2 = -\mathbb{1}$, $JD = +DJ$, and $J\mathcal{O}J^{-1} = \mathcal{O}$) all hold with bit-exact zero residual at every finite $n_{\max} \in \{1, 2, 3\}$ on both Weyl and full-Dirac sectors. The order-zero and order-one conditions (involving the abelian-multiplication identity that defines the classical limit of the algebra) fail at finite $n_{\max} \geq 2$ as finite-resolution artifacts of the same class as the operator-system non-multiplicative-closure already named by Connes–van Suijlekom. The offdiag CH Dirac fails $JD = +DJ$ structurally and is therefore not a spectral-action-foundational object; the truthful CH is the natural domain for both the present audit and the inner-fluctuation calculus relevant to the Track 3 almost-commutative extension (Sec. 6 discussion).

4.6 $\mathbb{Z}_2$-grading

Proposition 6 (Odd triple)**.** *The GeoVac spectral triple is* odd*: there is no chirality operator γ on $\mathcal{H}_{\mathrm{GV}}$ commuting with $\pi(\mathcal{A}_{\mathrm{GV}})$ and anticommuting with D_{GV}.*

Proof. S^3 is odd-dimensional and the Camporesi–Higuchi Dirac operator on S^3 has no chirality operator in the standard sense. The truncation inherits the same statement. □

This is correct, not a defect. Odd-dimensional spectral triples do not carry a $\mathbb{Z}_2$-grading; their spectral action is constructed from the absolute value $|D|$ rather than from D^2 projected onto chirality eigenspaces, and this is precisely the structure that Paper 28's QED-on-S^3 computations use.

4.7 Summary

Axiom	Status at finite $n_{\max}$	Status in continuum limit
Order zero (faithful $*$-rep)	Holds rigorously	Holds rigorously
Bounded commutators	Holds rigorously	Holds rigorously
Compact resolvent	Auto (finite dim)	Holds (CH summability)
Finite summability	Auto	3^+-summable
Order one	Holds trivially	Holds trivially
Real structure J	Holds (CH conjugation, audited)	Holds
$J^2 = -\mathbb{1}$	Holds exactly $\forall n_{\max}$	Holds
$JD = +DJ$ (truthful CH)	Holds exactly $\forall n_{\max}$	Holds
$J\mathcal{O}J^{-1} = \mathcal{O}$	Holds exactly $\forall n_{\max}$	Holds
Order zero $[a, JbJ^{-1}] = 0$	Fails at $n_{\max} \geq 2$ (artifact)	Holds
Order one	Fails at $n_{\max} \geq 2$ (artifact)	Holds
$\mathbb{Z}_2$-grading	Absent (odd)	Absent (odd)

Table 1: Axiom audit for the GeoVac triple $(\mathcal{A}_{\rm GV}, \mathcal{H}_{\rm GV}, D_{\rm GV})$ at signature $(3, 0)$ (Riemannian, KO-dimension 3). All standard axioms are satisfied; the triple is real, odd, and of KO-dimension 3 in the continuum limit. The Lorentzian-extension audit at $(m, n) = (4, 6)$ via Sprint L2-D extends this picture; see Table 2 below.

Lorentzian-extension audit at $(m, n) = (4, 6)$ (Sprint L2-B/C/D, 2026-05-16). The Riemannian audit above sits at signature $(s, t) = (3, 0)$ (KO-dim 3 in the standard Spin^c convention). The Lorentzian extension to $(s, t) = (3, 1)$ via the Bizi–Brouder–Besnard 2018 [48] Krein-space spectral-triple classification translates to the BBB index pair $(m, n) = (4, 6)$ (via $m \equiv t+s$, $n \equiv t-s$ (mod 8)). Sprints L2-B (Krein space, `geovac/krein_space_construction.py`), L2-C (Lorentzian Dirac, `geovac/lorentzian_dirac.py`), and L2-D (Connes axiom audit, `geovac/connes_axiom_audit_31.py`) verify four BBB-predicted-sign axioms bit-exactly at $n_{\max} \in \{1, 2, 3\}$ and $N_t \in \{1, 11, 21\}$ on the truthful Camporesi–Higuchi spatial Dirac. Per BBB Table 1 at $(4, 6)$: $\varepsilon = +1$, $\varepsilon'' = -1$, $\kappa = -1$, $\kappa'' = +1$ (verified directly against [48] v2 PDF). Sub-deliverables and verdicts are summarised in Sec. 8.6 and the subsections § 8.7, § 8.8, § 8.9.

Structural finding on χD. The universal BBB Sec 5(v) axiom $\chi D = -D\chi$ fails bit-exactly on the pair (γ^5_K, D_L) built in Sprints L2-B/C, with Frobenius residual $2\|D_{\rm GV}\|_F$ at $N_t = 1$ (numerical values $6.00, 18.33, 38.88$ at $n_{\max} = 1, 2, 3$ respectively). The mechanism is structural and follows from two locked Riemannian-side conventions: (i) the Sprint L2-B convention identifies `FullDiracLabel.chirality` with the γ^5 eigenvalue (the chiral-basis Peskin–Schroeder convention which keeps the Riemannian limit identification with $\mathcal{H}_{\rm GV}$ transparent); (ii) the truthful

Axiom	KO-dim 3 (Riemannian)	$(m, n) = (4, 6)$ Lorentzian
J^2	$-\nVdash$ (bit-exact, $\forall n_{\max}$)	$+\nVdash$ (bit-exact, BBB $\varepsilon = +1$)
JD	$+DJ$ (bit-exact, $\forall n_{\max}$)	$+DJ$ (bit-exact, BBB universal Sec 5(v))
$J\chi$	n/a (no χ in odd dim)	$\{J, \chi\} = 0$ (bit-exact, BBB $\varepsilon'' = -1$)
$J\eta$	n/a (no η in Riemannian)	$\{J, \eta\} = 0$ (bit-exact, BBB $\varepsilon\kappa = -1$)
$\eta\chi$	n/a	$\{\eta, \chi\} = 0$ (bit-exact, BBB $\varepsilon''\kappa'' = -1$)
χD	n/a (no χ)	$-D\chi$ (BBB universal); FAILS on truthful D_{GV} (structu
Order zero $[a, JbJ^{-1}]$	5–20% finite-resolution	$\leq 7\%$ finite-resolution (sample-of-3)
Order one $[[D, a], JbJ^{-1}]$	5–20% finite-resolution	$\leq 11\%$ finite-resolution (sample-of-3)

Table 2: Lorentzian-extension axiom audit at $(m, n) = (4, 6)$ via Sprint L2-B/C/D, 2026-05-16 (`debug/l2_{b,c,d}_*_memo.md`, `debug/data/l2_d_connes_axiom_audit_31.json`). Four BBB-predicted-sign axioms hold bit-exactly across the $(n_{\max}, N_t) \in \{1, 2, 3\} \times \{1, 11, 21\}$ panel. The BBB universal $\chi D = -D\chi$ axiom fails structurally on the truthful Camporesi–Higuchi D_{GV}: this is a structural feature of GeoVac, not a bug, and is the load-bearing scope finding of Sprint L2-D. Resolution path R1+R2 (truthful for axioms requiring $JD = +DJ$, offdiag CH for axioms requiring $\{\chi, D\} = 0$) mirrors the Riemannian-side R3.5/Paper 42 trade. WH1 PROVEN (Theorem 5) is NOT re-opened by these results; the Lorentzian audit extends the framework rather than re-tests its Riemannian foundation.

Camporesi–Higuchi Dirac is diagonal in chirality with eigenvalue chi $\cdot\, (n_{\mathrm{Fock}} + 1/2)$, so D_{GV} *commutes* with γ^5 rather than anticommuting with it. These two facts together force $\{\gamma^5, D_L\} = 2i\,(\gamma^5 D_{\mathrm{GV}}) \otimes I_{N_t} \neq 0$ in general. This is a structural property of GeoVac's chirality-diagonal D_{GV} in the Peskin–Schroeder chiral basis, not a basis-convention error and not a defect of the Lorentzian lift. Three resolutions are documented: **(R1) accept the structural finding** (truthful D_{GV} is Krein-self-adjoint with the BBB-predicted (4, 6) signs on the J-relations and is the right object for axioms requiring $JD = +DJ$, even though it does not form a full BBB indefinite spectral triple); **(R2) use the offdiag Camporesi–Higuchi Dirac** for axioms that require $\{\chi, D\} = 0$ (the same offdiag CH used as an SDP-bounding device in the Riemannian Connes-distance round 3 R3.5); **(R3) redefine γ^5 on $\mathcal{H}_{\mathrm{GV}}$** as an off-diagonal grading anticommuting with truthful D_{GV}, which breaks Sprint L2-B's chirality-as-γ^5 identification and is essentially a basis-convention rebuild. The recommended path for Sprint L2-E (Krein-level Paper 42 redo) is R1 + R2 in parallel, matching the Riemannian-side practice of using truthful CH for spectral-action and reality-condition tests while using offdiag CH for SDP-bounding Connes-distance computations.

Theorem 2 (Rigorous identification)**.** *The continuum limit of the GeoVac spectral triple is the canonical spectral triple of the unit Riemannian spin three-sphere with the Camporesi–Higuchi Dirac operator. In particular it is real, odd, and of* KO*-dimension* 3*.*

This is a structural theorem; no fitting parameters or empirical inputs. The audit in Table 1 establishes that the spectral-triple framing is not metaphorical.

5 Sub-sector Identification: Papers 25, 28, 30, 31

We now read the major framework papers as projections of the GeoVac triple.

5.1 Paper 28 (QED on S^3): the spectral action

Paper 28 [35] computes Tr $f(D^2/\Lambda^2)$ and related spectral sums on the Dirac operator D_{GV} in the continuum limit, with SO(4) vertex selection rules from the Hopf-bundle decomposition. In the language of Sec. 2:

- Theorem T9 (Paper 28 Sec. T9): the spectral zeta of D^2 is a two-term polynomial in π^2 with rational coefficients at every integer s. In spectral-triple language: the heat-kernel coefficients $a_k(D^2)$ on S^3 saturate at $k = 2$ (Paper 28's "SD two-term exactness theorem").
- Self-energy structural zero (Paper 28 Theorem 4): $\Sigma(n_{\mathrm{ext}} = 0) = 0$ on the continuum spectral sum, identically by vertex parity. This is the standard heat-kernel bosonic sector cancellation specialized to the Camporesi–Higuchi spectrum.
- Vector-photon QED on S^3 (Paper 28 §vector_photon_qed): adding photon (q, m_q) labels with Wigner $3j$ vertex coupling recovers 7/8 selection rules for scalar electrons and 8/8 for Dirac electrons. The $1/(4\pi)$ per-loop calibration is the S^2 Weyl exchange constant of the Hopf base.

In the spectral-triple language Paper 28 is the spectral-action computation Tr $f(D^2_{\mathrm{GV}}/\Lambda^2)$ at one loop; the Camporesi–Higuchi spectrum is the input data of D_{GV}; the SO(4) selection rules are the harmonic decomposition of the spinor bundle. Paper 28 is not a free-standing observation about S^3; it is the spectral action of the present triple.

5.2 Paper 25 (Hopf graph as lattice gauge): the Cartan torus

Paper 25 [34] introduces the signed node-edge incidence matrix B of the Fock graph and identifies $L_0 = BB^\top$ as the matter propagator and $L_1 = B^\top B$ as the discrete Hodge-1 Laplacian. In the spectral-triple language: extend $\mathcal{A}_{\mathrm{GV}}$ to $\mathcal{A}_{\mathrm{GV}} \otimes \mathbb{C}$ (trivial M_1 fiber) and consider inner fluctuations of D_{GV} in the abelian $U(1)$ inner-derivation algebra. The fluctuated Dirac operator $D_{\mathrm{GV}} + A$, with $A \in \Omega^1_D$ a self-adjoint 1-form in the D_{GV}-calculus, has L_1 as its quadratic kinetic term. This is the Cartan-subalgebra (maximal-torus) sector.

Paper 25's L_1 is the $U(1)$-fluctuated kinetic term of the GeoVac spectral action; its assignment as "photon propagator" in Paper 25 is exactly the spectral-action reading.

5.3 Paper 30 (SU(2) Wilson gauge): the full non-abelian sector

Paper 30 [36] builds the SU(2) Wilson lattice gauge theory on the same graph and proves: (a) the maximal-torus reduction reproduces Paper 25's $U(1)$ action exactly; (b) the weak-coupling kinetic term equals L_1 acting on $\mathfrak{su}(2)$-valued 1-cochains. In the spectral-triple language: extend $\mathcal{A}_{\mathrm{GV}}$ to $\mathcal{A}_{\mathrm{GV}} \otimes M_2(\mathbb{C})$ (a non-abelian M_2 fiber, equivalently a rank-1 non-abelian almost-commutative extension of the GeoVac triple). The order-one condition for this extension is non-trivial and constrains the form of fluctuations to be $\mathfrak{su}(2)$-valued connections, exactly Paper 30's link variables. The Wilson plaquette action is the lattice-gauge implementation of the spectral-action curvature term in this extension.

Paper 30 is the rank-1 non-abelian almost-commutative extension of the GeoVac triple; Paper 25 is the abelian projection of Paper 30; together they exhibit the standard Cartan torus / full group structure of the gauge sector of the present triple.

5.4 Paper 31 (universal/Coulomb partition): the $\mathcal{A}/D$ split

Paper 31 [37] partitions GeoVac results into a universal sector that depends only on the algebra $\mathcal{A}$ (angular sparsity, composed scaling, selection rules) and a potential-specific sector that depends on the Dirac operator D (spectrum, calibration κ, transcendental content, rigidity, α observation). In the spectral-triple language this is exactly the $\mathcal{A}/D$ split: properties of $\mathcal{A}_{\mathrm{GV}}$ alone (rank, dimension, representation content) transfer to any spectral triple with the same algebra; properties of D_{GV} specifically depend on the choice of Dirac operator.

Paper 31 is the structural reading of Paper 22 (angular sparsity theorem, an $\mathcal{A}$-statement) and Paper 23 (nuclear shell model, a different D-choice on the same $\mathcal{A}$): when one keeps $\mathcal{A}$ fixed and varies D, the universal results transfer; when one varies $\mathcal{A}$ (say, replace S^3 with S^5, Paper 24), the entire structural picture reorganizes.

5.5 Summary of sub-sector identification

Paper	Object in this paper's language	Sector
Paper 7	$\mathcal{A}_{\mathrm{GV}}$ continuum limit	Manifold sector
Paper 22	Selection rules from $\mathcal{A}_{\mathrm{GV}}$	$\mathcal{A}$ sector (universal)
Paper 23	D-replacement: nuclear shell model	Different D, same $\mathcal{A}$
Paper 24	Different $\mathcal{A}$: S^5 holomorphic	Distinct triple (HO)
Paper 25	$U(1)$-fluctuated $D_{\mathrm{GV}} + A$	Cartan-torus inner fluctuation
Paper 28	$\mathrm{Tr}\, f(D_{\mathrm{GV}}^2/\Lambda^2)$	Spectral action (one loop)
Paper 30	SU(2)-fluctuated, M_2 fiber	Non-abelian extension
Paper 31	$\mathcal{A}/D$ split	Structural partition

Table 3: Sub-sector identification of the major framework papers as projections of the GeoVac spectral triple.

Track NI as an explicit composed real-space spectral triple. The "Different D, same $\mathcal{A}$" row of Table 3 (Paper 23 [32], nuclear shell model) covers a second class of construction in Paper 23 §VI: the composed nuclear–electronic deuterium Hamiltonian, a 26-qubit Hilbert-space tensor product of a 16-qubit deuteron register (harmonic-oscillator basis) with a 10-qubit hydrogenic electronic register (Fock-projected S^3 lattice), coupled through a cross-register hyperfine $\mathbf{I}\cdot\mathbf{S}$ operator. This is structurally distinct from both the SM almost-commutative geometry (which couples a Riemannian factor with a finite *internal* factor) and from the standard spectral-truncation convergence machinery (which handles single triples or one-infinite-plus-one-finite products); it is an explicit Connes-style real-space multi-particle spectral triple, validated against the analytic 21 cm singlet–triplet hyperfine gap at 1.62×10^{-7} Ha. The positioning observation (that the canonical NCG / spectral-triple literature does not contain a directly comparable construction) is formalized in the standalone Track NI Zenodo memo (`papers/group4_quantum_computing/track_ni_spectral_triple_zer` The W2b cross-manifold structural blocker of Sec. 8.5 is exactly the convergence-theorem gap that surrounds this construction.

Paper 39 closes the same-manifold tensor-product case. The same-manifold sub-case of W2b — the tensor product $\mathcal{T}_{S^3}^{\lambda_a} \otimes \mathcal{T}_{S^3}^{\lambda_b}$ of two truncated Camporesi–Higuchi spectral triples on S^3 at distinct focal lengths $\lambda_a \neq \lambda_b$ — is closed in a companion standalone paper (Paper 39 [41]). The result extends Paper 38's five-lemma machinery factor-by-factor to the KO-dim-6 tensor product, with the joint Lipschitz comparison constant and the joint Lipschitz-distortion height term both controlled by the Connes–Marcolli graded anticommutation $\{D_a \otimes I, \gamma_a \otimes D_b\} = 0$ via a Pythagorean refinement of the triangle inequality on the graded module. This is the mathematical foundation for the GeoVac multi-focal-composition program, where two electrons in distinct hydrogenic shells live on S^3 at distinct focal lengths $\lambda = n/Z$; the convergence theorem of Paper 39 says that the cross-register joint operator system on $\mathcal{O}_{n_{\max,a}} \otimes \mathcal{O}_{n_{\max,b}}$ converges qualitatively to the continuum tensor algebra $C^\infty(S^3 \times S^3)$ in van Suijlekom's state-space Gromov–Hausdorff distance (the Latrémolière propinquity strengthening is the same open gap as for the single S^3 factor; Remark 43). The two remaining cross-manifold variants ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Bargmann}}$ and $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\text{Riem}}$) remain the W2b content of Sec. 8.5.

Remark 6 (Multi-center composition: the reducible–irreducible transition)**.** *The multi-focal program above concerns two electrons at distinct focal lengths on a single S^3; the complementary* multi-center *composition concerns the orthogonal projections P_A onto the single-center orbital subspaces of a polyatomic molecule in the shared Hilbert space with overlap metric G. The natural setting is the* $*$-representation *theory of orthogonal projections. For a* diatomic *the pair $\{P_A, P_B\}$ is* tame *(Type I): by the two-projection theorem (Halmos) the generated $*$-algebra decomposes into blocks of dimension ≤ 2, completely classified by the principal angles between the two subspaces—for LiH at R_{eq} the three angles $7.6°/44.7°/67.3°$, the middle one at the maximal-non-commutativity ceiling $\|[P_A, P_B]\| = \frac{1}{2}$. The $*$-algebra generated by* three or more *projections is, by contrast, $*$-wild in general; the passage from two to three centers is precisely this tame-to-wild threshold. What the molecular data adds is that the three-center configuration lands on an* irreducible *(Schurian) point of that wild family: for linear* BeH_2 *(the σ pair $\{1s, 2p_0\}$ per center) the three center-projections generate the* full *matrix algebra M_6 and act irreducibly (commutant $= \mathbb{C}\not{\Bbbk}$), welding the whole space into a single indecomposable block that admits no decomposition; every pair remains reducible.*

Tameness is not composability, and conflating the two misreads the obstruction. The diatomic pair is completely classified, but it emphatically does not commute—$\|[P_A, P_B]\| = \frac{1}{2}$ is the maximal possible value—and it is that two-*projection non-commutativity, not the three-projection wildness, which constitutes the multi-focal composition wall recorded above. The obstruction accordingly has two distinct rungs. At two centers the projections fail to commute while remaining completely classified by their principal angles: the obstruction is quantitative, and the cost of removing it is known in closed form. At three centers the classification itself fails, and no complete list of invariants exists. Forcing commutation at the first rung—Löwdin orthogonalization—is the framework's sparsity-destroying move; at the second rung there is no corresponding normal form available to force.*

The wildness is specific to the subspace configuration and is not *a generic consequence of non-commutativity. The framework's other non-commutative angular structure—$SO(3)/SU(2)$ coupling, the Gaunt and $6j$ machinery—is the representation theory of a* compact *group, hence type I by Peter–Weyl, hence tame however non-abelian it may be. The contrast is quantitative: on the menu $l = 0, 1, 2$ the $\mathfrak{su}(2)$ generators satisfy $\|[J_a, J_b]\| = 2$ yet their commutant is three-dimensional (one dimension per irreducible constituent, and $\sum_\lambda m_\lambda^2$ in general—four for $l=1$ at multiplicity two, five for $l=1 \oplus l=2 \oplus l=2$), whereas the three center-projections, at the* smaller *non-commutativity $\|[P_A, P_B]\| \approx 0.49$, have commutant $\mathbb{C}\not{\Bbbk}$. Non-abelianness and wildness are independent properties. This is the structural reason the framework's angular sparsity is graded, finite and potential-independent (Paper 22), and the reason no Gaunt-style graded reduction should be expected to rescue multi-center composition: arbitrary configurations of subspaces are not group representations and escape Peter–Weyl.*

Two invariants must be kept separate. (i) The categorical *statement—pairs reducible, triple irreducible—is robust: it survives enriching the per-center σ menu $\{1s, 2p_0\} \to \{1s, 2s, 2p_0\}$ ($M_6 \to M_9$, irreducible at both structural $Z = 1$ and physical $Z = 2$ charges) and holds at real molecular geometries. It is, however, a statement about center* count*: three coupled projections in general* *position are irreducible at essentially any bonding-relevant separation, the commutant reverting to reducible only once residual overlaps fall below resolution. (ii) The bonding-dependent content is instead the* magnitude *of the irreducible three-body coupling, measured by the nested commutator* $\|[[P_A, P_B], P_C]\|$ *(which cannot even be formed for a diatomic):* ≈ 0.35 *at bonding, decaying smoothly to zero as the centers separate.*

Geometry sharpens the picture. At exactly *linear geometry the in-plane basis* $\{s, p_x, p_z\}$ *per center decouples by axial symmetry into a* σ *and a* π *sector and the nine-dimensional algebra is reducible;* any *bend mixes the sectors and welds the full space into one irreducible* M_9 *block (bit-checked across* $160^\circ \to 90^\circ$ *and at the* H_2O *geometry* 104.5°*, for apex charges* $Z = 1, 2, 4$*)—the operator-algebra face of the Renner–Teller mechanism. As an internal observation the configuration operator* $F = \sum_i P_i$ *carries, over molecular geometry, a lattice of conical intersections each with a* π *(*$\mathbb{Z}_2$*) Berry phase and, under bending, a structural Renner–Teller promotion; this is the* generic *topology of a symmetric-triatomic degeneracy, distinct from the molecule's physical electronic conical intersection, not an identity with it. (Computed and regression-pinned in* ***tests/test_paper32_composition.py*** *on validated two-center overlaps, with the commutant routine self-validated on cases of known answer; chronicle in CHANGELOG v5.1.0.)*

Remark 7 (The configuration operator's Berry curvature: a flat $\mathbb{Z}_2$ bundle over nuclear geometry)**.** *The configuration operator* $F = \sum_i P_i$ *of the previous remark varies over the nuclear-geometry manifold* $\mathcal{M}$*, and its geometric-phase structure is the finished form of the "lattice of conical intersections" noted there. Because the* σ *orbitals are real,* F *is real-symmetric at every geometry, so its Berry connection is* flat *with only a* $\mathbb{Z}_2$ *(*$\pm$*) holonomy: the curvature is not a smooth field but a sum of* π*-flux Dirac deltas at the conical intersections,*

$$\Omega \;=\; \sum_{CI} \pi\, \delta^2(m - m_{CI}),$$

the Longuet–Higgins sign-change / molecular Aharonov–Bohm structure (Longuet-Higgins; Berry; Mead–Truhlar) realized here for F *rather than the electronic Hamiltonian. On the linear-stretch plane* (d_1, d_2) *the Fukui–Hatsugai–Suzuki lattice flux is exactly* 0 *or* π *on every plaquette (machine-exact), with three* π*-sources—the central intersection at* $(2.45, 2.45)$ *and an antisymmetric-stretch mirror pair— carried by the two crossing bands and absent on the other four. The holonomy is the* $\mathbb{Z}_2$ *representation of* $\pi_1(\mathcal{M} \setminus \mathrm{CI})$ *sending a loop to* $\pi \cdot (\#\,\textit{enclosed} \bmod 2)$*, cross-checked by direct parallel transport (*π *around one intersection and around all three).*

Each intersection is the real (equatorial) section of a charge-$\frac{1}{2}$ *Berry monopole: in a fixed real frame the local* 2×2 *model is* $F_0 + a_z\sigma_z + a_x\sigma_x$ *with the* σ_y *coefficient* identically zero*—the real family lives on the sphere's equator—and branching vectors* $g = \nabla a_z$ *along the symmetric stretch (the tuning mode) and* $h = \nabla a_x$ *along the antisymmetric stretch (the coupling mode), with nonzero winding. The* $\mathbb{Z}_2$ *(rather than* $U(1)$*) character is* forced by time-reversal symmetry*—the reality of the Coulomb overlaps—not by planarity: real orbitals give real overlaps in any geometry, so a non-coplanar real center leaves* F *real-symmetric and does not lift the phase. Breaking time reversal does. A magnetic (Peierls) phase* ϕ *threading the* H–Be–H *loop makes* F *complex-Hermitian, opens the empty* σ_y *axis, and turns each intersection into a genuine* $U(1)$ *Berry monopole of integer charge: the first Chern number over a small sphere enclosing* $(d^*, d^*, 0)$ *in* (d_1, d_2, ϕ) *is* ± 1 *(opposite on the two crossing bands), and the real* $\phi = 0$ *plane cuts the monopole at its equator, recovering the* π *holonomy. This is an internal observation about the geometry of the composition, distinct from the molecule's physical electronic conical intersection. (Regression-pinned in* ***tests/test_paper32_berry_curvature.py****; chronicle in CHANGELOG v5.1.0.)*

Remark 8 (Four centers: a continuous four-body modulus (the $\tilde{D}_4$ analog))**.** *The three-center geometric phase of Remark 7 is discrete ($\mathbb{Z}_2$ conical intersections). A* fourth *center adds a genuinely new,* continuous *four-body invariant. With $S_{ij} = Q_i^\top Q_j$ the metric-orthonormal overlaps between center subspaces, the loop holonomy of this overlap connection around the 4-cycle, $W = \mathrm{Tr}(S_{12}S_{23}S_{34}S_{41}) = \mathrm{Tr}(P_1P_2P_3P_4)$, is invariant under the per-center gauge $Q_i \mapsto Q_iO_i$ ($O_i \in O(r)$) and is therefore a physical configuration invariant. Two facts make it the four-center analogue of an elliptic modulus. First, a* threshold*: for four rank-one (single-orbital) subspaces W is a function of the pairwise and triple traces—the 4×4 Gram is the entire configuration, so there is no four-body degree of freedom—whereas for rank ≥ 2 it is not (a local Jacobian-nullspace test gives residual $\gtrsim 0.3$). A genuine four-body invariant appears exactly when each center carries a σ–π pair, mirroring the rank-one$\to$rank-two step in the tame four-subspace ($\tilde{D}_4$) problem. Second, over molecular shape W is a* smooth continuous modulus*: for a planar rhombus with $\{1s, 2p\}$ per center it runs monotonically $0.79 \to 0.04$ as the diagonal ratio varies —a one-parameter family of inequivalent four-center configurations, the continuous freedom the three-center case lacks. For real orbitals W is real (its phase is $\mathbb{Z}_2$); a magnetic (Peierls) flux through the 4-cycle makes* $\arg W$ *grow continuously from zero—the four-body $U(1)$ lift. The object is the $*$-representation (non-abelian holonomy) analogue of the $\tilde{D}_4$ cross-ratio: four* orthogonal *projections are $*$-wild rather than tame, so the modulus is a matrix loop holonomy, not a single scalar cross-ratio. It is a geometry-axis invariant—a function of nuclear shape—distinct from the density-axis cross-ratio of the three-center integral in Paper 59. (Regression-pinned in `tests/test_paper32_four_center_modulus.py`; chronicle in CHANGELOG v5.1.0.)*

Remark 9 (The configuration operator is the composed basis's overlap metric)**.** *The three preceding remarks treat $F = \sum_i P_i$ as an abstract configuration operator. It is not abstract. Let G be the Gram matrix of the union of the per-center bases, with blocks $G_{ij} = \langle \chi_i | \chi_j \rangle$, and suppose each* intra-center *block is the identity—automatic when each center's own functions are orthonormal, and exact for the Shibuya–Wulfman metric of Paper 60 [56]. Put $X = \mathrm{chol}(G)^\mathsf{T}$, so that $X^\mathsf{T} X = G$ and the columns of X are the basis vectors expressed in an orthonormal ambient frame; let W_k collect the columns belonging to center k. Then $W_k^\mathsf{T} W_k = G_{kk} = I$, so W_k has orthonormal columns, the pseudoinverse in $P_k = W_k W_k^+$ collapses to W_k^T, and*

$$F \;=\; \sum_k W_k W_k^\mathsf{T} \;=\; X X^\mathsf{T} \;=\; X\,(X^\mathsf{T} X)\,X^{-1} \;=\; X\,G\,X^{-1}.$$

*Hence $F \simeq G$—*similar*, not merely cospectral. Every spectral statement in the preceding remarks (the conical intersections, the band gaps, the $\mathbb{Z}_2$ holonomy of the eigenvectors) is therefore a statement about the* overlap metric of the composed basis *over nuclear geometry. Verified across the whole construction: linear* $\mathrm{BeH_2}$*, bent* $\mathrm{BeH_2}$ *(including the* $\mathrm{H_2O}$ *angle* $104.5°$*), and the four-center configuration, with* $\max|\operatorname{spec} F - \operatorname{spec} G| \leq 3 \times 10^{-15}$ *and* $\|FX - XG\|_{\max} \leq 9 \times 10^{-16}$*. The intra-center hypothesis is load-bearing rather than decorative: perturbing one intra-center block off the identity by* 0.3 *breaks the identity outright (spectral discrepancy* 6.7×10^{-2}*). The four-center holonomy of Remark 8 is the same statement read blockwise,* $\mathrm{Tr}(P_1P_2P_3P_4) = \mathrm{Tr}(G_{12}G_{23}G_{34}G_{41})$ *(residual* 3×10^{-16}*).*

Two consequences follow, one positive and one negative. (i) An orthogonalization criterion. *Since the intersections are degeneracies of the metric, the eigenvectors of G are double-valued around one: parallel transport of the near-degenerate eigenvector around a small loop enclosing the central intersection returns it with overlap* -1.0000*, against* $+1.0000$ *on a CI-free control. Any orthogonalization that* selects *metric eigenvectors—canonical orthogonalization with a drop threshold, the standard remedy for near-linear-dependence—is therefore path-dependent along a geometry scan* *through such a point, whereas symmetric (Löwdin) orthogonalization $G^{-1/2}$, being a function of the matrix rather than of its eigenvectors, returns bit-exactly.* (ii) It is not a conditioning statement. *The intersections are* mid-*spectrum degeneracies (crossing eigenvalue* ≈ 0.71 *against* $\lambda_{\min} \approx 0.13$*), and* $\mathrm{cond}(G)$ *passes through* d^* *monotonically (*$19.9 \to 16.2 \to 15.0$*); the metric-conditioning growth of Paper 60 is a different functional of the same object, and no kink at* d^* *should be expected or claimed. The distinction drawn in the preceding remarks between* F *and the molecule's* physical electronic *conical intersection is unchanged; what changes is that* F *is an invariant of the framework's own encoding rather than a generic operator introduced for the analysis. (Regression-pinned in* `tests/test_paper32_config_operator_metric.py`*; chronicle in CHANGELOG v5.1.2.)*

6 The $K = \pi(B + F - \Delta)$ Formula in Spectral-Action Language

Paper 2 [27] observed the numerical coincidence $K = \pi(B + F - \Delta) = 1/\alpha$ at 8.8×10^{-8}, with

$$B = 42, \quad F = \tfrac{\pi^2}{6}, \quad \Delta = \tfrac{1}{40},$$

and the Phase 4B–4I sprint series identified three independent spectral homes for the three pieces (CLAUDE.md §2):

- B is the finite Casimir trace at $m = 3$: $B = \sum_{n=1}^{3} (2l + 1)l(l + 1)|_{n,l} = 42$ (Paper 2 Eq. 17).
- $F = D_{n^2}(d_{\max} = 4) = \zeta_R(2)$, the Fock-degeneracy Dirichlet series at the packing exponent (Phase 4F α-J).
- $\Delta^{-1} = g_3^{\mathrm{D}} = 40$, the third single-chirality Camporesi–Higuchi Dirac mode degeneracy (Phase 4H SM-D).

In spectral-triple language, B, F, Δ live in three different sectors of the same triple:

- B is a *finite scalar Casimir trace* on the truncated algebra: in the language of Sec. 4, it is $\mathrm{tr}_{\mathcal{A}_{\mathrm{GV}}}(C \cdot \mathbb{1})$ for C the second Casimir of SO(4) at $n_{\max} = 3$.
- F is an *infinite scalar spectral zeta*: it equals $\zeta_{\Delta_{S^3}}(2)$ in the continuum limit, where Δ_{S^3} is the scalar Laplace–Beltrami operator on S^3 (equivalently, the eigenvalue-weighted Dirichlet of the Fock degeneracy).
- Δ^{-1} is a *single-level Dirac degeneracy*: it equals g_3^{D}, the multiplicity of the third Camporesi–Higuchi single-chirality eigenvalue $|\lambda_3^{\mathrm{D}}| = 9/2$.

These three are categorically different objects: a finite combinatorial trace, an infinite analytic Dirichlet series, and a single-level multiplicity. They sit in different sectors of the spectral triple (scalar/spinor; finite/infinite; trace/zeta/dimension) and Phases 4B–4I exhausted nine attempts to find a common spectral generator that produces all three simultaneously, without success.

Observation 1 (Three-sector composition)**.** *The Paper 2 combination rule* $K = \pi(B + F - \Delta)$ *is a sum across three structurally distinct sectors of the GeoVac spectral triple. Each piece has been individually identified as a clean spectral invariant; the combination rule itself remains a numerical observation.*

This rephrasing is what working hypothesis WH5 in the project register is built on: α is the conversion factor between three projection regimes that do not share a generator. The spectral-triple framing does not derive α; it makes precise where the obstruction is.

Remark 10 (Sprint A, $\pi^3\alpha^3$ residual)**.** *Sprint A (April 2026, CLAUDE.md §1.7) verified at 80 dps that the post-cubic residual $K - 1/\alpha - \alpha^2 = 1.2079 \times 10^{-5}$ matches $\pi^3\alpha^3 = 1.2049 \times 10^{-5}$ to within 0.25%. The structural reading is $\pi^3 = \mathrm{Vol}(S^5)$, hinting at a contribution from the Bargmann–Segal S^5 HO sector (Paper 24 [33]) at higher order. This is a structural hint, not a derivation; the cross-check (Paper 24's HO triple is a distinct spectral triple, not a sub-sector of the GeoVac triple) shows that the $\pi^3\alpha^3$ identification, if real, is a* cross-triple *relation and not internal to the GeoVac spectral-action expansion.*

7 The Coulomb/HO Asymmetry in Spectral-Triple Language

Paper 31 [37] formalized a three-layer asymmetry between the Coulomb spectral triple of the present paper and the harmonic oscillator spectral triple of Paper 24 [33], summarized here:

Layer	Coulomb (this paper)	Harmonic oscillator (Paper 24)
Manifold algebra $\mathcal{A}$	Functions on S^3 Fock graph	Functions on S^5 holomorphic Hardy graph
Operator order	Second-order (Laplace–Beltrami)	First-order (complex Euler)
Spectrum	Quadratic $\lvert\lambda_n\rvert = n^2 - 1$	Linear $E_N = N + 3/2$
Calibration constant	π (Hopf measure $\mathrm{Vol}(S^2)/4$)	None (π-free in exact arithmetic)
Wilson gauge content	Full SU(2) (Paper 30)	$U(1)$ only (Sprint 5 negative)
L_0 matter propagator role	Yes (eigenvalues = spectrum)	No (only adjacency, not spectrum)

Table 4: The Coulomb/HO asymmetry from Paper 31, in spectral-triple language. The two systems are distinct spectral triples sharing only the algebra type (functions on a graph) and the trace structure of the spectral action.

In spectral-triple language the asymmetry is structural: the Coulomb case is a Riemannian-spin spectral triple of KO-dimension 3, the HO case is a spectral triple of complex Euler type on the holomorphic Hardy space of S^5. These are different objects even though both are "almost-commutative" in the broad sense. Calibration π enters in the Coulomb case through the Hopf-bundle measure $\mathrm{Vol}(S^2)/4$ in the spectral-action coefficients; it does not enter the HO case because the HO spectral action is built from a linear (first-order) operator whose heat kernel does not produce π at the rational level of the truncation [33].

Observation 2 (Coulomb-specificity of π)**.** *The presence of π in the spectral-action coefficients of QED is specific to the Coulomb-projected S^3 triple of this paper, not a universal feature of GeoVac-style discretizations. The HO triple is π-free at the discrete level. This is the structural answer to "why does QED need π"—it needs π because it lives on the Coulomb-projection S^3 triple, where the calibration is the S^2 Hopf-base Weyl constant.*

Theorem 3 (G3 closure: Bertrand-rigidity category obstruction)**.** *There is no tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\mathrm{BL}}$ that simultaneously preserves:*

(a) the Riemannian-Dirac character of $\mathcal{T}_{S^3}$ (second-order spectral action, spinor bundle, Camporesi–Higuchi spectrum $\lvert\lambda_n\rvert = n + 3/2$); and

(b) the Bargmann character of $\mathcal{T}_{S^5}^{\mathrm{BL}}$ (π-free certificate, Hardy-sector tower structure, first-order Euler spectrum $E_N = N + 5/2$).

The obstruction is categorical and rests on three premises:

- Bertrand's theorem (1873): *the only central potentials with all bounded orbits closed are $V = -Z/r$ (Coulomb) and $V = \omega^2 r^2$ (harmonic oscillator).*
- Fock rigidity (Paper 23 [32], Thm. 4): *the S^3 conformal projection is unique to $-Z/r$; the projection is second-order (stereographic / conformal), yielding a Riemannian spectral triple with the Laplace–Beltrami operator.*
- HO rigidity (Paper 24 [33], Thm. 3): *the S^5 Bargmann–Segal projection is unique to $\omega^2 r^2$; the projection is first-order (complex-analytic / holomorphic), yielding a Berezin–Toeplitz construction with the Euler number operator on the Hardy space.*

Proof sketch. Step 1 (category identification). Fock rigidity and HO rigidity place the two potentials in different mathematical categories: Coulomb → second-order conformal → Riemannian spectral triple; HO → first-order holomorphic → Berezin–Toeplitz quantization.

Step 2 (exhaustion of tensor-product options). Three constructions are available: (i) the standard graded tensor product with the round-S^5 Riemannian Dirac—this preserves (a) but violates (b) (π re-enters via the Seeley–DeWitt coefficients on S^5); (ii) the tensor product with the Bargmann–Euler operator—this preserves (b) but violates (a) (E_{BL} is not a Dirac-type operator on S^5, being the radial part of the Laplacian restricted to the holomorphic sector, not an elliptic differential operator on the total space); (iii) a hypothetical framework-extending construction beyond standard Connes spectral triples and Berezin–Toeplitz quantization—no such framework exists in published mathematics.

Step 3 (irreducibility). Bertrand exhausts the potentials; the two rigidity theorems fix the projection type for each. The category mismatch is a structural feature of central-force quantum mechanics, not an artifact of the GeoVac construction. The five asymmetry layers of Paper 24 § V (spectrum-computing role of L_0; calibration π; Wilson gauge matter coupling; modular-Hamiltonian Pythagorean orthogonality; spectral-action gravity termination) are all downstream consequences of this single categorical obstruction. □

Remark 11 (G3 scope)**.** *Theorem 3 does not assert that S^3 and S^5 are unrelated — they are the two Bertrand geometries, deeply connected through the angular sparsity theorem (Paper 22). It asserts that their* spectral-triple-type structures *live in different mathematical categories, and no published framework bridges the gap while preserving both structural contents. Cross-manifold unification, if achievable, requires framework-extending mathematics beyond the scope of this paper and the published NCG literature.*

8 The π-Source Case-Exhaustion Theorem

The empirical principle of Paper 35 [39] (Prediction 1, graduated to load-bearing status on a cumulative 208/208 confirmation across sprints KG and TX-B, May 2026) states that a GeoVac observable contains π if and only if its evaluation includes a continuous integration over a temporal or spectral parameter promoted from the discrete graph spectrum. Read in the spectral-triple language of §3–§4, this principle has a structural underpinning: the only places in the GeoVac framework where such continuous integrations occur are anchored to three named mechanisms of the present triple. We state and prove the case-exhaustion.

Theorem 4 (π-Source Case-Exhaustion)**.** *Let $\mathcal{C} = (P_{i_k} \circ \cdots \circ P_{i_1})$ be any finite composition of projections drawn from the Paper 34 list [38] § III.1–§ III.15, applied to a Layer-1 datum on $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ or its Bargmann–Segal sibling triple. The following are equivalent:*

(i) The converged value $\mathcal{C}(\text{datum})$ *contains a transcendental of the form* π, $\sqrt{\pi}$, π^k *for* $k \in \mathbb{Q}$, *or* π *in a denominator.*

(ii) The chain $\mathcal{C}$ *contains at least one projection* P_{i_j} *whose evaluation pipeline includes a continuous integration over a parameter promoted from the discrete graph spectrum (heat-kernel time, Matsubara frequency, Mellin contour, Hopf* S^2*-base angular measure,* $\mathrm{SU}(N)$ *Haar measure, or analytic-continuation parameter).*

(iii) The chain $\mathcal{C}$ *engages at least one of three spectral-triple mechanisms:*

M1 *(Hopf-base measure): introduces* $\pi = \mathrm{Vol}(S^2)/4$ *via the Hopf bundle* $S^3 \to S^2 \times S^1$, *or its higher-rank analogues [34, 36].*

M2 *(Seeley–DeWitt heat-kernel coefficient): introduces* $\sqrt{\pi}$ *via heat-kernel coefficients on the Camporesi–Higuchi Dirac spectrum on unit* S^3 *(Paper 28 Theorem T9 [35]:* $\zeta_{D^2}(s)$ *is a two-term polynomial in* π^2 *with rational coefficients at every integer* $s \geq 1$*). The sub-mechanism for the spinor sector is the half-integer Hurwitz* $\zeta(s, 3/2)$*.*

M3 *(vertex-topology Dirichlet L-content): introduces Catalan* $G = \beta(2)$ *and Dirichlet* $\beta(s)$ *via quarter-integer Hurwitz shifts* $\zeta(s, 3/4)$, $\zeta(s, 5/4)$ *at two-loop sunset vertices (Paper 28 Theorem 3:* $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$*).*

Proof sketch. (i) $\Leftrightarrow$ (ii) is Paper 35 Prediction 1, verified across 208 individual checks (sprints KG, TX-B, May 2026 [39]).

(ii) $\Rightarrow$ (iii): the proof body that follows enumerates the fifteen Paper 34 projections in §III.1–§III.15 (the corpus state at this theorem's drafting; Paper 34 has since grown to twenty-eight projections [38], and the theorem's claim is here restricted to those first fifteen pending an extension of the case-by-case enumeration to projections 16–28). Each of the fifteen Paper 34 projections is either π-free at the projection step (projections 1, 3, 4, 5, 8, 9, 12, 13, 14: Fock conformal, Bargmann–Segal, Stereographic, Sturmian, Wigner $3j$, Wigner D, Molecule-frame, Drake–Swainson, Rest-mass) or π-bearing through a continuous integration that reduces to one of M1, M2, M3 by direct identification (Hopf bundle, Spectral action, Camporesi–Higuchi spinor lift, Wilson plaquette, Vector-photon promotion, Observation/temporal-window). The seven π-free projections live in either Paper 31's universal sector (algebra-only, transfers to any spherical-fermion system) or Paper 22's potential-independent angular sparsity tier; neither sector contains π. The eight π-bearing projections are tagged in `debug/track_ts_e1_theorem_promotion_memo.md` §2 to one or more of M1/M2/M3, with closed-form identification for each (Paper 25 for M1, Paper 28 T9 theorem for M2, Paper 28 Theorem 3 for M3).

(iii) $\Rightarrow$ (i) is closed-form for each mechanism: M1's $\mathrm{Vol}(S^2)/4 = \pi$ identity, M2's $a_0(S^3) = \sqrt{\pi}$ heat-kernel identity, M3's $\beta(s) - \beta(s-2)$ closed form via the χ_{-4} Dirichlet character. Each forces π to appear in any chain that engages it. $\square$

Corollary 2 (M1 sub-mechanism lies in the localised pure-Tate sub-ring generated by π)**.** *The M1 sub-mechanism (the Hopf-base measure factor* $\mathrm{Vol}(S^2)/4 = \pi$ *on the* $S^3 \to S^2$ *Hopf bundle) produces period values in the localised pure-Tate sub-ring* $\mathbb{Q}[\pi, \pi^{-1}]$ *of mixed Tate periods over* $\mathbb{Q}$, *at depth* 0 *and Tate weight* $\in \mathbb{Z}$. *Every M1 closed form witnessed across the corpus (Paper 25 Hopf base* $\mathrm{Vol}(S^2)/4 = \pi$*; Paper 38 §L2 asymptotic propinquity rate* $4/\pi = \mathrm{Vol}(S^2)/\pi^2$*; Paper 40 Thm 1 universal* $4/\pi$ *across compact Lie groups; Paper 43 §10.2 Pythagorean orthogonality* $1/\pi^2$*; Paper 50 §3 F-theorem* $-3\zeta(3)/(16\pi^2)$ *as a M1×M3 cross-product) is a* $\mathbb{Q}$*-linear combination of* π^n *for* $n \in \mathbb{Z}$, *with* $|n| \leq 2$ *on the empirical witnesses logged so far. Higher integer powers of* π *enter only through M2 (Seeley–DeWitt) or M3 (vertex-parity sums). The M1 ring is convention-free (unlike M2's* $\sqrt{\pi}$ *vs* π^2 *volume-normalisation choice), because the Hopf-base measure is itself a volume* *integral and does not engage the heat-kernel asymptotic. Sprint M1 Pure-Tate (June 2026, memo* ***debug/sprint_m1_pure_tate_memo.md****).*

Corollary 3 (M2 sub-mechanism inherits Fathizadeh–Marcolli mixed-Tate classification, with pure-Tate refinement)**.** *At the standard volume-normalized Connes–Chamseddine convention, the M2 sub-mechanism Seeley–DeWitt coefficients on S^3 are mixed-Tate periods over $\mathbb{Q}$ by direct inheritance from Fathizadeh–Marcolli [6] (the static-R–W sub-case $a(t) \equiv 1$ on $\mathbb{R} \times S^3$). The GeoVac S^3 M2 sector is in fact the strictly smaller pure-Tate sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ of even-weight Tate periods, with no $\zeta(3), \zeta(5)$, or MZV content appearing at the SD level. This refines F–M's continuum classification in two structural ways: (i) the Dirac SD expansion is two-term exact ($a_k^{D^2} = 0$ for $k \geq 2$, Paper 51 [53] Cor. 2.1), not generically infinite; (ii) the scalar Laplacian SD coefficients have the closed-form geometric-series collapse $a_k^{\Delta} = 2\pi^2/k!$. The M3 sub-mechanism (vertex-parity Hurwitz, Catalan G, $\beta(s)$) lives in cyclotomic mixed Tate at level 4 (mixed Tate over $\mathbb{Z}[i]$, not over $\mathbb{Q}$) and does* not *contaminate the SD coefficients, by the operator-order partition of Paper 18 [30] § III.7 ($k = 2$ for SD vs $k = 1$ for vertex parity). Sprint Mixed-Tate Test (June 2026, memo* ***debug/sprint_mixed_tate_test_memo.md****).*

Corollary 4 (M3 sub-mechanism inherits Deligne–Glanois cyclotomic mixed-Tate classification at level ≤ 4)**.** *The M3 sub-mechanism (vertex-parity Hurwitz / Dirichlet L / depth-k loop tower) on the discrete Camporesi–Higuchi S^3 spectral triple lies in the cyclotomic mixed-Tate period ring at level ≤ 4 over $\mathcal{O}_4[1/4] = \mathbb{Z}[i, 1/2]$ at depth equal to the loop order, by direct application of Deligne [7] (proven for $N \in \{2, 3, 4, 8\}$) and the explicit basis of Glanois [8]. Three sub-rings are realised:*

$$\mathcal{MT}(\mathbb{Z}) \subset \mathcal{MT}(\mathbb{Z}[1/2]) \subset \mathcal{MT}(\mathbb{Z}[i, 1/2]),$$

selected by (i) un-restricted vs vertex-restricted summation, and (ii) loop order = motivic depth. Specifically:

- *Un-restricted Dirac Dirichlet $D(s)$ at integer $s \geq 2$ (Paper 28 [35] Thm 2) lies in $\mathcal{MT}(\mathbb{Z})$ at depth 1;*
- *Vertex-restricted $D_{\mathrm{even}} - D_{\mathrm{odd}} = 2^{s-1}(\beta(s) - \beta(s-2))$ at integer $s \geq 2$ (Paper 28 [35] Thm 3) lies in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level 4, depth 1, via Hurwitz at quarter-integer shifts; Catalan $G = \beta(2)$ and $\beta(4)$ are level-4 Dirichlet L-value periods in this ring;*
- *Two-loop irreducible $S_{\min}$ (Paper 28 §$S_{\min}$) lies in $\mathcal{MT}(\mathbb{Z}[1/2])$ at depth 2; the depth-k tower (Paper 28 Prop. depth_k) generates new irreducible periods at depth k = loop order in $\mathcal{MT}$ at level ≤ 4.*

The vertex-parity-as-χ_{-4} Dirichlet character mechanism IS the Deligne–Glanois Galois descent from level 4 to level 2 (Glanois [8] Cor. 1.2) realised on the framework's natural QED observables. Combined with Corollary 3, the master Mellin engine of Theorem 4 now has a complete period-theoretic stratification: M1 in $\mathbb{Q}[\pi, 1/\pi]$, M2 in the pure-Tate sub-ring $\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$, and M3 in $\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4. Specific motivic identification of irreducible constants at depth ≥ 2 ($S_{\min}$, $S^{(3)}$, ...) within the Deligne–Glanois basis is an open question at each depth. Sprint M3 Cyclotomic Mixed-Tate (June 2026, memo ***debug/sprint_m3_cyclotomic_mixed_tate_memo.md****).*

Remark 12 (Joint engagement)**.** *M1, M2, M3 are not set-theoretically disjoint as constructions: the spectral action invokes both M1 (Hopf-base measure on the spinor bundle) and M2 (Seeley–DeWitt coefficients) simultaneously, and the vacuum-polarization coefficient $\Pi = 1/(48\pi^2)$ (Paper 28 § V) contains one factor of π from each mechanism. The theorem requires only that* at least one *of the three be engaged; multiple engagement is the typical case at higher loop order. The mechanisms are role-disjoint in the Paper 18 § III.7 Mellin-engine sense (M1 = packing-tier* ω_d*, M2 = heat-kernel via Mellin/Gaussian, M3 = vertex-parity quarter-integer Hurwitz character mechanism). The Track TS-E3 sprint (May 2026) further sharpened this by showing that M1, M2, M3 are all sub-cases of a single master Mellin-engine mechanism* $\mathrm{Tr}(D^k \cdot e^{-tD^2})$ *on a compact Riemannian manifold, with* $k \in \{0, 1, 2\}$ *selecting the sub-mechanism.*

Remark 13 (Status of $K = \pi(B + F - \Delta)$)**.** *Paper 2's combination* $K = \pi(B + F - \Delta) \approx 1/\alpha$ *matches its target at* 8.8×10^{-8}*. Under Theorem 4, K's chain (Fock ∘ Hopf∘ spinor) engages M1 (the explicit prefactor* π*, identified by Sprint A as* $\mathrm{Vol}(S^2)/4$ *[27]) and M2 (in* $F = \pi^2/6 = D_{n^2}(d_{\max})$ *via the Mellin transform of the scalar Fock-degeneracy). The* π*-content is therefore consistent with case-exhaustion. The remaining anomaly is at the* depth-prediction *layer of Paper 34 § VIII.3 (depth predicts* $\sim 3\%$ *residual at three projections; K matches at* 10^{-8}*): a quantitative-error puzzle, distinct from the qualitative case-exhaustion of the present theorem, and unaffected by it.*

Remark 14 (Falsification target)**.** *A counter-example to Theorem 4 would be a sixteenth projection (or a finite chain of existing projections) introducing* π *via a fourth mechanism not in {M1, M2, M3}. Three natural candidates were investigated in Track TS-E3 (May 2026) and each REDUCED to combinations of the three existing mechanisms: continuum-gravity coupling (M1+M2), anomaly-class projections (M1+M2+M3 via the eta-invariant's first-order Mellin), and non-trivial principal-bundle projections (M1 normalisations against integer integrands). The strongest remaining computational falsification target is the discrete first-Chern-class invariant on the Hopf* S^3 *graph at* $n_{\max} = 3$*, predicted integer-valued by the case-exhaustion; if a* π *appears unexpectedly, a fourth mechanism M4 would need to be named.*

Remark 15 (Mechanism-as-domain sharpening (Sprint MR-A/B/C, May 2026))**.** *A three-track follow-up sprint (MR-A: Dirac propinquity rate; MR-B: spectral-action modular residual; MR-C: L2 next-order constant) tested whether the master Mellin engine of Paper 18 § III.7 is also* predictive *for convergence rates, not only classifying for* π*-sources. The combined verdict sharpens the case-exhaustion theorem into a mechanism-as-domain partition.*

- *M1 (*$k = 0$*) is the natural domain of state-space propinquity rates. The L2 sprint (2026-05-06) measured the asymptotic constant of the SU(2) scalar central Fejér kernel as* $4/\pi = \mathrm{Vol}(S^2)/\pi^2$.
- *M2 (*$k = 2$*) is the natural domain of heat-kernel and spectral-action convergence. Sprint MR-B derived a closed form for the Camporesi–Higuchi modular residual:*

$$\varepsilon(t) \;=\; \sum_{m \geq 1} (-1)^m \sqrt{\pi}\, e^{-m^2\pi^2/t} \left[t^{-3/2} - 2m^2\pi^2\, t^{-5/2} - \tfrac{1}{2}\, t^{-1/2} \right].$$

  Verified at > 100 *digits at every tested* t*; every coefficient in* $\sqrt{\pi}\,\mathbb{Q} \oplus \pi^2\,\mathbb{Q}$*; modular exponent exactly* π^2 *via Jacobi* ϑ_2*.*

- *M3 (*$k = 1$*) is the natural domain of vertex-restricted parity-character observables (Paper 28 § QED-vertex: Catalan G,* $\beta(4)$*). Sprint MR-A demonstrated that asking for M3 from a half-integer-only Peter–Weyl propinquity is a* category error*: the half-integer-spin truncation cannot span the constant function* χ_0 *on* $\mathrm{SU}(2)$*, so the kernel does not localize and the moment is identically* $\gamma_n^{\mathrm{Dirac}} = \pi$ *for every* $n_{\max} \geq 1$*. The L2 scalar rate's* $(4/\pi)\log(n)/n$ *decay is an integer×half-integer interference effect; removing the integer sector kills it. This is exactly what the M1 mechanism predicts when the sub-bundle lacks the constant function.*

The reading is therefore: the index $k \in \{0,1,2\}$ classifies both *the sub-mechanism and the natural domain of observable from which its transcendental signature can be extracted. The case-exhaustion theorem 4 classifies* which *transcendentals appear; the domain partition refines to* which observables *can produce them. Sprint MR-C extracted the L2 next-order constant $c = 4.1093214674877940927\ldots$ to ≥ 80 dps via Richardson extrapolation but found no PSLQ identification in $M_1 \cup M_2 \cup M_3 \cup \{\gamma_E, \log 2, G, \zeta(3)\}$ at coefficient ceiling 10^4. This is consistent with c being a downstream Stein–Weiss IBP derivative of the M1 mechanism rather than a direct Mellin output, and is a partial negative against extending the predictive engine to all asymptotic-expansion coefficients along subsequent analytical pipelines. The engine is predictive* within *each domain. Source:* `debug/mr_a_dirac_propinquity_*`, `debug/mr_b_spectral_action_rate_*`, `debug/mr_c_l2_subleading_*`.

Remark 16 (CH-substrate-specificity of the chirality-parity factorization (Sprint Q5'-S5-falsifier, 2026-06-05))**.** *The bit-exact $\mathbb{Z}/2$ chirality-parity factorization of the master Mellin source on the truncated Camporesi–Higuchi spectral triple (Sprint Q5'-CH-1) is CH-specific: it requires both spectral $\pm$-symmetry and chirality alignment $\gamma_{ii} = \chi_i$, $\Lambda_{ii} = \chi_i|\lambda_i|$ of the CH Dirac. The Bargmann–Segal Hardy-on-S^5 triple (Paper 24) has neither at any finite $N_{\max}$: the Hardy spectrum is strictly positive (no $\pm$-partner) and the $(N,0)$ $\mathrm{SU}(3)$ tower carries no canonical chirality grading (the two natural candidates $\chi_N = (-1)^N$ and $\chi_l = (-1)^l$ coincide because $(N,0)$ branching forces l-parity $\equiv$ N-parity, and neither reproduces the CH-1 even-j zeros). The ω^{tri} enrichment program of Paper 55 § subsec:open_m2_m3 correctly targets the CH triple as substrate; the case-exhaustion classification 4 remains universal in scope but its bit-exact* finite-cutoff *witness form is CH-substrate-specific. This places a sixth layer on the Coulomb/HO asymmetry of Paper 24 § V. Source:* `debug/sprint_q5p_s5_falsifier_memo.md`.

Remark 17 (Stage-1 cochain-level witness of the master-Mellin partition (Sprint Q5'-Stage1-JLO-nmax2, 2026-06-05))**.** *On the truncated Camporesi–Higuchi spectral triple at $n_{\max} = 2$, the master-Mellin partition (M_1, M_2, M_3) is bit-exactly visible at the cyclic-cochain level as the symbol $\omega^{\mathrm{tri}}(\mathcal{T}_2) = (16, 84, 36) \in \mathbb{Z}^3$. The first two slots live inside a single JLO degree-0 cochain: M_1 at t^0 of $\phi_0^{\mathrm{odd}}(1;t) = \mathrm{Tr}\, e^{-tD^2}$, M_2 at $-t^1$ of the same. M_3 lives outside the JLO framework on commutative $\mathcal{A} = \mathbb{C}^5$ and instead in the Connes–Moscovici residue framework as the leading coefficient of $\mathrm{Tr}(\gamma\, D\, e^{-tD^2})$. The first non-trivial JLO cochain degree on commutative $\mathcal{A}$ is $n = 2$; structural vanishing $\phi_1(a_0, a_1; t) \equiv 0$ is proved bit-exactly for all idempotent pairs (a_0, a_1). The cutoff sweep (Sub-Sprint 2a) extends the symbol bit-exactly to $n_{\max} \in \{1,2,3,4\}$ via the closed forms $M_1(n) = 2n(n+1)(n+2)/3$, $M_2(n) = n(n+1)(n+2)(2n+1)(2n+3)/10$, $M_3(n) = n(n+1)^2(n+2)/2$, all re-derived from the chirality-balanced shell degeneracy $2n(n+1)$ and the CH eigenvalue $n+1/2$. See Paper 55 § subsec:open_m2_m3 for the Stage-1 Q5' framing. Source:* `debug/sprint_q5p_jlo_nmax2_memo.md`, `debug/sprint_q5p_2a_jlo_nmax_sweep_memo.md`.

Remark 18 (Mellin extraction points of the master-Mellin partition (Sprint Q5'-Stage1-2b, 2026-06-05))**.** *The master Mellin partition (M_1, M_2, M_3) corresponds to three categorically different extraction points of the formal Mellin transform of the JLO degree-0 cochain on a compact 3-dimensional spectral triple:*

- *M_1 at the spectral-dimension pole $s = d/2 = 3/2$ of $\Gamma(s)\,\zeta_{D^2}(s)$: the Weyl-law short-time leading $\mathrm{Tr}(e^{-tD^2}) \sim \sqrt{\pi}/2 \cdot t^{-3/2}$ supplies the Hopf-base measure π residue.*
- *M_2 at integer-s regular points of the same $\Gamma(s)\,\zeta_{D^2}(s)$: Seeley–DeWitt pure-Tate continuum values (e.g., $\zeta_{D^2}(2) = \pi^2 - \pi^4/12$ on unit CH) reproduce the Sprint Q5'-CH-2 panel bit-exactly.*
- *M_3 at the analogous extraction on a structurally different Mellin complex, the η-style pairing $\Gamma(s)\,\mathcal{M}[\mathrm{Tr}(\gamma\, D\, e^{-tD^2})](s)$: vertex-parity Hurwitz content at quarter-integer shifts.*

At finite $n_{\max}$ the spectral zeta is bit-exact rational (e.g., $\zeta^{(2)}_{D^2}(1) = 1\,978\,465\,196\,515\,232/532\,976\,619\,280\,625$ for the full Dirac at $n_{\max} = 2$, $\kappa = -1/16$); the $s \to 0$ residue is exactly $\dim \mathcal{H} \in \mathbb{Z}$ (no π). Master-Mellin transcendentals enter only in the $n_{\max} \to \infty$ continuum limit; the cosmic-Galois Stage 2 motivic-Galois action acts at continuum, not at finite cutoff. Source: `debug/sprint_q5p_2b_phi0_mellin_memo.md`

Remark 19 (Continuum residue identification of the master-Mellin M_1 Hopf-base coefficient (Sprint Q5'-Stage1-2b-continuum, 2026-06-05))**.** *The Mellin extraction-points reading of Remark 18 is closed at theorem grade for the M_1 component on the truncated Camporesi–Higuchi spectral triple of dimension $d = 3$. The continuum spectral zeta $\zeta^{\text{cont}}_{D^2}(s) = 2\,\zeta(2s-2, 3/2) - \frac{1}{2}\,\zeta(2s, 3/2)$ has a simple pole at the spectral-dimension value $s = d/2 = 3/2$ with bit-exact meromorphic residue 1 (verified two independent ways: from the Hurwitz pole at $u = 2s - 2 = 1$ with residue 1, and from direct Laurent expansion via the digamma identity). The Mellin residue of $\Gamma(s)\,\zeta^{\text{cont}}_{D^2}(s)$ at $s = 3/2$ is therefore $\Gamma(3/2) \cdot 1 = \sqrt{\pi}/2$, which matches the standard heat-kernel Weyl-law leading coefficient $r\,\mathrm{Vol}(S^3)/(4\pi)^{3/2}\big|_{r=2}$ ([21, 22]) bit-exactly. The three published instances of the M_1 Hopf-base measure ($\mathrm{Vol}(S^2)/4 = \pi$ in Paper 18 § III.2; $\mathrm{Vol}(S^2)/\pi^2 = 4/\pi$ in Paper 38 § VIII as the L2 asymptote rate constant; $\Gamma(d/2) = \sqrt{\pi}/2$ here at the Mellin pole) are sibling normalizations of the same spectral object, related by exact factors of $\pi^2/4$ and $\pi^{3/2}/8$ with no spectral-data input. The M2 continuum panel at integer $s \in \{1, 2, 3, 4, 5\}$ reproduces the Sprint Q5'-CH-2 Seeley–DeWitt panel bit-exactly (5/5 matches). Tauberian rate uniformity in a neighborhood of $s = 3/2$ is the Stage-2-relevant open question, reachable from Karamata [23] and Korevaar [24]. Source:* `debug/sprint_q5p_continuum_mellin_memo.md`.

Remark 20 ($\mathrm{HP}^{\text{even}}$ class identification on the bicomplex of the truncated CH triple (Sprint Q5'-Stage1-2c-Bicomplex, 2026-06-05))**.** *The cyclic (b, B) bicomplex of the truncated Camporesi–Higuchi spectral triple at $n_{\max} = 2$ has been constructed bit-exactly. The load-bearing degree-1 entire-cyclic cocycle condition $b\phi_0 + B\phi_2 = 0$ holds bit-exactly on the full panel of 36 idempotent-pair inputs at three t-orders t^0, t^1, t^2, both even and odd flavors (216 bit-exact zero residuals). The explicit $\mathrm{HP}^{\text{even}}$ class on the Morita-trivial baseline $\mathbb{Q}^5 = \mathrm{HP}_0(\mathbb{C}^5)$ is*

$$[\phi_{\text{JLO}}]^{\mathrm{HP}^{\text{even}}} \;=\; (+2, -2, +2, +2, -4) \in \mathbb{Q}^5,$$

the sector-resolved McKean–Singer index summing to $\mathrm{Tr}(\gamma) = 0 = \chi(S^3)$. The Sub-Sprint 1 symbol $(16, 84, 36)$ is therefore a JLO cocycle representative, not a coboundary. A structural finding (honest scope): at degree 3, $(b\phi_2 + B\phi_4)$ has bit-exact residual $\pm 1/(3 \cdot 2^{16}) = \pm 1/196\,608$ on specific palindromic 4-tuples, a JLO-tower truncation artifact tied to the structural vanishing of odd-degree cochains $\phi_1 \equiv \phi_3 \equiv 0$ on commutative $\mathcal{A}$. The CM-residue cocycle for M_3 is single-cochain and immune to this artifact; the JLO/CM-residue distinction is Stage-2-relevant. Source: `debug/sprint_q5p_2c_bicomplex_memo.md`.

Remark 21 (CM-residue η-class identification, Sprint Q5'-Stage1-CM-Bicomplex, June 2026)**.** *The second span of the Stage-1 cosmic-Galois U^* bridge — the Connes–Moscovici η-pairing residue cocycle on the truncated Camporesi–Higuchi spectral triple at $n_{\max} = 2$ — has been constructed bit-exactly in* sympy.Rational *(*`debug/sprint_q5p_cm_bicomplex_memo.md`*,* `debug/data/sprint_q5p_cm_bicomplex.js`*). The CM-η cochain $\psi^\eta_n(a_0, \ldots, a_n; t) := \int_{\Delta_n} \mathrm{Tr}(\gamma D\, a_0\, e^{-s_0 t D^2}\, [D, a_1] \cdots [D, a_n]\, e^{-s_n t D^2})\, ds$ replaces the JLO leftmost insertion γ by γD, and shares the bicomplex structure $(b + B)\psi = 0$ with JLO via the same formulas. The degree-1 cocycle condition $b\psi^\eta_0 + B\psi^\eta_2 = 0$ holds bit-exactly on the full panel of 36 idempotent-pair inputs at three t-orders both flavors (216 bit-exact zero residuals), with NO truncation artifact of the JLO $\pm 1/196608$ type. The CM-η class on the Morita-trivial baseline $\mathbb{Q}^5 = \mathrm{HP}_0(\mathbb{C}^5)$ is explicitly*

$$[\psi^{\eta,\text{even}}_{\text{CM}}]^{\mathrm{HP}_\eta} = (3, 3, 5, 15, 10) \in \mathbb{Z}^5,$$

summing to $36 = M_3(n_{\max} = 2)$ *(the Sub-Sprint 1 master-Mellin M3 ground truth). The sector-resolved structure is the chirality-symmetrized eigenvalue magnitude per sector* $\eta_s = \dim_s \cdot (n_s + 1/2)$*, in clean dual to the JLO* $\chi_s = \dim(\gamma_+ e_s \mathcal{H}) - \dim(\gamma_- e_s \mathcal{H})$ *chirality balance per sector. The two classes share the same* K_0 *sector-decomposition* $\{[e_s]\}_{s=0..4}$ *but encode distinct cohomological content: JLO captures the zeroth heat-kernel supertrace coefficient (McKean–Singer index* $\chi(S^3) = 0$*), CM-*η *captures the leading* t^0 η*-density supertrace coefficient (*M_3 *mechanism leading). Cross-check at* $n_{\max} = 3$*:* $M_3(3) = 120$ *bit-exact, CM-*η *degree-1 cocycle bit-exact zero on all 9 diagonal sectors. See Paper 55 § subsec:open_m2_m3 for the Stage-1 Q5' construction context.*

Remark 22 (Q5' Stage 1 pro-system functoriality of the JLO and CM-η classes (Sprint Q5'-Stage1-Prosystem, June 2026))**.** *The Stage-1 cocycle classes form a strictly compatible inverse system across the cutoff pro-system* $\{\mathcal{T}_{n_{\max}}\}_{n_{\max}\geq 1}$ *(`debug/sprint_q5p_prosystem_memo.md`, `debug/data/sprint_q5p_prosystem.json`). Sector idempotents are indexed by* (n, l) *with* $1 \leq n \leq n_{\max}$*,* $0 \leq l \leq n$*, giving* $N(n_{\max}) = n_{\max}(n_{\max} + 3)/2$ *sectors at each cutoff (2, 5, 9, 14 at* $n_{\max} = 1, 2, 3, 4$*, with strict nesting); the truncation* $P_{n+1\to n} : \mathcal{O}_{n+1} \to \mathcal{O}_n$ *is sector projection (algebra homomorphism), the algebraic Berezin* $B_{n\to n+1}$ *is sector inclusion. Both the JLO* $\mathrm{HP}^{\mathrm{even}}$ *class* $\chi_{(n,l)}$ *and the CM-*η *class* $\eta_{(n,l)}$ *are* sector-local *— the per-sector value depends only on the sector label, not on the cutoff:*

$$\chi_{(n,l)} = \begin{cases} +2 & l < n \\ -2n & l = n \end{cases}, \qquad \eta_{(n,l)} = \begin{cases} (2l+1)(2n+1) & l < n \\ n(2n+1) & l = n \end{cases}.$$

Bit-exact pull-back identity $P^*_{n+1\to n}[\psi^{(n+1)}] = [\psi^{(n)}]$ *holds across all three consecutive cutoff pairs for both classes (six inverse-image identities, each with bit-zero residual); algebraic Berezin compatibility holds with the same bit-exactness on the common sectors. Total sums match the Sub-Sprint 2a polynomial closed forms* $M_1(n_{\max}) = 2n_{\max}(n_{\max} + 1)(n_{\max} + 2)/3$*,* $\sum_s \chi_s = 0$ *(McKean–Singer index* $\chi(S^3) = 0$ *preserved cutoff-uniformly),* $\sum_s \eta_s = M_3(n_{\max}) = n_{\max}(n_{\max} + 1)^2(n_{\max} + 2)/2$ *bit-exactly at all four cutoffs. The pro-system is the cleanest possible inverse-system lift of the cohomological Stage-1 data: strict at the class level (no twist, no coboundary modification), consistent with the sector-locality of the Camporesi–Higuchi shell decomposition. The strict-strong-form question — pro-system functoriality at the level of* cochain morphisms *in the entire-cyclic bicomplex (vs the class level closed here) — is sharpened to a bit-exact dichotomy in Remark 24; the deep NCG-mathematics sub-question on the published Connes 1994 Ch. IV / Loday § 1.4 gap remains open, beyond sprint scale, independent of the Stage-1 closure here. See Paper 55 § subsec:open_m2_m3 for the Q5' Stage 1 construction context.*

Remark 23 (Q5' Stage 2 candidate Hopf algebra substrate, Sprint Q5'-Stage2-Hopf, June 2026)**.** *The first scoping step of Stage 2 of the cosmic-Galois* U^* *program produces a concrete bit-exact finite-cutoff candidate Hopf algebra* $\mathcal{H}_{\mathrm{GV}}^{(n_{\max})} = \mathrm{Sym}_{\mathbb{Q}}(V_{n_{\max}})$ *on the pro-system substrate of Remark 22, where* $V_{n_{\max}}$ *is the* $\mathbb{Q}$*-vector space of primitive generators* $x_{(n,l),k}$ *indexed by* $(n, l) \in \mathcal{S}_{n_{\max}}$ *and Mellin slot* $k \in \{0, 1, 2\}$ *(with* $k = 0, 1, 2$ *corresponding to M1, M3, M2 of §8, respectively). With primitive coproduct* $\Delta(x_g) = x_g \otimes 1 + 1 \otimes x_g$*, counit* $\varepsilon(x_g) = 0$*, and antipode* $S(x_g) = -x_g$ *on generators (extended as algebra homomorphisms),* $\mathcal{H}_{\mathrm{GV}}$ *satisfies all five Hopf axioms bit-exactly at* $n_{\max} \in \{2, 3\}$ *(269 bit-exact zero residuals on the axiom panel: 52 coassociativity, 104 counit-left/right, 104 antipode-left/right, 9 bialgebra-compatibility), and the k-grading is preserved by* Δ *so that* $\mathcal{H}_{\mathrm{GV}} = \mathcal{H}_{\mathrm{GV}}^{[0]} \otimes \mathcal{H}_{\mathrm{GV}}^{[1]} \otimes \mathcal{H}_{\mathrm{GV}}^{[2]}$ *factors as a tensor product of three Hopf sub-algebras (42 bit-exact generator-level k-preservation checks). The pro-system truncation* $P_{n+1\to n}$ *of Remark 22 lifts bit-exactly to a Hopf-algebra homomorphism* $\mathcal{H}_{\mathrm{GV}}^{(n+1)} \to \mathcal{H}_{\mathrm{GV}}^{(n)}$ *(126 bit-exact zero residuals on the truncation panel at* $(n_{\max}, n_{\max} - 1) \in \{(2, 1), (3, 2)\}$*). The candidate motivic Galois group at finite* *cutoff is the additive affine group* $U^{*(n_{\max})}_{\mathrm{GeoVac}} = \mathbb{G}_a^{3N(n_{\max})}$ *(dimensions 15 and 27 at* $n_{\max} = 2, 3$*), with the Mellin-slot factorisation* $U^{*(n_{\max})}_{\mathrm{GeoVac}} = \prod_{k=0}^{2} \mathbb{G}_a^{N(n_{\max})}$ *structurally enforcing the M1/M2/M3 partition of Theorem 4 at the Stage-2 cosmic-Galois level. The candidate is the* abelian primitive-cospan *of the Connes–Kreimer Hopf algebra of Feynman graphs (Connes–Kreimer 1998, arXiv:hep-th/9808042): same generator-and-grading shape, same pro-system truncation structure, but the sub-graph coproduct of CK is replaced by the primitive coproduct here because the CH Fock sector idempotents are sector-disjoint (no proper sub-sector nesting). The natural continuation toward Stage-2's full Tannakian construction (Connes–Marcolli 2007, arXiv:math/0409306) — a substantial open program — is to enrich the coproduct with a sub-sector or nested-Hopf-tower structure that introduces non-abelian pro-unipotent content. See Paper 55 § subsec:open_m2_m3 for the cosmic-Galois narrative.*

Remark 24 (Q5' Stage 1 strict-strong-form cochain-morphism drift, Sprint Q5'-Stage1-StrictStrong, June 2026)**.** *Sharpening the v3.60.0 honest-scope caveat of Remark 22: while pro-system functoriality at the level of cocycle CLASSES is bit-exactly strict across the cutoff pro-system* $\{\mathcal{T}_{n_{\max}}\}_{n_{\max}\geq 1}$ *on commutative* $\mathcal{A}$*, pro-system functoriality at the COCHAIN-MORPHISM level of the entire-cyclic bicomplex FAILS bit-exactly on the same algebra (`debug/sprint_q5p_strict_strong_memo.md`, `debug/data/sprint_q5p_strict_strong.json`). The degree-3 closure residual* $(b\phi_2 + B\phi_4)$ *on the load-bearing* (e_2, e_3) *palindromic 4-tuples carries bit-exact values*

$$n_{\max} = 2 : \pm 1/(3 \cdot 2^{16}) \quad \to \quad n_{\max} \geq 3 : \pm 1/2^{16} \quad (\textit{bit-exact fixed point at } n_{\max} = 3, 4),$$

with the pull-back cochain increment $\Delta(B\phi_4)_{3\to 2} = -1/(3 \cdot 2^{14})$ *closing the residual identity bit-exactly:* $1/196608 + (-1/49152) = -1/65536$*. The CM-*$\eta$ *cocycle tower exhibits the same structural pattern at degree 3 with structurally distinct quantitative values (the v3.59.0 reading that the CM-*η *tower is the cochain-clean alternative is restricted to degree 1 and does not extend to degree 3). The pro-system functoriality verdict is therefore a sharp dichotomy: strict at the class level (Remark 22), bit-exact non-strict with* $n_{\max} = 3$ *fixed-point structural drift at the cochain-morphism level (this remark). The drift is consistent with a boundary-vs-interior shell partition:* $n_{\max} = 2$ *is the boundary cutoff (no shell above* $n = 2$ *in the* ϕ_4 *matrix-element channel),* $n_{\max} \geq 3$ *are interior cutoffs (stabilized). Empirical confirmation at bit-exact precision of the published-open Connes 1994 Ch. IV / Loday §1.4 gap on entire-cyclic functoriality under algebra truncations. See Paper 55 § subsec:open_m2_m3 for the Stage-1 Q5' construction context.*

Remark 25 (Tauberian rate uniformity in a neighborhood of $s = 3/2$ (Sprint Q5'-Stage1-Tauberian, 2026-06-05))**.** *The pointwise rate of approach in Remark 19 is sharpened to a uniform rate on compact above-pole neighborhoods of the spectral-dimension pole* $s = d/2 = 3/2$ *via the standard uniform Tauberian theorem of Karamata [23] (foundational), Korevaar [24] § III.4 (modern treatment with boundary-saturation Remark 3), and Tenenbaum [25] § II.7 Thm II.7.7 (uniform rates of approach with explicit constants). The four hypotheses transport bit-exactly to the CH spectral zeta: (H1) positivity* $a_n = 2n(n+1) > 0$*; (H2) polynomial growth* $a_n = O(n^2)$*; (H3) meromorphic continuation with unique simple pole at* $\sigma_0 = 3/2$ *of residue* 1 *(the v3.59.0 Track 2 closure); (H4) vertical-strip boundedness from standard Hurwitz convexity. Conclusion: on the compact above-pole neighborhood* $U_> = [3/2 + \epsilon,\, 3/2 + 1/4]$, $\sup_{s\in U_>} |\zeta^{(N)}_{D^2}(s) - \zeta^{\mathrm{cont}}_{D^2}(s)| = O(N^{-2\epsilon+\delta})$ *uniformly in* $U_>$*, for any* $\delta > 0$*. Boundary-saturation: the supremum is attained at the boundary point closest to the pole, verified bit-exactly at* $n_{\max} \in \{2, 3, 4, 5\}$ *on the six-grid panel* $\{5/4, 11/8, 23/16, 25/16, 13/8, 7/4\}$ *(sup attained at* $s = 25/16$ *at every cutoff; empirical slope* -0.092 *at* $n_{\max} \in \{2, \ldots, 5\}$ *monotonically approaches the predicted* $-1/8$*, hitting* -0.123 *at* $n_{\max} = 100$*). Source: `debug/sprint_q5p_tauberian_memo.md`.*

Remark 26 (Continuum residue identification of the master-Mellin M_3 η-tower (Sprint Q5'-Stage1-M3-Continuum 2026-06-05))**.** *The continuum M_3 η-Mellin object on the truncated Camporesi–Higuchi spectral triple of dimension $d = 3$ is closed at theorem grade with parity-of-s stratification. The continuum η-Dirichlet series $\eta_D(s) := \sum_{n\geq 0} g_n|\lambda_n|^{1-s}$ admits the Hurwitz reduction $\eta_D(s) = 2\zeta(s-3, 3/2) - \frac{1}{2}\zeta(s-1, 3/2)$ (equivalently $\eta_D(s) = D(s-1)$ via Paper 28's Dirac Dirichlet series), with TWO simple poles at $s = 4$ and $s = 2$ of bit-exact meromorphic residues 2 and $-1/2$ respectively (verified two independent routes: Hurwitz pole rule at $u = 1$ AND direct Laurent via digamma). The corresponding Mellin residues $\mathrm{Res}_{s=4}\Gamma(s)\eta_D(s) = 12$ and $\mathrm{Res}_{s=2}\Gamma(s)\eta_D(s) = -1/2$ are rational integer values, NOT in the M3 cyclotomic ring; the genuine M3 cyclotomic content lives in the CHIRALITY-RESOLVED η-Mellin $(D_{\mathrm{even}} - D_{\mathrm{odd}})(s-1)$, which inherits the Sprint Q5'-CH-3 even-s M3 stratification via the $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$ Galois descent. Specifically at $s = 3$ the chirality discriminant equals $-1+2G$ (Catalan G; genuine M3 depth 1); at $s = 5$ it equals $8\beta_4 - 8G$; at $s = 7$ it equals $-32\beta_4 + 32\beta_6$ (both genuine M3 depth 1+). Odd-$(s-1)$ values collapse to M_1 via $\beta(2k+1) \in \pi^{\mathrm{odd}}\mathbb{Q}$. The discrete-side Karamata Tauberian signature at the $s = 4$ pole holds exactly: partial sums on the v3.60.0 sector-local substrate satisfy $S(n_{\max})/\log(n_{\max}) \to 2$ at $n_{\max} \geq 200$ (meromorphic residue; no Jacobian since the natural η-Mellin variable IS the spectral $|\lambda| = n + 1/2$). This closes the M_3 component of the continuum-residue identification triad started by Remark 19. Structural sharpening: unlike M_1 (which admits three exact-factor sibling normalizations π, $4/\pi$, $\sqrt{\pi}/2$ of a single generator), M_3's ring $\mathcal{MT}(\mathbb{Z}[i, 1/2], 4)$ is depth-graded with generators $\{1, G, \beta_4, \beta_6, \ldots\}$ at successive depths with no π-power reduction at any depth ≥ 1 (Apéry-style irrationality of Catalan G vs $\pi^{2k}\mathbb{Q}$).* This asymmetry IS the operational content of the master-Mellin partition M_1 *at $k = 0$ depth 0 vs M_3 at $k = 1$ depth 1+. Source:* `debug/sprint_q5p_m3_continuum_memo.md`.

Remark 27 (Q5' Stage 2 substrate enrichment via $J^*(S^3)$, Sprint Q5'-J-Star-S3, June 2026)**.** *A first scoping step of the nested-Hopf-tower substrate enrichment of Remark 23 (a substantial open program) replaces the disjoint sector-idempotent substrate $\mathbb{C}^{N(n_{\max})}$ with the SU(2) Peter–Weyl matrix-coefficient substrate*

$$J^*_{j_{\max}}(S^3) = \mathrm{span}_{\mathbb{Q}}\{\pi^j_{mn} : 0 \leq j \leq j_{\max},\ -j \leq m, n \leq j\},$$

with the matrix-coefficient coproduct $\Delta\pi^j_{mn} = \sum_p \pi^j_{mp} \otimes \pi^j_{pn}$. This coproduct is bit-exactly non-primitive *on every $j > 0$ generator (verified at $j_{\max} \in \{1/2, 1, 3/2\}$ on 29 generators with $j > 0$); coassociativity and counit hold bit-exactly in the free polynomial algebra (147 zero residuals); the antipode axiom requires passing to the standard $\mathcal{O}(SU(2)) \cong \mathcal{O}(SL_2)$ quotient by SU(2) unitarity relations (textbook construction; Klimyk–Schmüdgen 1997 §1.3.2); the pro-system truncation $P_{j_{\max}+1/2\to j_{\max}}$ lifts bit-exactly to a Hopf-algebra homomorphism (132 zero residuals). The candidate motivic Galois group at the quotient is $U^{*(j_{\max})}_{\mathrm{GeoVac},J^*} = SL_2$ for every $j_{\max} \geq 1/2$ — a non-abelian semisimple algebraic group, achieving the qualitative goal of breaking the abelian primitive shape of Remark 23. The trade is axis-orthogonal: $J^*(S^3)$ has non-abelian content but sacrifices the Mellin-slot k-partition; v3.61.0's $\mathcal{H}_{\mathrm{GV}}$ has the Mellin partition but is abelian. The natural continuation, beyond sprint scale, is a synthesis combining both — either (a) tensor product $\mathcal{H}_{\mathrm{v3.61}} \otimes \mathcal{H}^{J^*}$ giving $U^* = \mathbb{G}_a^{3N} \times SL_2$, or (b) Mellin-slot-decorated Peter–Weyl $\pi^{j,k}_{mn}$ giving $U^* = SL_2^3$. The $SL_2 \times \mathbb{G}_a^d$ form would be the Levi decomposition of a general algebraic group (semisimple times pro-unipotent), suggesting the synthesis substrate is the right Stage-2 target. See Paper 55 § subsec:open_m2_m3.*

Remark 28 (Q5' Stage 2 cross-shell off-diagonal Dirac enrichment scoping, Sprint Q5'-OffDiag-Dirac, June 2026)**.** *The first scoping step of the cross-shell off-diagonal Dirac perturbation enrichment — itself a substantial open program — (second of three structural ingredients flagged in Remark 23)* *promotes the v3.61.0 pro-system substrate to track the off-diagonal Dirac perturbation κA via transition generators $T_{s'\to s} := e_s \cdot (\kappa A) \cdot e_{s'}$ (non-zero for E1-Gaunt-selected sector pairs). At $n_{\max} = 2$ on the truncated CH spectral triple, the enriched substrate breaks the sector-locality structural condition that v3.61.0 Track A identified as forcing the abelian primitive coproduct: bit-exact in* sympy.Rational*, 12 of 12 single-step transitions carry non-vanishing η-class values in $\kappa^2 \cdot \mathbb{Z}$ (the M3 cocycle slot non-vanishes on inter-sector content), 12 of 12 chain compositions land on idempotents $T_2 \cdot T_1 = c \cdot e_s$ with non-zero $2\eta(T_1)\eta(T_2)$ cross-term cocycle image, and an additional 18 chain compositions land on two-step transitions outside the basic single-step generator basis (algebra-closure failure). The lowest-order non-trivial commutators $[T_a, T_b] \neq 0$ appear at 24 of 66 = 36% of transition pairs — explicit non-abelian Lie generators. The non-primitivity content has denominator family $\kappa^2 \cdot \mathbb{Z} = 2^{-8} \cdot \mathbb{Z}$ times inverse path counts, structurally aligned with the bit-exact $\pm 1/65536$ closure drift residual of Remark 24 modulo a 2^3 JLO simplex normalization factor; the exact bit-exact bridge identification between the two phenomena is a feasible follow-on. The enriched substrate's candidate motivic Galois group $U^{*(n_{\max})}_{\mathrm{GeoVac,enriched}}$ has both an abelian (idempotent-generated) part and a unipotent (transition-generated) part, structurally targeting the pro-unipotent factor of Connes–Kreimer–Marcolli's cosmic-Galois U^* (arXiv:hep-th/9808042; arXiv:math/0409306) at the Stage-2 level. See Paper 55 § subsec:open_m2_m3 for the cosmic-Galois narrative.*

Remark 29 (Q5' Stage 2 Cuntz-extension enrichment: structural incompatibility, Sprint Q5'-Cuntz, June 2026)**.** *The first scoping step of the long-range Cuntz-extension enrichment ingredient flagged in Remark 23 returns a structural negative. Replace the commutative algebra $\mathcal{A}_{\mathrm{GV}}^{(n_{\max})} = \mathbb{C}^{N(n_{\max})}$ of sector idempotents by the Cuntz algebra $\mathcal{O}_N$ generated by $N = 3 \cdot N(n_{\max})$ isometries $S_{(n,l),k}$ (one per sector (n,l) and Mellin slot $k \in \{0,1,2\}$). The two natural candidate coproducts on $\mathcal{O}_N$ are bit-exactly incompatible with the Cuntz relations: (i) the primitive coproduct $\Delta(S_i) = S_i \otimes 1 + 1 \otimes S_i$ extended as a $*$-algebra homomorphism fails $S_i^* S_j = \delta_{ij}$ on 0/225 pairs at $n_{\max} = 2$ with $N = 15$; (ii) the diagonal coproduct $\Delta(S_i) = S_i \otimes S_i$ passes the first Cuntz relation on 225/225 pairs but fails $\sum_i S_i S_i^* = 1$ by exactly $N(N-1) = 210$ missing cross-terms. Net: $\mathcal{O}_N$ is structurally a Hopf*-comodule *(over the gauge $U(1)$ scaling each S_i by a phase) rather than a Hopf-algebra. The M1/M2/M3 partition survives only at the trivial comodule level via fixed-k generator labels, structurally weaker than the clean Hopf tensor product $\mathcal{H}_{\mathrm{GV}} = \otimes_k \mathcal{H}_{\mathrm{GV}}^{[k]}$ established for the commutative case in Remark 23. The JLO degree-1 cochain ϕ_1 is structurally non-zero on $\mathcal{O}_N$ (the commutative-case vanishing theorem requires sector orthogonality $e_s e_t = \delta_{st} e_s$ which Cuntz isometries violate), but extraction at theorem grade requires the full CH-adjacency-extended Dirac representation through the Pimsner–Voiculescu sequence (Pimsner–Voiculescu 1980,* J. Operator Theory *4; Pimsner 1997, Fields Inst. Commun. 12), genuinely long-range. The Cuntz route is therefore the structurally most disruptive of the three enrichment ingredients flagged in Remark 23: it does not enrich the Hopf substrate but shifts the framework into the categorically different Hopf-comodule-over-gauge-group setting (Cuntz–Pimsner extension). The remaining two ingredients (nested-Hopf-tower $J^*(S^3)$ closed at finite cutoff in Remark 27; cross-shell off-diagonal Dirac closed at finite cutoff in Remark 28) preserve the Hopf-algebra category at the substrate level and are correspondingly more tractable. See Paper 55 § subsec:open_m2_m3 for the cosmic-Galois program narrative.*

Remark 30 (Q5' Stage 2 substrate HEADLINE: Levi-decomposition tensor synthesis, Sprint Q5'-Levi-Synthesis, June 2026)**.** *The HEADLINE Stage-2 substrate construction combines the v3.61.0 abelian primitive substrate of Remark 23 with the v3.62.0 Peter–Weyl substrate of Remark 27 into the tensor-product Hopf algebra*

$$\mathcal{H}_{\mathrm{GV}}^{\mathrm{Levi},(n_{\max},j_{\max})} := \mathcal{H}_{\mathrm{GV}}^{(\mathrm{v3.61}),n_{\max}} \otimes_{\mathbb{Q}} \mathcal{H}_{\mathrm{GV}}^{J^*,j_{\max}}.$$

At finite cutoff $(n_{\max}, j_{\max}) \in \{(2,1/2),(2,1),(3,1/2)\}$, all five Hopf axioms (coassociativity, counit, antipode), bialgebra compatibility, the M1/M3/M2 Mellin-slot k-grading preservation on the first factor, and the non-cocommutativity on the second factor (encoding non-abelian SL_2 content) are verified bit-exactly (882 zero residuals across the joint axiom + truncation panel: 636 axiom residuals + 246 truncation residuals). The Mellin partition lifts to $\mathcal{H}^{\mathrm{Levi}} = \bigotimes_{k=0}^{2} \mathcal{H}^{[k]}_{\mathrm{v3.61}} \otimes \mathcal{H}^{J^*}$. The candidate motivic Galois group at the standard $\mathcal{O}(SL_2)$ quotient is

$$\boxed{U^{*(n_{\max},j_{\max})}_{\mathrm{GeoVac,Levi}} = \mathbb{G}_a^{3N(n_{\max})} \times SL_2}$$

by the Tannakian theorem (tensor product of Hopf algebras = direct product of affine group schemes; Waterhouse 1979 §1.4; Sweedler 1969 Ch. IV; Deligne 1990 Catégories tannakiennes*). The semidirect product $\mathbb{G}_a^{3N(n_{\max})} \rtimes SL_2$ reduces to a direct product because the Levi action on the unipotent radical is trivial: the v3.61.0 generators $x_{(n,l),k}$ carry no $SU(2)$ representation content (they are abelian primitives indexed by sector and Mellin slot), so any SL_2 element acts on them as the identity (Hochschild 1981 Ch. VII Levi-decomposition direct-product reduction). The pro-unipotent $\mathbb{G}_a^{3N(n_{\max})}$ factor carries the master-Mellin engine M1/M3/M2 partition (Remark 23); the semisimple SL_2 factor carries the Peter–Weyl non-abelian content (Remark 27). The Levi-decomposition shape is the published structural target for the cosmic-Galois U^* in the Connes–Marcolli motivic Galois machinery (Connes–Marcolli 2007 arXiv:math/0409306; Connes–Marcolli 2008 book Ch. 4): pro-unipotent times semisimple. The long-range Stage-2 Tannakian construction reduces to (a) full Tannakian closure of the pro-unipotent factor, and (b) verification that the cosmic-Galois U^* of Connes–Marcolli 2007 acts on $\mathcal{H}^{\mathrm{Levi}}_{\mathrm{GV}}$ in the expected way. See Paper 55 § subsec:open_m2_m3.*

Remark 31 (Q5' Stage 2 substrate enrichment via Mellin-slot-decorated Peter–Weyl, Sprint Q5'-Decorated-PW, June 2026)**.** *The alternative synthesis flagged in Remark 27 option (b) — the Mellin-slot-decorated Peter–Weyl substrate*

$$\mathcal{H}^{(j_{\max})}_{\mathrm{dec}} = \mathrm{span}_{\mathbb{Q}}\{\pi^{j,k}_{mn} : 0 \le j \le j_{\max},\ -j \le m,n \le j,\ k \in \{0,1,2\}\}$$

with k-preserving coproduct $\Delta\pi^{j,k}_{mn} = \sum_p \pi^{j,k}_{mp} \otimes \pi^{j,k}_{pn}$ — is constructed and verified bit-exactly at $j_{\max} \in \{1/2, 1\}$ with $n_k = 3$ Mellin slots. All Hopf axioms hold bit-exactly with the standard counit and per-slot $SU(2) \to SL_2$ quotient (411 zero residuals: coassoc 57, counit L+R 114, antipode at quotient 57, k-grading preservation 57, pro-system Hopf-hom 126). The substrate factorises as $\mathcal{H}_{\mathrm{dec}} = \mathcal{O}(SL_2)^{\otimes 3}$ with the three Mellin mechanisms M1, M2, M3 carried by three independent non-abelian semisimple SL_2 factors. The candidate motivic Galois group at the $SU(2)^{\otimes 3}$ quotient is $U^*_{\mathrm{dec},(a)} = SL_2^3$ for every $j_{\max} \ge 1/2$. The alternative k-summing coproduct $\Delta\pi^{j,k}_{mn} = \sum_{k_1+k_2=k \pmod 3} \sum_p \pi^{j,k_1}_{mp} \otimes \pi^{j,k_2}_{pn}$ passes coassociativity and a $\mathbb{Z}/3$-augmented counit $\varepsilon_{\mathrm{aug}}(\pi^{j,k}_{mn}) = \delta_{mn}\delta_{k,0}$ but FAILS the antipode axiom at the per-slot $SU(2)$ quotient by a structural factor of $n_k = 3$ — fixing the normalisation requires $1/n_k \in \mathbb{Q}(1/3)$ breaking the discrete-for-skeleton discipline. The two synthesis routes (this remark's SL_2^3 vs Remark 30's tensor synthesis $\mathbb{G}_a^{3N} \times SL_2$) capture different aspects of the Stage-2 substrate: the tensor synthesis is the Levi-decomposition shape with explicit v3.61.0 abelian primitive content; this remark's decorated-PW is the Mellin-decorated symmetry shape with the M1/M2/M3 partition realised as independent SL_2 factors. Both substrates are valid candidates; the strategic choice between them is the next sprint-scale decision in the long-range Stage-2 program. See Paper 55 § subsec:open_m2_m3.

Remark 32 (Q5' Stage 2 bridge identity closing the T3b umbrella, Sprint Q5'-Bridge-Id, June 2026)**.** *The bit-exact bridge identification flagged as a feasible follow-on in Remark 28 closes (`debug/sprint_q5p_br`* *debug/data/sprint_q5p_bridge_id.json). On the load-bearing (e_2, e_3, e_3, e_2) palindrome at the strict-strong-form cochain-morphism interior cutoff $n_{\max} \geq 3$, the closure drift residual is bit-exactly $-\kappa^4 = -1/2^{16}$, equivalent in four closed forms:*

$$\mathrm{drift}_{n_{\max}\geq 3} = -\kappa^4 = -\frac{1}{4!}\kappa^4\, T^{\mathrm{int}}_{\mathrm{path}} = -\frac{2\eta(T_{(2,0)\to(2,1)})\eta(T_{(2,1)\to(2,0)})}{40} = -\frac{5/8192}{40},$$

with $T^{\mathrm{int}}_{\mathrm{path}} = 24$ the chirality-weighted two-step E1 path count at $n_{\max} \geq 3$ and the integer $40 = 2\cdot 10\cdot 2$ the palindromic two-step path-count product on the OffDiag substrate $T_{(2,0)\to(2,1)}, T_{(2,1)\to(2,0)}$ of Remark 28. The boundary cutoff $n_{\max} = 2$ value $+\kappa^4/3 = +1/196608$ uses $T^{\mathrm{bdy}}_{\mathrm{path}} = 8$, with the bit-exact ratio $T^{\mathrm{int}}_{\mathrm{path}}/T^{\mathrm{bdy}}_{\mathrm{path}} = 24/8 = 3$ identifying the -3 boundary-vs-interior factor of Remark 24 as an explicit integer path-count ratio. The pullback increment closes the integer arithmetic: $32 = 24 + 8$. The denominator-scale mismatch 2^{13} (T3b $\eta \otimes \eta$) vs 2^{16} (Track B drift) is bit-exactly the 2^3 piece of the JLO simplex factor $1/4! = 1/(3 \cdot 2^3)$; the leftover numerator 5 in $5/8192$ is half the path-count product $40 = 5 \cdot 8$. The T3b umbrella conjecture ("the two findings are aspects of one Stage-2 substrate-enrichment story with denominator scales aligned modulo a 2^3 JLO simplex factor") is therefore bit-exactly correct, and the bridge identity is constructive at finite cutoff. Long-range follow-ons (closed-form $T_{\mathrm{path}}(n_{\max})$, Lie-algebra structure constants, continuum-limit Mellin lift) remain open.

Remark 33 (Q5' Stage 2 OffDiag algebra-closure dimension at $n_{\max} = 3$, Sprint Q5'-OffDiag-Closure-nmax3, June 2026)**.** *The first long-range scoping step of the algebra-closure dimension at general $n_{\max}$ (named follow-on (1) of Remark 28) identifies the closure of the OffDiag-enriched substrate $\mathcal{A}^{(n_{\max})}_{\mathrm{OD}} = \mathbb{Q}\langle e_s, A_{\mathrm{off}}\rangle$ under matrix multiplication as the path algebra of the sector quiver realised in $M_{\dim H}(\mathbb{Q})$ via the canonical adjacency representation $A_{\mathrm{off}} = D - \Lambda = \kappa A_{\mathrm{graph}}$. Bit-exact in* sympy.Rational *at $n_{\max} \in \{2, 3\}$: $\dim \mathcal{A}^{(2)}_{\mathrm{OD}} = 84$ and $\dim \mathcal{A}^{(3)}_{\mathrm{OD}} = 592$, with closure decomposing as a sum of per-block ranks over all $N(n_{\max})^2$ sector pairs (all 25 of 25 blocks at $n_{\max} = 2$, all 81 of 81 at $n_{\max} = 3$ realised non-trivially — the sector quiver is fully connected via multi-step E1 dipole paths). Fill-fractions $84/256 = 21/64$ and $592/1600 = 37/100$ slowly increasing; asymptotic limit requires $n_{\max} = 4$ for honest identification. Lie subalgebra of single-step transitions does NOT close: 82 of 378 commutator pairs lie outside the 28-dim transition span at $n_{\max} = 3$, generalising the 24 of 66 at $n_{\max} = 2$. Named sprint-scale follow-on: extend bit-exact closure to $n_{\max} = 4$ for third data point + asymptotic fill-fraction identification. Long-range: closed-form per-block rank formula from quiver combinatorics, and embedding of $\{\mathcal{A}^{(n_{\max})}_{\mathrm{OD}}\}_{n_{\max}}$ into the inductive system whose limit hosts the conjectural pro-unipotent factor of the cosmic-Galois U^* Hopf algebra (Connes–Marcolli 2007, arXiv:math/0409306). See Paper 55 § subsec:open_m2_m3.*

Remark 34 (Q5' Stage-2 combined substrate L5 = L1 categorically, Sprint Q5'-Combined-Substrate, June 2026)**.** *The first long-range scoping step of combining the T3a Peter–Weyl substrate (Remark 27) and T3b cross-shell off-diagonal Dirac substrate (Remark 28) on a SINGLE combined substrate closes via categorical reduction. The combined substrate $\mathcal{A}^{\mathrm{combined}} = \mathcal{A}^{J^*}_{j_{\max}} \otimes \mathcal{A}^{\mathrm{OD}}_{n_{\max}}$ reduces to the categorical TENSOR PRODUCT of Remark 30 at the basic substrate level, not a smash product. Two structural obstructions block the smash-product upgrade: (i) Peter–Weyl matrix coefficients π^j_{mn} are polynomial* functions *on SL_2, while OffDiag transitions $T_{s'\to s}$ are operators on $\mathcal{H}$ with* scalar *cocycle data (η-class values in $\kappa^2\cdot\mathbb{Z}$) — categorically different objects; no SL_2 group element evaluates the 4 spin-1/2 matrix coefficients to the 12 OffDiag η-rationals simultaneously. (ii) The natural $SU(2)$ z-rotation U_θ does NOT preserve the basic 5-sector OffDiag basis: bit-exact symbolic computation (debug/compute_q5p_combined_substrate_part2.py) gives three distinct phase* *factors* $\{1, e^{i\theta}, e^{-i\theta}\}$ *on the conjugate* $U_\theta T U_\theta^{-1}$ *— the result lives in the span of* m_J*-resolved sub-transitions, not in the span of the basic basis. The candidate motivic Galois group at the quotient is therefore the Levi-decomposition product*

$$U^{*(n_{\max},j_{\max})}_{\text{combined}} = SL_2 \times \mathbb{G}_a^{N_{\text{OD}}(n_{\max})},$$

exactly the long-range synthesis target predicted by Remark 30. ***L5 reduces to L1 at the substrate level****; the genuinely new content (a non-trivial smash product) lives at the* m_J*-resolved refinement of the OffDiag basis, which loses the v3.61.0 sector structure as load-bearing data and constitutes a long-range continuation. See Paper 55 § subsec:open_m2_m3.*

Remark 35 (Q5' Stage-2 substrate decision: L1 is forced via the primitive-space invariant, Sprint Q5'-L1-vs-L2-Diagnostic, June 2026)**.** *The strategic question flagged in Remark 34 and in Sprint Q5'-Levi-Arc § 5 – whether the v3.61.0 abelian primitives* $x_{(n,l),k}$ *in* $\mathcal{H}_{\text{GV}}^{(n_{\max})}$ *are reducible to Peter–Weyl matrix coefficients in disguise (so that the simpler* SL_2^3 *substrate of Remark 31 would suffice) or carry independent content (so that the Levi-decomposition substrate of Remark 30 is forced) – is resolved bit-exactly at finite cutoff by the primitive-space invariant* $\dim \mathrm{Prim}(\mathcal{H}) = \dim \mathrm{Lie}(\mathrm{Spec}\,\mathcal{H})$*, a Hopf-isomorphism invariant. For* $\mathcal{H}_{\text{GV}}^{(n_{\max})}$ *the Milnor–Moore structure theorem for connected cocommutative Hopf algebras over* $\mathbb{Q}$ *identifies* Prim *with the generating vector space* $V_{n_{\max}}$*, giving* $\dim \mathrm{Prim}(\mathcal{H}_{\text{GV}}) = 3N(n_{\max}) \in \{6, 15, 27, 42, 60, 81\}$ *at* $n_{\max} \in \{1, \dots, 6\}$*. For* $\mathcal{H}_{\text{dec}}^{(j_{\max})}$ *the standard identification* $\mathrm{Prim}\,\mathcal{O}(SL_2)^{\otimes 3} = \mathfrak{sl}_2 \oplus \mathfrak{sl}_2 \oplus \mathfrak{sl}_2$ *gives* $\dim \mathrm{Prim}(\mathcal{H}_{\text{dec}}) = 9$ *independent of* $j_{\max}$ *(higher* $j_{\max}$ *adds symmetric-square Clebsch–Gordan polynomials of degree* ≥ 2 *in the* $j = 1/2$ *generators, contributing zero new dimensions to* $\mathfrak{m}/\mathfrak{m}^2$ *at the augmentation ideal). At every* $n_{\max} \geq 2$ *the dim defect is strictly positive* $(6, 18, 33, 51, 72)$*, so no Hopf isomorphism exists between the two substrates; L1 is forced. The substrate-dim coincidences flagged in Remark 31 (*$15 = 15$ *at* $(n_{\max}, j_{\max}) = (2, 1/2)$ *and* $42 = 42$ *at* $(4, 1)$*) are small-cutoff artifacts breaking at* $n_{\max} \in \{1, 3, 5, 6\}$*. An independent Hopf-isomorphism invariant (cocommutativity) separates the substrates at every* $n_{\max} \geq 1$ *–* $\mathcal{H}_{\text{GV}}$ *is cocommutative (every generator primitive) while* $\mathcal{H}_{\text{dec}}$ *is not on* $j > 0$ *matrix coefficients. Bit-exact panel: 258 zero residuals / consistent checks (driver* ***debug/compute_q5p_l1_vs_l2_diagnostic.py****; data* ***debug/data/sprint_q5p_l1_vs_l2_diagnostic.json****). The L2* SL_2^3 *substrate of Remark 31 remains valid as a complementary semisimple-only reading of the M1/M2/M3 partition; the Levi-decomposition* $U^*_{\text{GeoVac,Levi}} = \mathbb{G}_a^{3N(n_{\max})} \times SL_2$ *of Remark 30 is the structurally-correct Stage-2 target, matching the published Connes–Marcolli motivic-Galois shape. The canonical long-range next step of the Q5' program is the Tannakian closure of the pro-unipotent factor* $\mathbb{G}_a^{3N(n_{\max})}$ *in the Connes–Marcolli machinery. See Paper 55 § subsec:open_m2_m3.*

Remark 36 (Q5' Stage-2 closed-form generating function for $T^{\text{abs}}_{\text{path}}(n_{\max})$ on the OffDiag substrate, with cutoff-independence + E1 selection + McKean-Singer-on-A^2 closures, Sprint Q5'-T-Path-Generating (T2), June 2026)**.** *The chirality-weighted two-step path count*

$$N^{(2)}_{s' \to s} \;:=\; \mathrm{Tr}(\gamma\, A\, e_s\, A\, e_{s'})/\kappa^2$$

on the OffDiag substrate (Remark 28) is bit-exactly cutoff-independent*: every realised sector pair* (s', s) *has the same* $N^{(2)}$ *value at every* $n_{\max}$ *where both sectors are present. Hypothesis verified bit-exactly across* $n_{\max} \in \{1, 2, 3, 4, 5\}$*, including boundary pairs. The E1 selection rule* $|\Delta n|, |\Delta l| \leq 1$ *holds for all realised transitions. The total signed sum vanishes identically:*

$$T^{\text{signed}}_{\text{path}}(n_{\max}) \;=\; \sum_{(s',s)} N^{(2)}_{s' \to s} \;=\; \mathrm{Tr}(\gamma A^2)/\kappa^2 \;\equiv\; 0,$$

extending McKean-Singer ($\chi(S^3) = 0$, McKean-Singer on $A^0 = I$) to the chirality-weighted A^2-trace. The total absolute path count admits the bit-exact closed form

$$\boxed{T^{\rm abs}_{\rm path}(n_{\max}) \;=\; \frac{(n_{\max}^2 + 9n_{\max} - 4)(n_{\max}^2 + 13n_{\max} - 6)}{6}}$$

verified at panel $(8, 72, 224, 496, 924)$ *for* $n_{\max} \in \{1, 2, 3, 4, 5\}$, *predicting* $T^{\rm abs}_{\rm path}(6) = 1548$. *Bit-exact verification (driver* `debug/compute_q5p_t_path_generating.py`; *data* `debug/data/sprint_q5p_t_path_generatin` *memo* `debug/sprint_q5p_t_path_generating_memo.md`*). Unlocks the continuum Mellin lift of Remark 37.*

Remark 37 (Q5' Stage-2 continuum Mellin lift of the discrete bridge identity to the spectral zeta side with M2/M3 decomposition, Sprint Q5'-Mellin-Lift-Bridge (T1), June 2026)**.** *Using the quartic closed form of Remark 36, the continuum Mellin lift of the v3.63.0 L3 discrete bridge identity* $\mathrm{drift}_{n_{\max}\geq 3} = -\kappa^4$ *(Remark 32) is the generating function*

$$F(s) \;:=\; \sum_{n\geq 1} T^{\rm abs}_{\rm path}(n)\cdot n^{-s} \;=\; \tfrac{1}{6}\zeta(s-4) + \tfrac{11}{3}\zeta(s-3) + \tfrac{107}{6}\zeta(s-2) - \tfrac{53}{3}\zeta(s-1) + 4\,\zeta(s).$$

$F(s)$ admits the bit-exact decomposition by shift parity

$$F(s) \;=\; F_{\rm M2}(s) + F_{\rm M3}(s),$$

$$F_{\rm M2}(s) = \tfrac{1}{6}\zeta(s-4) + \tfrac{107}{6}\zeta(s-2) + 4\,\zeta(s) \quad (\textit{even shifts, M2 Seeley–DeWitt}),$$
$$F_{\rm M3}(s) = \tfrac{11}{3}\zeta(s-3) - \tfrac{53}{3}\zeta(s-1) \quad (\textit{odd shifts, M3 vertex-parity-Hurwitz}).$$

Bit-exact verification: $F - F_{M2} - F_{M3} = 0$ *symbolically; 5 simple poles at* $s \in \{1, 2, 3, 4, 5\}$ *with residues* $\{4, -53/3, 107/6, 11/3, 1/6\}$; *integer-s panel exhibits* mixed M2 (π^{2k}) + M3 ($\zeta(\mathrm{odd})$) content *at every test point (e.g.* $F(6) = -53\zeta(5)/3 + \pi^2/36 + 4\pi^6/945 + 11\zeta(3)/3 + 107\pi^4/540$*). F is structurally distinct from the v3.62.0 M3 $\eta_D(s)$ Hurwitz Mellin (shares only the odd-shift structure with $\mathbb{Q}$ vs $2^s\,\mathbb{Q}$ coefficients). This is the* first algebraic-side realisation of the M1/M2/M3 master Mellin engine partition appearing on a single Hopf-algebra-level invariant — *the Mellin engine extends from the trace-evaluation level (case-exhaustion theorem 4) to the substrate-side Mellin generating function. Driver* `debug/compute_q5p_mellin_lift_bridge.py`; *data* `debug/data/sprint_q5p_mellin_lift_` *memo* `debug/sprint_q5p_mellin_lift_bridge_memo.md`.

Remark 38 (Q5' Stage-2 m_J-resolved smash-product foundation: bit-exact $\Delta m_J \in \{-1, 0, +1\}$ grading and $U(1)$ z-rotation diagonalisation at $n_{\max} = 2$, Sprint Q5'-mJ-Smash-Product (T4), June 2026)**.** *The m_J-resolved OffDiag substrate at $n_{\max} = 2$ admits a bit-exact $\mathbb{Z}_3$-grading by* $\Delta m_J = m_{J,\rm to} - m_{J,\rm from} \in \{-1, 0, +1\}$ *(Wigner-Eckart selection for the vector operator $A = \kappa A_{\rm graph}$) with symmetric dimension count*

$$\dim \mathcal{A}_{-1} = 30, \quad \dim \mathcal{A}_0 = 40, \quad \dim \mathcal{A}_{+1} = 30, \quad \textit{total} = 100,$$

over all 100 non-zero off-diagonal Dirac entries. The $U(1)$ z-rotation $U_\theta|n, \kappa, m_J\rangle = e^{i m_J \theta}|n, \kappa, m_J\rangle$ *acts diagonally:* $U_\theta T_{m'_J\to m_J} U_\theta^{-1} = e^{i\Delta m_J \theta} T_{m'_J \to m_J}$, *exactly reproducing the three phase factors* $\{1, e^{i\theta}, e^{-i\theta}\}$ *that Remark 34 identified as broken on the basic substrate — the m_J-resolved refinement* diagonalises *the $U(1)$ action. Critical scope finding: the j-content at $n_{\max} = 2$ includes $j = 3/2$ states (from $\kappa = \pm 2$ in sectors $(2, 1)$ and $(2, 2)$), so the decorated-PW substrate of Remark 31 at $j_{\max} = 1/2$ is* insufficient *for the full $SU(2)$ action on the m_J-resolved OffDiag;* L2 must be extended to $j_{\max} \geq 3/2$ *as the sprint-scale prerequisite to the long-range smash-product construction* $\mathcal{A}^{m_J\text{-}OD} \rtimes \mathcal{O}(SL_2)$ *— the only path to a non-trivial Hopf extension beyond Levi-decomposition. Driver* `debug/compute_q5p_mj_smash_product.py`*; data* `debug/data/sprint_q5p_mj_smash_product.json`*; memo* `debug/sprint_q5p_mj_smash_product_memo.md`*.*

Remark 39 (Q5' L2 substrate extended to $j_{\max} = 3/2$: sprint-scale prerequisite for the long-range smash-product construction closed, Sprint Q5'-FO1 (v3.66.0), June 2026)**.** *The sprint-scale prerequisite flagged by Remark 38 (T4 of v3.65.0) for the long-range smash-product construction* $\mathcal{A}^{m_J\text{-}OD} \rtimes \mathcal{O}(SL_2)^{\otimes 3}$ *at* $n_{\max} = 2$ *is closed bit-exactly. The L2 decorated-PW substrate of Remark 31 extends to* $j_{\max} = 3/2$ *with all 7 Hopf axioms passing: dim* $\mathcal{H}^{(3/2)}_{\text{dec}} = 90$ *(per slot:* $\sum_{2j \leq 3}(2j+1)^2 = 1 + 4 + 9 + 16 = 30$*), 621 bit-exact zero residuals across coassociativity (*$90/90$*), counit L+R (*$180/180$*), antipode at SU(2)*$^{\otimes 3}$ *quotient (*$90/90$ *structural per slot), k-grading preservation (*$90/90$*), and two pro-system Hopf-homomorphism truncations* $P_{3/2\to 1}$ *(*$126/126$*) and* $P_{3/2\to 1/2}$ *(*$45/45$*). The substrate factorisation* $\mathcal{O}(SL_2)^{\otimes 3}$ *and candidate motivic Galois group* SL_2^3 *are unchanged from v3.63.0* $j_{\max} \in \{1/2, 1\}$ *panel — the* $j_{\max}$ *parameter controls only which irrep matrix coefficients are explicitly included; the underlying group structure is* $j_{\max}$*-invariant. This unblocks the long-range smash-product construction at the sprint-scale prerequisite level. Driver* `debug/compute_q5p_fo1_l2_extension_jmax32.py`*; data* `debug/data/sprint_q5p_fo1_l2_extension_jmax32.js` *memo* `debug/sprint_q5p_fo1_l2_extension_jmax32_memo.md`*.*

Remark 40 (Q5' Interpretation C closure: $F(s)$ integer-s panel sits in MT period ring at level ≤ 4 over $\mathbb{Z}[i, 1/2]$, AND U^* acts on $\chi, \eta, F(s)$ cocycle classes via period-pairing, Sprint Q5'-FO2+FO3 (v3.66.0), June 2026)**.** *T5 of v3.65.0 identified three interpretations of "*U^* *acts on* $\mathcal{H}_{\text{Levi}}$*": (A) Hopf-coaction, (B) Hopf-automorphism, (C) period-pairing on the image* $\pi(\mathcal{H}_{\text{Levi}}) \subset \mathbb{C}$ *via motivic Galois conjugation. Interpretation C is the cleanest sprint-scale closure. This remark records its bit-exact closure for all named cocycle-class families. T5 SQ1+SQ2+SQ5 combined: (i) every* $\chi_{(n,l)}$ *value and every* $\eta_{(n,l)}$ *value (60 + 60 entries up to* $n_{\max} = 4$*) is an integer in* $\mathbb{Z}$ *at depth 0 in the MT period ring; (ii) every term of* $F(s)$ *(Remark 37) at integer* $s \in \{6, 7, 8, 9, 10\}$ *(5 cells × 5 terms = 25 term classifications) is either pure-Tate* $\pi^{2k}\,\mathbb{Q}$ *(M2 component of v3.65.0 T1) or odd-Riemann* $\zeta(\text{odd})\,\mathbb{Q}$ *(M3 component); both sit in* $\mathrm{MT}(\mathbb{Q}, 1)$*, which embed in* $\mathrm{MT}(\mathbb{Z}[i, 1/2], 4)$ *trivially. The Paper 18 §III.7 M1/M2/M3 master Mellin engine partition aligns bit-exactly with the standard MT depth/weight grading at integer-shifted* ζ *values:* $k = 0$ *M1 (Hopf-base measure, depth 0, weight 2 in* π*),* $k = 2$ *M2 (Seeley–DeWitt, depth 0, weight* $2k'$ *in* $\pi^{2k'}$*),* $k = 1$ *M3 (vertex-parity-Hurwitz, depth 1, weight* $2k' + 1$ *in* $\zeta(2k' + 1)$*). The cosmic-Galois* U^* *acts on* χ, η *trivially (depth 0 fixed-points), on the M2 component of* $F(s)$ *as the Tate subgroup (invariant up to* π^{2k} *scaling), and on the M3 component as the standard motivic Galois action on* $\mathrm{MT}(\mathbb{Q}, 1)$ *odd-zeta classes (Brown 2012, Glanois 2015). Total bit-exact zero residuals: 145 (25 + 60 + 60). Driver* `debug/compute_q5p_fo2_fo3_mt_period_containment.py`*; data* `debug/data/sprint_q5p_fo2_fo3_mt_period.js` *memo* `debug/sprint_q5p_fo2_fo3_mt_period_memo.md`*. Honest scope: Interpretation C closes for the named cocycle classes (*$\chi, \eta, F(s)$ *panel); Interpretations A (Hopf-coaction) and B (Hopf-automorphism) remain open as named sprint-scale (B via SQ3) and long-range (A) follow-ons. The L1 follow-on (a) Tannakian closure of the pro-unipotent factor in the Connes–Marcolli machinery remains the canonical long-range next step.*

Remark 41 (Rational-residue extension of the case-exhaustion theorem: the Bernoulli ladder and the functional-equation reading of the two-layer split (Sprint GB, 2026-05-29))**.** *The case-exhaustion theorem 4 classifies the* π*-content of every GeoVac observable as* $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$*. The companion question is where the framework's* rational *coefficients come from. A diagnostic on the gravity sector's small rationals places them on a single Bernoulli ladder, with each rung* B_{2n} *glued across* *two windows by the Riemann functional equation:*

$$\zeta(2n) = (-1)^{n+1}\frac{B_{2n}(2\pi)^{2n}}{2(2n)!} \in \pi^{2n}\,\mathbb{Q} \qquad \Longleftrightarrow \qquad \zeta(1-2n) = -\frac{B_{2n}}{2n} \in \mathbb{Q}.$$

The left (positive-even) window is the M2 transcendental ring of Layer 2 observables; the right (negative-integer) window is the π-free rational residue carried by Layer 1 skeleton coefficients. Verified placements (all exact in `sympy`*):*

- *Conical-defect tip coefficient* $\frac{1}{12} = -\zeta(-1) = B_2/2$ *(§4.15); per-t UV target* $\frac{1}{24} = -\zeta(-1)/2 = B_2/4$ *(the string c/24); scalar* S^3 *Casimir* $\frac{1}{240} = +\zeta(-3)/2$ *(*B_4 *rung); the* α*-observation's* $F = \pi^2/6 = \pi^2 B_2 = \zeta(2)$ *(Paper 2; the M2 window of the same* B_2 *whose skeleton window is the conical tip).*
- Non-trivial internal check*: the replica-entropy derivative is forced, not free —* $\frac{d}{d\alpha}\big[\frac{1}{12}(\frac{1}{\alpha} - \alpha)\big]\big|_{\alpha=1} = -\frac{1}{6} = -B_2 = 2\zeta(-1)$*, the* α*-derivative of the tip's* $B_2/2$*.*
- Controls behave*:* $\kappa = -1/16$ *(the Fock Jacobian* Ω^{-4}*) has no clean* ζ *form; the Dirac Casimir* $17/480$ *misses the scalar* B_4 *rung by exactly* $1/32$ *and requires the half-integer Hurwitz shift* $\zeta(-3, \frac{1}{2})$*, i.e. the scalar/spinor distinction is the integer/half-integer (M2/M3) split.*

The reading is the rational-residue extension of Theorem 4: the rational coefficients are the negative-integer analytic continuation of the same *Mellin object whose positive-even values are the Layer-2 transcendentals, and the empirical two-layer split (Paper 34/35) is the Riemann functional-equation reflection* $\zeta(s) \leftrightarrow \zeta(1-s)$ *instantiated per Bernoulli order.*

Honest scope. *This is a structural correspondence at the same epistemic level as Theorem 4, not a forcing proof; it concerns* ζ *at* integer *arguments (special values), not the non-trivial zeros, so it makes no contact with the critical-line structure (cf. WH6 and the three classical-RH walls RH-N/O/M, which remain in force). The* B_2*,* B_4*,* and B_6 *rungs are now populated (Sprint GB-3): the* B_6 *rung's falsifiable target was met by the conformal-scalar Casimir on* S^5 *(a GeoVac manifold via the Bargmann–Segal lattice, Paper 24), which evaluates to* $-31/60480 = \frac{1}{24}(\zeta(-5) - \zeta(-3))$ *— the* B_6 *rung* $\zeta(-5) = -1/252$ *enters with clean coefficient* $+\frac{1}{24}$*, alongside* B_4 *at* $-\frac{1}{24}$*, with* B_2 absent*. The* $\zeta(-5)-\zeta(-3)$ *spreading (rather than a pure* $\zeta(-5)$*) is the* same *quadratic-multiplicity mechanism that obstructs the functional equation in Paper 28's functional-equation obstruction remark (the* S^5 *harmonic degeneracy is quartic), now seen on the Casimir side from an independent observable; the method reproduces the* S^3 *control* $1/240 = \frac{1}{2}\zeta(-3)$ *(Paper 35). The integer/half-integer (M2/M3) split recurs: the conformal scalar gives the clean ladder value, the half-integer HO/Bargmann* S^5 *Casimir gives* $-17/3840$ *(the "17" echoing the Dirac-*S^3 $17/480$*). Next rung:* S^7 *conformal Casimir is predicted to spread across* $\zeta(-7), \zeta(-5), \zeta(-3)$*. The ladder is thus supported across three rungs and two observable types, but remains a structural correspondence, not a forcing theorem. Source:* `debug/sprint_gb_bernoulli_ladder_memo.md`*,* `debug/sprint_gb3_b6_rung_memo.md`*.*

Remark 42 (Bisognano–Wichmann reading of the M1 sub-mechanism on temporal compactifications (Track D, May 2026))**.** *The M1 sub-mechanism of the master Mellin engine,* $\mathcal{M}[\mathrm{Tr}(D^0\,e^{-tD^2})]$*, produces a* Vol/Mellin-measure *factor on the spatial* S^3 *Hopf base (*$\mathrm{Vol}(S^2)/\pi^2 = 4/\pi$*, the L2 propinquity rate constant; Sprint MR-A/B in Remark 15) and on a temporal* S^1_β *factor (the bare circumference* β *itself; Sprint TD Track 1,* $T_{S^3}\otimes T_{S^1_\beta}$*). Sprint TD Track 4 (*`debug/sprint_td_track4_memo.md`*, May 2026) extends the temporal-side instantiation to the Euclidean Schwarzschild cigar, where smoothness at* $r = 2M$ *forces the* τ*-circle period* $\beta_{\mathrm{cigar}} = 8\pi M$*, reproducing the Hawking temperature* $T_H = 1/(8\pi M)$ *from M1 alone (sympy residual zero). Track D (*`debug/bisognano_wichmann_track_d_memo.md`*,* *May 2026) makes explicit the dual physical reading of M1 on temporal compactifications: under the standard Wick-rotation chain (Hartle–Hawking [45]; Sewell [44]; Bisognano–Wichmann [43]), the framework's M1 2π on S^1_τ IS the period of the modular-flow / Lorentz-boost orbit of a stationary observer outside the Schwarzschild horizon. Equivalently, the M1 sub-mechanism on temporal compactifications is the Wick-rotated image of the Bisognano–Wichmann modular-flow mechanism. This is a* structural correspondence*, not a literal identification: two mathematical objects (a Hopf-bundle measure factor on the Riemannian side; a modular-automorphism period on the Lorentzian side) are equated by the published Wick-rotation prescription (Bisognano–Wichmann 1976; Sewell 1982; Hartle–Hawking 1976), not by an operator-level proof inside the truncated spectral triple $\mathcal{T}_{n_{\max}}$. The natural follow-up that would lift the correspondence to literal identification — compute the modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted Camporesi–Higuchi state and verify $\sigma_{i\cdot 2\pi}$ = identity at finite $n_{\max}$, then take the GH limit using the lemmas of Theorem 5 — is named as the principal falsifier in Track D §3.1; estimated beyond sprint scale; priority MEDIUM-HIGH. The case-exhaustion theorem 4 itself is unchanged: M1 now has* two physical interpretations *(Hopf-bundle measure / boost-orbit period via Bisognano–Wichmann), unified by Wick rotation, not two independent constraints on the master Mellin engine partition. The corollary on Rindler wedges (Unruh temperature $T_U = a/(2\pi)$ from the same M1 mechanism with $\beta_{\text{Rindler}} = 2\pi/a$) is a pendant follow-up flagged in Track D §5.4; not pursued in the sprint window.*

Unruh pendant (May 2026). *The Track D §5.4 follow-up was executed as a pendant sprint (`debug/unruh_pe` May 2026). Wick-rotating Rindler coordinates (η, ρ) for an observer at proper acceleration a produces Euclidean polar (η_E, ρ) with conical regularity at $\rho = 0$ forcing η_E-period 2π, hence proper-time period $\beta_U = 2\pi/a$ and Unruh $T_U = a/(2\pi)$ (Unruh [46]). Sprint TD Track 1's* `matsubara_spectrum()` *applied verbatim with $\beta = 2\pi/a$ reproduces the Unruh-thermal Matsubara spectrum bit-identically (sympy residual zero on $T_U = 1/\beta_U$, lowest bosonic mode = a, lowest fermionic mode = $a/2 = \pi T_U$). The single 2π in β_U is the same M1 Hopf-base measure /* $\mathrm{Vol}(S^1)$ *signature that drove the cigar identification, instantiated on the Rindler η_E-circle instead of the cigar τ-circle. The reading is identical: under the Wick-rotation chain, the framework's M1 2π on $S^1_{\eta_E}$ IS the period of the modular-flow / boost-orbit of the accelerated observer. Hawking, Bisognano–Wichmann, and Unruh are therefore three faces of one Wick-rotation theorem with one M1 sub-mechanism, all of which the master Mellin engine tracks at $k = 0$; the published Wick-rotation prescription supplies the bridge to the Lorentzian-side modular-flow vocabulary in each case. The* operator-level falsifier *(modular Hamiltonian on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted state has analytic period 2π) named in Track D §3.1 covers all three faces simultaneously, since the bridge runs through the same KMS-Tomita-Takesaki algebra in every case; lifting the period-closure check on any one face lifts the structural correspondence on Hawking, Bisognano–Wichmann, and Unruh together. Same scope (structural correspondence, not literal identification); same falsifier; same MEDIUM-HIGH priority and scale.*

Sprint L1 closure (2026-05-16): literal identification at the operator-system level (Riemannian). *The principal falsifier named above is now closed positively at the strongest possible level. Sprint L1 (`debug/l1_modular_hamiltonian_results_memo.md`) constructs the modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$ for the hemispheric-wedge-restricted (W1 of L1-A) Camporesi–Higuchi state and verifies the period-closure $\sigma_{2\pi}(O) = O$ for all four physical witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh) at every tested $n_{\max} \in \{2, 3, 4, 5\}$. Maximum periodicity residual scales as $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\text{machine}})$: 1.8×10^{-16} at $n_{\max} = 2$ ($\dim_H = 16$), 4.8×10^{-16} at $n_{\max} = 3$ ($\dim_H = 40 = g_3^{\text{Dirac}} = \Delta^{-1}$), 9.3×10^{-16} at $n_{\max} = 4$ ($\dim_H = 80$), 1.6×10^{-15} at $n_{\max} = 5$ ($\dim_H = 140$). Cross-witness collapse is bit-exact (residuals match across witnesses at machine precision, max consistency residual = 0.0 at every tested $n_{\max}$): the period closure is independent of κ_g because the geometric BW-α realization $K = J_{\text{polar}}$ (the rotation generator pre-*

serving the wedge) has integer spectrum (odd integers two_m_j), so $\mathrm{e}^{\mathrm{i}\cdot 2\pi\cdot n} = 1$ *holds for all integer* n*. The closure is finite-cutoff exact, not a limit-statement, and does not require the Gromov–Hausdorff machinery of Theorem 5 (although it is consistent with it). This lifts the BW reading from "structural correspondence" to "literal identification at the operator-system level (Riemannian)" at every finite* $n_{\max}$*. The four-witness Wick-rotation theorem (HH + Sew + BW + Unruh) becomes a framework-internal theorem at signature* $(3,0)$*. The Lorentzian extension to signature* $(3,1)$ *via the Bizi–Brouder–Besnard 2018 [48] Krein-space spectral triple is the named Sprint L2 follow-up (long-range scale per Sprint L0 audit* `debug/lorentzian_l0_audit_memo.md`*). The L1-A architecture obstructions §§10a–10e of* `debug/l1_modular_hamiltonian_architecture_memo.md` *were addressed: §10a operator-system leakage is zero by construction (*$\sigma_{2\pi}(O) = O$ *exactly, so* $\sigma_{2\pi}(O)$ *trivially stays in* $\mathcal{O}_{n_{\max}}$*); §10b/c chirality / M3 trivialization consistency: the closure is purely M1-content (the* 2π *in the period is the Hopf-base measure* $\mathrm{Vol}(S^1)$*, identified per Sprint TS-E1's master Mellin engine case-exhaustion theorem 4); §10d Gibbs phenomenon: the hemispheric projector* P_W *uses a discrete* $m_j \to -m_j$ *reflection (no smooth cutoff needed, exact involution); §10e* J_{TT}*-vs-*J_{GV} *compatibility: the Tomita conjugation is constructed as a fresh class* `geovac.modular_hamiltonian.TomitaConjugation` *with verified* $J_{\mathrm{TT}}^2 = +I$*, categorically distinct from the Connes KO-dim 3 charge conjugation* J_{GV} *of* `geovac.real_structure` *(which has* $J_{GV}^2 = -I$*).*

Sprint L1-tighten (2026-05-16 evening): unified BW-α/BW-γ closure. *The L1 closure above used the BW-*α *geometric* $K = J_{\mathrm{polar}}$ *as the load-bearing object; the Tomita-Takesaki BW-*γ *realization was only an expectation-level KMS cross-check. Sprint L1-tighten (*`debug/l1_tighten_tomita_results_m` *rebuilds the BW-*γ *construction at the full Tomita-Takesaki level on the GNS Hilbert-Schmidt space* $H_{\mathrm{GNS}} = M_{\dim_W}(\mathbb{C})$*, extracts* $K_{\mathrm{TT}} = -\log\Delta$ *from the polar decomposition* $S = J_{\mathrm{TT}}\Delta^{1/2}$*, and verifies* $\sigma_{2\pi}^{\mathrm{TT}}(O) = O$ *at the operator-action level for all four physical witnesses at every tested* $n_{\max} \in \{2,3,4,5\}$*. Result:* UNIFIED_STRONG closure *at all tested cutoffs and witnesses. Maximum periodicity residual at the Tomita-Takesaki level:* 1.09×10^{-16} *(*$n_{\max} = 2$*),* 1.07×10^{-15} *(*$n_{\max} = 3$*),* 3.03×10^{-15} *(*$n_{\max} = 4$*),* 4.79×10^{-15} *(*$n_{\max} = 5$*). The BW-*α *and BW-*γ *flows are operator-action-level conjugates:* $\sigma_t^{\mathrm{TT}}(a) = \sigma_{-t}^{\alpha}(a)$ *bit-exactly at general* t *(verified at* $t = 1$ *with residual* $< 10^{-16}$*), and identical at the period* $t = 2\pi$ *because* $e^{\pm i2\pi n} = 1$ *for any integer* n*. The state-dependent* $J_{\mathrm{TT}}(X) = \rho^{1/2}X^*\rho^{-1/2}$ *from the polar decomposition satisfies* $J_{\mathrm{TT}}^2 = +I$ *to machine precision at every* $n_{\max}$ *(the L1 verification only checked the canonical* $U = I$ *sanity case; L1-tighten verifies the* $\rho^{1/2}$*-weighted genuine Tomita conjugation). The construction uses the natural local Hamiltonian* $H_{\mathrm{local}} := K_\alpha^W/\beta$*, making the wedge KMS state* $\rho_W = e^{-K_\alpha^W}/Z$ *the operator-system analog of the Wightman BW vacuum* $\rho_W \propto e^{-2\pi K_{\mathrm{boost}}}$ *on the Rindler wedge. The four-witness Wick-rotation theorem is closed at signature* $(3,0)$ *via* both *the geometric BW-*α *and the Tomita-Takesaki BW-*γ *readings, not just one; the L1 "BW-*α*-only" caveat is removed.*

Theorem 5 (Gromov–Hausdorff convergence on S^3)**.** *Let* $\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3,\Sigma), D_{\mathrm{CH}})$ *be the round-*S^3 *metric spectral triple with the Camporesi–Higuchi Dirac, and let* $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$ *be the Connes–van Suijlekom truncated metric spectral triple at cutoff* $n_{\max}$ *(operator system from sprint R2.1; truthful Camporesi–Higuchi Dirac restricted to the truncation). Then the truncated triples converge to* $\mathcal{T}_{S^3}$ *in van Suijlekom's state-space Gromov–Hausdorff distance (the strengthening to the Latrémolière quantum propinquity remains an open named gap; see Remark 43):*

$$d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; C_3 \cdot \gamma_{n_{\max}} \xrightarrow[n_{\max}\to\infty]{} 0,$$

where $C_3 = 1$ *is the Lipschitz comparison constant and* $\gamma_{n_{\max}}$ *is the L2 mass-concentration moment of the central spectral Fejér kernel on SU(2). The bound is qualitative-rate; the asymptotic rate*

$\gamma_{n_{\max}} = O(\log n/n)$ *is consistent with but not rigorously proved by the small-n closed forms (deferred to a quantitative-rate sub-track).*

Proof sketch. The proof is a Latrémolière tunneling-pair assembly of five lemmas closed in WH1 sprint R2.5 (April–May 2026):

L1′ (Sprint R3.5, 2026-05-04; `debug/wh1_r35_full_dirac_memo.md`): the offdiag Camporesi–Higuchi operator system substrate has every non-forced cross-shell Connes-distance pair finite; this provides the Connes-vS-compatible metric sub-structure on which the propinquity is defined.

L2 (Sprint R2.5/L2, 2026-05-04; `debug/r25_l2_proof_memo.md`): the positive central spectral Fejér kernel $K_{n_{\max}} = Z_{n_{\max}}^{-1} |\sum_{j \le j_{\max}} \sqrt{2j+1}\, \chi_j|^2$ on SU(2) is normalized, central, with mass-concentration moment $\gamma_{n_{\max}} \to 0$ (closed-form algebraic values $\gamma_2, \gamma_3, \gamma_4$, monotone decrease), Plancherel symbol $\hat{K}_{n_{\max}}(N) = N/Z_{n_{\max}}$ on the Fock-shell index, and central-multiplier cb-norm $\|T_K\|_{\mathrm{cb}} = 2/(n_{\max}+1)$ on the central subalgebra (Bożejko–Fendler / Pisier 2001 Ch. 8 transcription).

L3 (Sprint R2.5/L3, 2026-05-04; `debug/r25_l3_proof_memo.md`): the Lipschitz comparison $\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}} \le C_3 \|\nabla f\|_{L^\infty}$ holds with $C_3 = 1$ on the natural Avery–Wen–Avery test panel at $n_{\max} \in \{2, 3, 4\}$, with theoretical bound $(N-1)/\sqrt{N^2-1} \nearrow 1$ asymptotically. The "torsion-and-curvature correction" anticipated in the Track-A master memo is identified concretely as the spectral-gap quantization $|n - n'|$ of the truthful Camporesi–Higuchi Dirac.

L4 (Sprint R2.5/L4, 2026-05-06; `debug/r25_l4_proof_memo.md`): the Berezin-type reconstruction map $B_{n_{\max}} : C(S^3) \to \mathcal{O}_{n_{\max}}$ defined by $B_{n_{\max}}(f) = \sum_{N \le n_{\max}} \sum_{L,M} \hat{K}_{n_{\max}}(N)\, c_{NLM}\, M_{NLM}$ (using L2 Plancherel weights and Avery–Wen–Avery multipliers from sprint R3.1) is positive (preserves $f \ge 0$), contractive ($\|B_{n_{\max}}(f)\|_{\mathrm{op}} \le \|f\|_{C(S^3)}$), an approximate identity ($\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\mathrm{op}} \le \gamma_{n_{\max}} \|\nabla f\|_{L^\infty}$), and L3-compatible ($\|[D_{\mathrm{CH}}, B_{n_{\max}}(f)]\|_{\mathrm{op}} \le \|\nabla f\|_{L^\infty}$). The map is the SU(2) analog of the Connes–van Suijlekom 2021 §3 Fejér-kernel reconstruction on S^1, equivalently the non-Kähler analog of the Hawkins 2000 Berezin map for Kähler coadjoint orbits.

L5 (Sprint R2.5/L5, 2026-05-06; `debug/r25_l5_proof_memo.md`): the tunneling pair $(B_{n_{\max}}, P_{n_{\max}})$ between $\mathcal{T}_{S^3}$ and $\mathcal{T}_{n_{\max}}$ has reach $\le \gamma_{n_{\max}}$ in both directions (L4(c) approximate identity + the dual estimate via L2(g) Bożejko–Fendler), height contributions bounded constants (L4(b) contractivity + $\mathrm{height}_P = 0$ as P is a projection), giving $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \le C_3 \cdot \gamma_{n_{\max}} \to 0$.

The full lemma chain is verified numerically at $n_{\max} \in \{2, 3, 4\}$ with monotone decrease of the bound (2.075, 1.610, 1.322); ratio $\Lambda(n_{\max} = 4)/\Lambda(n_{\max} = 2) = 0.637$. Module `geovac/gh_convergence.py` packages the assembly with 39 unit tests passing, integrating with the L1′–L4 modules verbatim. □

Remark 43 (Status and named gaps (2026-06-09))**.** *Three gaps in the five-lemma proof chain behind Theorem 5 are named here. (i) The L2 cb-norm statement conflated the kernel's Plancherel* mass distribution *with its convolution*-multiplier symbol*: the smoothing map is unital completely positive with cb-norm exactly* 1*, and the quantity* $2/(n_{\max}+1)$ *is the mass-distribution maximum, not a cb-norm; the rate content (the mass-concentration moment* $\gamma_{n_{\max}}$*, its closed forms, and the* $4/\pi$ *asymptote) is unaffected. (ii) The* reach_P *dual estimate in L5 is a* named gap*: a forward cb-norm controls no partial inverse; the correct mechanism is an antiderivative-transference lemma in the style of Leimbach–van Suijlekom (Adv. Math.* ***439*** *(2024) 109496); for scalar sectors on compact metric groups the conclusion is covered by Gaudillot-Estrada–van Suijlekom [15]. (iii) The distance* *proved is van Suijlekom's state-space Gromov–Hausdorff distance (arXiv:2005.08544 lineage); the Latrémolière propinquity label is an open named gap — the cited Latrémolière framework builds tunnels from quantum isometries, not UCP pairs. (iv) The kernel condition $L(a) = 0 \iff a \in \mathbb{C}\mathbf{1}$ holds for the engineered offdiag Camporesi–Higuchi Dirac (kernel dimension $1/14$ at $n_{\max} = 2$, $1/55$ at $n_{\max} = 3$, verified) but* fails *for the truthful chirality-diagonal CH Dirac ($10/14$, $26/55$). Items (ii)–(iv) are resolved (2026-06-10) by the translation-seminorm reframing of the companion S^3 paper: the truncated state space is metrized by the left-translation Lipschitz seminorm (equal to the Lipschitz constant on the continuum), whose kernel is exactly the scalars on the* truthful *substrate (Schur multiplicity-one plus per-band injectivity of band-limited compression); the dual direction is supplied by an exact-fit spinor lifted state (the Fejér vector $h \otimes \chi$, lying in the window by the $V_j \otimes V_{j\pm1/2}$ block decomposition of the Camporesi–Higuchi eigenspaces), and both almost-inverse defects are Fejér smoothings at the common moment $\gamma_{n_{\max}}$ — no partial inverse or transference estimate remains. The convergence theorem is unconditional in van Suijlekom's state-space framework at rate $(4/\pi + o(1)) \log n_{\max}/n_{\max}$.*

Remark 44 (Limit identification)**.** *A companion proposition (Kantorovich–Rubinstein duality on the van Suijlekom state-space distance) identifies the limit of $(\mathcal{T}_{n_{\max}}, d_{D_{n_{\max}}})$ with the Wasserstein–Kantorovich state space $(P(S^3), d_{\mathrm{Wass}})$ against round-S^3 geodesic distance. The argument is bookkeeping at the L4(a) positivity + L4(b) contractivity level (Rieffel 1999/2004; D'Andrea–Lizzi–Martinetti 2014) and is recorded in `debug/r25_l5_proof_memo.md` §7. No new analytical content.*

Remark 45 (WH1 keystone closure)**.** *Theorem 5 closes the WH1 keystone (CLAUDE.md §1.7). The structural claim of WH1 — that GeoVac is an almost-commutative spectral triple with Connes–van Suijlekom truncations converging to the round-S^3 spectral triple — is now* proved *at the qualitative-rate level. The "torsion-and-curvature correction" anticipated in the Track-A master memo is identified concretely as the spectral-gap quantization $|n - n'|$ of the truthful Camporesi–Higuchi Dirac, with per-multiplier ratio bounded by $N - 1$ from the SO(4) selection rule, and the corresponding asymptotic ratio $(N-1)/\sqrt{N^2-1} \nearrow 1$ giving the L3 constant $C_3 = 1$. The remaining open questions (Higgs-gap analysis on the four-way S^3 coincidence; full real structure J audit at finite $n_{\max}$, scoping memo `debug/real_structure_finite_nmax_memo.md`) are downstream of this theorem. The proof chain was corrected in June 2026 (Remark 43): the theorem stands unconditionally in van Suijlekom's state-space framework under the translation-seminorm metrization.*

Remark 46 (Pythagorean M1 closure of the L2-E residual)**.** *The case-exhaustion theorem 4 classifies the transcendental ring in which the π-content of a finite chain of Paper 34 projections must sit. A post-Sprint-L2-E PSLQ probe (`debug/h_local_residual_pslq_memo.md`, 2026-05-17) provides a concrete operator-level instantiation of the theorem at $k = 0$ (the M1 sub-mechanism / Hopf-base measure sector). On the truncated Lorentzian Krein spectral triple (Paper 43 [52] §10.2), the Bisognano–Wichmann-aligned local Hamiltonian $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$ and the wedge-restricted Lorentzian Dirac D_W^L are Hilbert–Schmidt orthogonal,*

$$\langle H_{\mathrm{local}}, D_W^L \rangle_{\mathrm{HS}} \;=\; 0 \qquad \textit{(bit-exact at every tested } (n_{\max}, N_t)\textit{)},$$

and the squared residual decomposes as a Pythagorean sum in the M1 ring:

$$\left\| H_{\mathrm{local}} - D_W^L \right\|_F^2 \;=\; \frac{\kappa_g^2 \cdot S(n_{\max})}{4\pi^2} \;+\; D(n_{\max}),$$

with $S, D \in \mathbb{Q}[n_{\max}]$ given by cumulative wedge Casimir traces (Paper 43 §10.2, Cor. cor:pythagorean_orthogonality therein). The sole transcendental in the residual is $1/\pi^2 = 1/\mathrm{Vol}(S^1)^2$, the squared Hopf-base-measure signature, exactly the M1 output the case-exhaustion theorem predicts for a $k = 0$ chain that engages no M2 heat-kernel content and no M3 vertex-parity Hurwitz content.

This sharpens the engine's predictive content along a complementary axis to Remark 15. Sprint MR-A/B/C established the mechanism-as-domain partition (M1 owns propinquity-rate observables, M2 owns heat-kernel residuals, M3 owns vertex-restricted parity observables). The L2-E result adds a fourth empirical instantiation: M1 owns the modular-Hamiltonian / spectral-action generator-orthogonality residual on the hemispheric-wedge KMS state, with the specific ring decomposition $\mathbb{Q}[n_{\max}] \oplus \pi^{-2}\,\mathbb{Q}[n_{\max}]$ *that the case-exhaustion theorem allows at* $k = 0$*. The engine does not just classify which* π*-powers can appear; it now correctly predicts the exact M1 ring for the modular-Hamiltonian generator distinction observed in Paper 42 §7.2 and Paper 43 §10.1.*

The orthogonality is universal across all six modular witnesses (Paper 42 [51] Cor. cor:single_construction; Paper 43 §10.2), via κ_g*-linearity of* H_{local} *and* κ_g*-independence of* D_W^L*. The structural mechanism is the diagonal/off-diagonal subspace decomposition of* $\mathcal{B}(\mathcal{H}_W)$*:* H_{local} *is real and diagonal in the unfolded wedge spinor basis (eigenvalues* $\mathrm{two\_m}_j$*),* D_W^L *is off-diagonal in the same basis (intertwines* $\pm m_j$ *chirality partners). A formal proof of the subspace decomposition at general* $(n_{\max}, N_t)$ *is a named follow-on; the empirical verification across the 18-cell six-witness panel is strong evidence the decomposition is structural rather than coincidental.*

Cross-references. *Paper 43 §10.2 Cor. cor:pythagorean_orthogonality (closed form and witness-universality); Paper 42 §8 Rem. rem:pythagorean_underlies_collapse (the six-witness-collapse statement at the inner-product level); Paper 24 §5.3 four-layer Coulomb/HO asymmetry (the structural finding that the Pythagorean orthogonality is* S^3*-specific to the construction — the* S^5 *Hardy sector has no clean analog, joining the modular-Hamiltonian level as a fourth Coulomb/HO asymmetry layer alongside spectrum-computing* L_0*, calibration* π*, and Wilson matter coupling).*

Remark 47 (BH Phase 0 — clean negative on wedge KMS area-law)**.** *The modular-Hamiltonian construction closed by Sprint L1 (the operator-system-level four-witness Wick-rotation theorem, this section) and by Sprint L2-E (the Lorentzian extension at finite cutoff) reproduces the Hartle–Hawking and Unruh* temperatures *via the M1 mechanism (§ 42). A natural follow-up question is whether the same construction also reproduces the Bekenstein–Hawking* entropy $S_{BH} = A/(4G)$ *at finite cutoff via the von Neumann entropy of the wedge KMS state* $\rho_W = e^{-K_\alpha^W}/Z$*. Sprint BH-Phase0 (`debug/bh_phase0_diagnostic_memo.md`, 2026-05-22) tests this directly at* $n_{\max} \in \{2, 3, 4, 5, 6, 7\}$.

Result. $S(\rho_W) \approx 1.944\,\log(n_{\max}) + 0.565$ *at* $R^2 = 0.99991$ *across the panel. No polynomial scaling appears in any of the four candidate fits (log-linear, power-law, poly-2, poly-3). An s-deformation* $\rho(s) = e^{-sK_\alpha^W}/Z$ *across* $s \in \{0.01, 0.05, 0.1, 0.5, 1, 2, 2\pi\}$ *shows power-law exponents* $\alpha \in [0.64, 0.72]$*, never close to the* $\alpha = 2$ *that a naive Bekenstein–Hawking area-law* $S \sim n_{\max}^2$ *would predict.*

Mechanism. *The structural reading is*

$$S(\rho_W) \;\approx\; \log\!\big(n_{\text{equator}}\big), \qquad n_{\text{equator}} = n_{\max}\,(n_{\max}+1),$$

where n_{equator} *is the count of wedge states with smallest* $|m_j| = 1/2$ *(the equator analog). The BW canonical state is effectively the maximally mixed state on the lowest-*K_α *shell, with exponentially suppressed contributions* $\sim e^{-(2k+1)}$ *from shells* $k = 1, 2, \ldots$ *. The two asymptotic limits saturate bit-exactly:* $S(s \to 0) = \log(\dim W)$ *(hot, maximally mixed) and* $S(s \to \infty) = \log(n_{\text{equator}})$ *(cold, lowest shell only). The BW canonical* $s = 1$ *sits between, closer to the cold limit.*

The slope ≈ 2 *equals* $\dim(\partial W) = \dim S^2_{\text{equator}}$ *(the dimension of the wedge boundary manifold), giving a Cardy–Calabrese-style log entropy along the* $d = 2$ *boundary. This is structurally weaker than the QFT Bekenstein–Hawking area-law* $S \sim A \cdot \Lambda^2/4$*, which would scale as a power of the UV cutoff times the boundary dimension.*

Structural reading. *The operator-system truncation that closes the four-witness Wick-rotation theorem at machine precision (§ VIII.F, Sprint L1) does* not *preserve the continuum QFT vac-*

uum entanglement that would give the Bekenstein–Hawking area-law. The modular-flow period closure (M1 Hopf-base measure mechanism) and the Bekenstein–Hawking area-law are structurally distinct features of the gravitational thermodynamics chain; the GeoVac framework captures the former at finite cutoff and not the latter. Consistent with the structural-skeleton-scope pattern (CLAUDE.md §2, sprints H1, LS-8a, HF-3/4/5, multi-focal-composition wall, W3): the framework determines the algebraic skeleton of modular flow but does not autonomously generate the calibration data (here: Newton's constant) that links wedge-entropy to physical area.

Cross-references: Paper 34 §V.B Bekenstein–Hawking probe row; Sprint TD Track 4 (Hawking temperature, machinery-witness, M1 closure); the May-16 BH scoping memo (`debug/bekenstein_hawking_sprint` full Connes–Chamseddine cigar route) remains the longer-route alternative if a literal $S_{BH} = A/(4G)$ closure is pursued, but with the honest scope statement that Newton's constant is calibration input not derivation.

8.1 The unified $U(1) \times SU(2) \times SU(3)$ gauge content and its limits

The case-exhaustion theorem of Sec. 8 identifies M1 (Hopf-base measure) as the mechanism by which the Wilson plaquette-Haar projection injects π into spectral-triple observables. This subsection records a complementary structural fact about the family of GeoVac sub-triples that engage M1: they realise all three Standard-Model gauge groups as Wilson lattice constructions, on three structurally distinct sub-manifolds, with a single unified weak-coupling coefficient $1/(4\,N_c)$. The construction is honest about what it is and is not: a unified *catalogue* of three Wilson-Yang–Mills instances of the Marcolli–van Suijlekom gauge-network framework, not a unified spectral triple in the Connes–Chamseddine sense [4].

Three Wilson constructions, one kinetic coefficient

Three independent sprints have produced gauge-invariant, finite, Haar-normalisable non-abelian Wilson lattice gauge theories on GeoVac sub-graphs:

- *Paper 25* [34] constructs the abelian $U(1)$ Wilson–Hodge structure on the Fock-projected S^3 Coulomb graph (Paper 7), with phases χ_v on nodes and A_e on edges covariant under the standard incidence operator.
- *Paper 30* [36] constructs the non-abelian SU(2) Wilson lattice gauge theory on the same $S^3 = \mathrm{SU}(2)$ graph, with link variables $U_e \in \mathrm{SU}(2)$, plaquettes from primitive non-backtracking walks, and Haar measure dU. Paper 30 §5.2 verifies the weak-coupling kinetic coefficient symbolically and numerically.
- *Sprint ST-SU3* (May 2026, modules `geovac/su3_wilson_s5.py`, `tests/test_su3_wilson_s5.py`, 39/39 pass) constructs the non-abelian SU(3) Wilson lattice gauge theory on the Bargmann–Segal S^5 graph of Paper 24 [33], with link variables $U_e \in \mathrm{SU}(3)$ in the fundamental representation (independent of the shell index N), plaquettes from primitive non-backtracking walks, and SU(3) Haar measure. Sprint ST-SU3 §4 verifies Theorem 2 (kinetic coefficient $1/12$) symbolically and numerically at $N_{\max} = 2, 3$ (7 765 plaquettes), with maximal Cartan reduction to a coupled $U(1) \times U(1)$ theory verified at machine precision.

The three constructions are summarised in Table 5. The columns are not independent discoveries: each is a specialisation of the gauge-networks framework of Marcolli–van Suijlekom [5], with the Perez-Sanchez correction [10] that the continuum limit of such gauge networks is Wilson Yang–Mills *without* a Higgs sector.

Construction	G	**Host manifold**	**Vertex Hilbert space** $\mathcal{H}_v$	$\frac{1}{4N_c}$
Paper 25	$U(1)$	S^3 Hopf base	$\mathbb{C}$ (scalar)	$\frac{1}{4}$
Paper 30	SU(2)	$S^3 = \mathrm{SU}(2)$	$\mathbb{C}^2$ (Dirac)	$\frac{1}{8}$
ST-SU3	SU(3)	S^5 Bargmann–Segal	$\bigoplus_N \mathbb{C}^{\dim(N,0)}$	$\frac{1}{12}$

Table 5: The three Wilson lattice gauge constructions as instances of the Marcolli–van Suijlekom gauge-network framework on GeoVac sub-manifolds. The kinetic coefficient $\frac{1}{4N_c}$ in the rightmost column is the weak-coupling expansion coefficient of the Wilson action $S_W = \beta \sum_P \big(1 - \frac{1}{N_c} \mathrm{Re}\,\mathrm{tr}\, U_P\big)$ around the trivial vacuum, in the standard normalisation $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. The three rows verify this coefficient independently: Paper 30 §5.2 and Sprint ST-SU3 §4 to machine precision; the abelian Paper 25 case is the trivial $N_c = 1$ limit of either non-abelian construction.

In each construction the edge Laplacian $L_1 = B^\top B$, where B is the signed node–edge incidence matrix, plays the role of weak-coupling kinetic operator on $\mathfrak{g}$-valued 1-cochains. Paper 25's "discrete Hodge–1 Laplacian" is therefore a single combinatorial object across all three constructions; only its target Lie algebra changes ($\mathfrak{u}(1)$, $\mathfrak{su}(2)$, $\mathfrak{su}(3)$). This is the single mathematical identity unifying the three sectors at the kinetic level.

The two-step SU(3) story

The SU(3) row of Table 5 requires a specific clarification, because an earlier Sprint 5 Track S5 (April 2026, memo `debug/s5_gauge_structure_memo.md`) returned a clean negative when asking whether the S^5 Bargmann–Segal lattice carries a non-abelian gauge structure analogous to Paper 25's $U(1)$. The two findings are not in tension: they answer two different questions, and read together they give a two-step honest picture.

- *Step 1 (Sprint 5 Track S5, negative).* An adjacency-preserving SU(3) action on the $(N, 0)$ tower of the Bargmann graph fails Wilson's fixed-group-on-every-link requirement, because the transitions $(N, 0) \to (N + 1, 0)$ are Clebsch–Gordan intertwiners between distinct SU(3) irreducible representations of growing dimension $\dim(N, 0) = (N + 1)(N + 2)/2$, not group elements of a single SU(3). The natural shell-respecting gauge group of the Bargmann graph is $U(1)$, and this is recorded in Paper 25 §VII.A as a structural asymmetry between the S^3 and S^5 sectors.
- *Step 2 (Sprint ST-SU3, positive at the gauge level).* A *shell-independent* Wilson lattice gauge theory with link variables $U_e \in \mathrm{SU}(3)$ in the fundamental representation (not in any of the $(N, 0)$ irreps) is a structurally different object that bypasses the Step 1 obstruction. The links live as external gauge degrees of freedom on edges; they do not act on shell labels. Sprint ST-SU3 verified gauge-invariance of the Wilson action under node-local $U_e \mapsto g_{\mathrm{src}(e)} U_e g^\dagger_{\mathrm{tgt}(e)}$ to machine precision and computed the kinetic coefficient $1/12$ both symbolically and numerically.
- *Matter coupling (Sprint ST-SU3 Theorem 3, mixed).* When one tries to couple this Wilson SU(3) to the natural shell matter on the $(N, 0)$ tower, the original Step 1 obstruction is rediscovered: a fundamental-representation link cannot uniformly couple to representations of growing dimension, and the matter coupling reverts to the Clebsch–Gordan intertwiner structure. Wilson SU(3) on the Bargmann graph is therefore self-consistent as a *pure-gauge* theory but does not admit a natural matter coupling in the sense of Paper 30's spin-1/2 Dirac fermions on the Coulomb graph.

The two-step picture is consistent with Paper 24's Coulomb/HO asymmetry and sharpens it from three layers to four: to the prior asymmetries (spectrum-computing role of L_0; calibration π; Wilson gauge-with-natural-matter) one now adds a fourth layer at which the S^3 Coulomb side is special: the gauge group acts in the same representation as the matter, and matter–gauge coupling is structurally available. On the S^5 Bargmann side the combinatorial Wilson–Hodge vocabulary is universal but the gauge–matter coupling is not.

Marcolli–vS lineage of the unified content

Each construction in Table 5 is a specific gauge-networks instance in the sense of Marcolli and van Suijlekom [5], with the Perez-Sanchez correction [10] that the continuum limit of such a gauge network is Wilson Yang–Mills *without* a Higgs. The vertex algebra is $\mathcal{A}_v = \mathbb{C}$ (or $C(V)$ globally) in all three rows; the vertex Hilbert space and the gauge group differ row-by-row, as listed in the table. In each row the gauge-network spectral action reduces to the standard Wilson action plus a quadratic kinetic term in the weak-coupling expansion, with coefficient $1/(4N_c)$.

The lineage placement is identical to that of Sec. 9, and each Wilson construction inherits the same status: each is a published-machinery instance with a specific manifold and gauge group; what is not duplicated in the literature is the *simultaneous* existence of all three SM gauge groups as siblings across GeoVac sub-manifolds.

What this is not: structural distance to a Standard Model

Table 5 is a catalogue, not a unified theory, and the rhetoric rule of CLAUDE.md §1.5 forbids treating it as the latter. The structural distance from this catalogue to Connes' SM construction lives in four named gaps.

- **G1** (Three generations of matter.) Each construction in Table 5 carries a single matter species: Paper 30 has one generation of Dirac fermions (Paper 14 Tier 2 [29]); Sprint ST-SU3 has one tower of $(N, 0)$ irreps. Sprint 4H Track SM-B (April 2026) tested the naive shell-to-generation map $\sum_f N_c Q_f^2 = 8 = |\lambda_3|$ and returned a clean negative: the per-generation contribution is charge-universal across the three SM generations whereas every per-shell S^3 invariant varies with n, so no shell→generation map is consistent with charge-universal per-generation structure. The $8 = 8$ equality was a numerical coincidence with no representation-theoretic content. There is no GeoVac mechanism currently selecting a generation tripling of the Hilbert space.
- **G2** (Higgs as inner fluctuation of D.) In Connes' SM, the Higgs field arises as the off-diagonal block of an inner fluctuation $D \mapsto D + \omega + \epsilon' J\omega J^{-1}$ ($\omega = \sum_i a_i[D, b_i]$) for $a_i, b_i \in \mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ in an almost-commutative algebra. Per Perez-Sanchez 2024, the Marcolli–vS gauge-network spectral action gives YM *without* a Higgs sector by default: gauge networks supply only edge-Dirac hopping data with no internal fiber Dirac, so the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ Yukawa block that turns inner fluctuations into a Higgs is absent. The natural extension for GeoVac is $\mathcal{A}_{\text{GV}} \otimes \mathcal{A}_F$ with the electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ (deferring color), and with a finite Dirac D_F on the four-dimensional electroweak fiber whose off-diagonal block encodes the Yukawa coupling matrix. Inner fluctuations on this extension formally split into a gauge-1-form sector (which must reproduce the unified $U(1) \times \mathrm{SU}(2)$ content of Papers 25–30) and a candidate Higgs scalar sector living in the off-diagonal $(\mathbb{C}, \mathbb{H})$ component. Whether the required D_F can be selected from GeoVac structure rather than imposed is the load-bearing open question; we report the result of this test (Sprint H1) below in Sec. 8.2. Note that the

operator-system non-multiplicative-closure of Remark 1 (witness pair $M^2_{2,1,0}$, 14.9% residual against $\mathcal{O}_{n_{\max}=2}$) is a separate, finite-resolution feature of the truncated operator system itself; inner fluctuations are defined operationally as $\omega + J\omega J^{-1}$ with $\omega = \sum_i a_i[D, b_i]$, which uses ALGEBRA elements (the multiplier matrices, which are an algebra over $\mathbb{C}$ at the level of $M_{\dim H}(\mathbb{C})$ even though their span $\mathcal{O}_{n_{\max}}$ is not closed under multiplication), so the $a \cdot b \notin \mathcal{O}$ feature does not block the fluctuation calculus. Sprint H1 (Sec. 8.2) closes this gap up to the Yukawa-undetermined verdict; Sprint G3 (Sec. 8.3) supplies the structural reason that Y is undetermined — γ_{GV} and γ_F are independent commuting $\mathbb{Z}_2$ gradings, the Yukawa lives on the γ_F flip, and no GeoVac-side data couples to that flip. G2 and G3 are therefore one structural fact, not two.

G3 (Hypercharge / electroweak unification.) The SM relation $Q = T_3 + Y_W/2$ requires co-located $U(1)_Y$ and $\mathrm{SU}(2)_L$ on a common Hilbert space with a left–right chiral asymmetry. Paper 25's $U(1)$ and Paper 30's SU(2) are co-located on the same S^3 graph (this is the most concrete near-term unification target, see below), but the Camporesi–Higuchi Dirac (Sec. 3 Definition 3) is non-chiral by default. The chirality grading introduced in Sprint TS Track E2 (R3.5, full Dirac with both chiralities) is a $\mathbb{Z}_2$-grading of the spinor bundle, not the SM weak-isospin chirality, and it does not distinguish left- and right-handed sectors of a charged matter field. Sprint G3 (Sec. 8.3) closed this in the negative on S^3 alone: γ_{GV} and γ_F are independent commuting $\mathbb{Z}_2$ gradings on $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$, with operator-norm residual exactly 2 at every $n_{\max} \in \{1, 2, 3\}$. Identification is structurally impossible by tensor-product factorisation, and the electroweak co-location route is therefore pushed to G4.

G4 (Cross-manifold obstruction, the dominant gap.) Connes' SM uses a *single* spectral triple $(\mathcal{A}_{\mathrm{SM}}, \mathcal{H}_{\mathrm{SM}}, D_{\mathrm{SM}})$ on a single base manifold, with all three SM gauge groups emerging from inner automorphisms of one almost-commutative algebra. The GeoVac construction has, at present, three separate spectral triples on two different graphs (the S^3 Coulomb graph for $U(1)$ and SU(2); the S^5 Bargmann graph for SU(3)). There is no GeoVac operator that geometrically unifies the S^3 and S^5 graphs into a single spectral triple containing both as commuting factors: at best the Bargmann and Coulomb graphs can be encoded as orthogonal factors of a tensor-product Hilbert space, but the resulting object is a direct sum of two Wilson-YM instances rather than a unification. This is the most consequential gap. It is also the most directly tied to the four-layer Coulomb/HO asymmetry of Sec. 7: the absence of cross-manifold unification is a structural symptom of the same asymmetry that forbids a calibration-π analog and a spectrum-computing L_0 on S^5. Sprint G4 scoping (Sec. 8.4) sharpens this gap into G4a (Connes SM on $\mathcal{T}_{S^3}$ alone with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, bounded sprint sequence, predicted positive-thin) and G4b (genuine cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}$, blocked at the NCG-framework level by the Riemannian-vs-Hardy-sector asymmetry). After G3 NEGATIVE, G4a is the only sprint-scale electroweak-unification path remaining.

A footnote on the GUT-embedding direction: Sprint 4H Track SM-C (April 2026) tested whether $1/40 = \Delta$ in Paper 2's $K = \pi(B + F - \Delta)$ formula appears naturally in SU(5), E_8, or Spin(10) as a Chern number, η-invariant, or Chern–Simons level, and returned a clean negative: the dual-Coxeter rewrite $1/40 = (3/8)/15 = \sin^2 \theta_W^{\mathrm{GUT}} / \dim(\bar{\mathbf{5}} \oplus \mathbf{10})$ is tautological, recycling the integers 5 and 8 already present in Paper 2. The naive shell-to-generation embeddings $\mathrm{shell}_n \to \bar{\mathbf{5}}$ fail at the $\mathrm{SU}(3) \times \mathrm{SU}(2)$ branching level. GeoVac therefore has no current contact with the GUT-embedding lineage of the SM, beyond the tautological numerical coincidences.

The most concrete near-term unification target

Of the four gaps, G4 (cross-manifold) is the most consequential and G3 (electroweak unification) is the most reachable. Paper 25's $U(1)$ and Paper 30's SU(2) already live on the same underlying graph (the Fock-projected S^3 Coulomb graph) and share the same Camporesi–Higuchi Dirac operator, the same incidence structure B, and the same edge Laplacian $L_1 = B^\top B$. The two constructions differ only at the gauge-group level, and Paper 30 §5.1 (Result 1) establishes that Paper 25 is exactly the maximal-torus reduction of Paper 30. The natural unified construction is therefore an $\mathrm{SU}(2)\times U(1)$ Wilson lattice gauge theory on the same graph, with link variables $U_e \in \mathrm{SU}(2)\times U(1)$ and a chosen embedding of hypercharge as the $U(1)$ phase. Whether such a construction admits a Higgs-style inner fluctuation through an almost-commutative extension (closing G2 within the electroweak sector) is an open structural question; whether the Camporesi–Higuchi Dirac admits a chirality grading compatible with weak-isospin (closing G3) is the second open structural question. This is a clean S^3-only target that does not require any progress on the S^5 side and that would, if successful, take GeoVac from "three Wilson-YM instances on three sub-manifolds" to "unified electroweak Wilson construction on S^3, plus a separate SU(3) instance on S^5."

We do not claim such a unification has been achieved, or even that its component pieces have been verified. The status of this subsection is: a recorded structural fact (the unified $1/(4N_c)$ coefficient and the lineage placement are real); a recorded honest gap analysis (G1–G4 are real); and a recorded open direction (S^3 electroweak co-location is the most concrete target). Per CLAUDE.md §1.5, no claim is made that GeoVac contains the Standard Model; the claim is only that it contains the three SM gauge groups as separate Wilson lattice gauge theories on its own sub-manifolds, with a unified weak-coupling kinetic coefficient and a unified gauge-network lineage.

8.2 Sprint H1: Higgs as inner fluctuation on the GeoVac electroweak almost-commutative extension

The G2 gap of Sec. 8.1 (Higgs as inner fluctuation) admits a constructive sprint-scale test: build the minimal electroweak almost-commutative extension $\mathcal{T}_{\mathrm{AC}} = \mathcal{T}_{\mathrm{GV}} \otimes \mathcal{T}_F$ with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ on the truthful Camporesi–Higuchi GeoVac triple, compute the inner fluctuation $D \mapsto D + \omega + \epsilon' J\omega J^{-1}$ explicitly at finite $n_{\max}$, and ask whether the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ Yukawa block of the finite Dirac D_F that turns Marcolli–vS-without-Higgs into a Higgs construction can be *selected from GeoVac structure* rather than imposed. This is the H1 sprint reported in `debug/h1_ac_extension_memo.md`; we summarize its content here.

Construction. Following Connes–Marcolli 2008 Ch. 13 [2], the finite electroweak Hilbert space is doubled into matter and antimatter sectors: $\mathcal{H}_F = \mathcal{H}_F^{\mathrm{matter}} \oplus \mathcal{H}_F^{\mathrm{antimatter}} = \mathbb{C}^4 \oplus \mathbb{C}^4$, with $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ acting on the matter sector only (the opposite-algebra action on antimatter is realized via J). The 8 by 8 finite Dirac is $D_F = \mathrm{block\_diag}(M, \bar{M})$ where the matter-sector M is the 4 by 4 Yukawa-Dirac with off-diagonal $L \leftrightarrow R$ block $Y = \mathrm{diag}(y_\nu, y_e)$. The combined Dirac is $D = D_{\mathrm{GV}} \otimes \mathbb{1}_F + \gamma_{\mathrm{GV}} \otimes D_F$ with $\gamma_{\mathrm{GV}} = \mathrm{sign}(D_{\mathrm{GV}})$ (Track 2 closure): this commutes with D_{GV} on the truthful CH Dirac. The combined real structure is $J = J_{\mathrm{GV}} \otimes J_F$ with J_F swapping matter and antimatter via complex conjugation; matter-antimatter doubling makes $J_F^2 = +\mathbb{1}$ (KO-dim 6 of $\mathcal{A}_F$), which combined with KO-dim 3 of the GV factor gives KO-dim $3 + 6 = 9 \equiv 1 \pmod 8$ for the combined triple, hence $J^2 = -\mathbb{1}$ and $JD = +DJ$. Both are verified to machine precision at $n_{\max} \le 3$ (`tests/test_almost_commutative.py`, 38 tests passing).

Inner fluctuations split cleanly into gauge and Higgs sectors. For $\omega = \sum_i a_i[D, b_i]$ with $a_i, b_i \in \mathcal{A}_{\rm GV} \otimes \mathcal{A}_F$, we have

$$\omega = \underbrace{\sum_i a_i^{\rm GV}[D_{\rm GV}, b_i^{\rm GV}] \otimes a_i^F b_i^F}_{\text{gauge piece}} + \underbrace{\sum_i a_i^{\rm GV} b_i^{\rm GV} \gamma_{\rm GV} \otimes a_i^F [D_F, b_i^F]}_{\text{Higgs piece}}. \tag{4}$$

The gauge piece reproduces Papers 25 ($U(1)$ from the $\mathbb{C}$-summand) and 30 (SU(2) from the $\mathbb{H}$-summand) at the operator level. The Higgs piece is non-zero exactly when $[D_F, b_i^F]$ has off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ matrix elements — which happens *iff* the imposed Yukawa Y is non-zero. At $n_{\rm max} = 1, 2, 3$, with three test choices of Y:

D_F choice	$\lVert\Phi\rVert_{\rm max}$	$\lVert\omega_{\rm gauge}\rVert_{\rm max}$	**Falsifier**
$Y = 0$	0.000	0.000–0.580	**HOLDS**
$y_e = 0.1$, $y_\nu = 0$	0.046–0.080	0.000–0.580	FAILS
$y_e = 0.3$, $y_\nu = 0.2$	0.171–0.271	0.000–0.580	FAILS

(50 random generators per cell; data `debug/data/h1_falsifier.json`).

Verdict: positive-thin. The Higgs construction is *well-defined* on the GeoVac AC extension at finite $n_{\rm max}$: the algebra, Hilbert space, Dirac, and real structure satisfy the relevant Connes axioms, and inner fluctuations decompose into a gauge sector recovering Papers 25/30 plus a Higgs sector Φ that is non-trivial whenever the imposed Yukawa Y is non-zero.

However, GeoVac structure does **not autonomously select** a non-trivial Y. Two candidate sources for D_F off-diagonal data were flagged in the H1 scoping memo (`debug/almost_commutative_scoping_memo.` candidate A used the offdiag CH chirality bridging variant of the GeoVac Dirac, but Track 2 (`debug/real_structure_finite_nmax_memo.md` §3.D) ruled this out by showing $JD = +DJ$ fails at residual ~ 2.0 on offdiag CH; candidate B was a Sturmian / DUCC inter-shell coupling whose physical interpretation as a Yukawa is itself speculative and was not pursued.

The strong falsifier of the scoping memo §5 ("every Hermitian D_F derivable from GeoVac structure forces the Higgs sector to vanish") therefore does not fire as a clean negative. The construction admits a Higgs given any Y; the question is which Y, and GeoVac data is silent. This is exactly the positive-thin reading of the scoping memo §0: GeoVac sits on the *Marcolli–vS-without-Higgs* side of the 2014/2024 distinction at the structural level — not because inner fluctuations cannot define a Higgs (they can), but because no GeoVac mechanism selects the Yukawa.

What this means for G2. Gap G2 of Sec. 8.1 is sharpened to the more precise statement: *the Higgs construction goes through, but the Yukawa is a free input*, not closed to either "GeoVac contains a Higgs" (which would require a structural selection of Y) or "GeoVac excludes a Higgs" (which would require a structural obstruction to non-zero Y). This places G2 in the same status as G1 (three generations are formally definable but not selected by GeoVac structure): definable but not derivable.

What this means for G3. Gap G3 (hypercharge / electroweak unification, requiring identification of the GV chirality grading $\gamma_{\rm GV}$ with the SM weak-isospin chirality) is *not* closed by H1. In our construction $\gamma_{\rm GV}$ is the sign of $D_{\rm GV}$ on the spinor bundle, while the SM weak-isospin chirality is a separate $\mathbb{Z}_2$-grading on the matter sector of $\mathcal{H}_F$. The two are added independently; no GeoVac mechanism currently identifies them.

What this means for the operator-system non-multiplicative-closure caveat. Remark 1's observation that $\mathcal{O}_{n_{\max}}$ is not closed under multiplication (witness pair $M^2_{2,1,0}$, 14.9% residual at $n_{\max} = 2$) is a separate, finite-resolution feature of the operator system itself. Inner fluctuations are defined operationally as $\omega + J\omega J^{-1}$ with $\omega = \sum_i a_i[D, b_i]$, where the multiplier matrices a_i, b_i are individual elements of $M_{\dim H}(\mathbb{C})$ (which is an algebra over $\mathbb{C}$); the lack of closure of *their span* $\mathcal{O}_{n_{\max}}$ under multiplication does not block the fluctuation calculus. Order-zero $[a, JbJ^{-1}] = 0$ holds automatically by the matter–antimatter doubling (a acts on matter, JbJ^{-1} on antimatter, sectors are disjoint).

Scope and limitations. H1 establishes the constructive existence of the AC extension at finite $n_{\max}$ on the truthful CH GeoVac triple. H1 does *not* establish a Higgs vev as a derived quantity (that would require a continuum spectral-action computation; deferred to Track 1 R2.5 L4 closure for the rigorous interpretation), does *not* include the SM color factor $M_3(\mathbb{C})$ (deferred per Sec. 8.1 G4), and does *not* identify the GV and weak-isospin chiralities (G3 stays open). Per CLAUDE.md §1.5, no claim is made that GeoVac contains the Higgs: the claim is that it admits a Higgs given an imposed Yukawa, and that the Yukawa is not derived from GeoVac structure.

Files. Implementation `geovac/almost_commutative.py`; tests `tests/test_almost_commutative.py` (38 passing); data `debug/data/h1_falsifier.json`; memo `debug/h1_ac_extension_memo.md`.

Thermal extension of the H1 verdict (Sprint TD Track 3, May 2026). The same Yukawa-undetermined structural reading extends to the thermal Higgs effective potential $V_{\text{eff}}(\phi, T)$ on the combined triple $T_{\text{GV}} \otimes T_{S^1_\beta} \otimes T_F$. At one-loop, the thermal mass squared takes the form

$$m_T^2(T) \;=\; \frac{T^2}{12} \cdot \operatorname{Tr}_{\text{inner}}\big(Y^2_{\text{diag}}\big) \;+\; (\text{gauge contribution}),$$

where the outer Stefan–Boltzmann backbone $T^2/12$ comes from Sprint TD Track 1's $M_1 \times M_2$ factorisation (Paper 35, tensor-product verification section), and the inner $\operatorname{Tr}_{\text{inner}}(Y^2_{\text{diag}})$ is a $k = 2$ Dirichlet sum in the Yukawa eigenvalues (Paper 18 § IV.6; η-trivialization-theorem-style filtering applies). Any candidate T_F whose squared Yukawa diagonal reproduces the empirical SM spectrum reproduces the SM-data-fit value $m_T^2(v, T_C) \approx (160 \text{ GeV})^2$; GeoVac admits any compatible spectrum but selects none.

Empirical falsification of the strict $Y = 0$ reading. The strict Marcolli–vS-without-Higgs case ($Y = 0$) predicts $\operatorname{Tr}_{\text{inner}}(Y^2_{\text{diag}}) = 0$ and hence no high-T symmetry-restoring quadratic correction to $V_{\text{eff}}(\phi, T)$. This is empirically falsified by the electroweak phase transition / sphaleron crossover at $T_C \approx 160$ GeV inferred from baryogenesis phenomenology. The empirical universe forces $Y_F > 0$. This does not derive Y_F, but it closes the H1 verdict's residual ambiguity: at finite T, the framework plus observed sphaleron phenomenology rules out the strict $Y = 0$ slice. The "thin" part of the positive-thin verdict is now thinner — five thermal observables (Sprint TD Track 3 catalogue) are individual probes of distinct inner-factor structures (Yukawas, color algebra, generation count, charge assignment, Higgs sector), and the inner factor is required to deliver a specific value at each. Three further candidates are *forbidden* by the η-trivialization theorem (Paper 18 § IV.6) on any Krajewski-class even spectral triple — a positive structural prediction: axial chemical potential thermodynamics, parity-odd transport, and finite-T EDMs vanish identically on the inner factor of any Connes–Chamseddine-compatible AC extension.

Modular propinquity reformulation of the H1 falsifier (Sub-sprint M-H1, May 2026). The dual modular propinquity construction [57] provides a natural test of whether the Higgs sub-bimodule structure imposes a constraint on the Yukawa Y beyond what the Connes axioms already enforce. The bimodule reformulation of the H1 verdict (Sub-sprint M-H1, 2026-05-23) closes *negatively* on this question: the dual modular duality requirement reduces to Hermiticity of D_F plus matter–antimatter doubling, both already imposed by Connes' axiom system; the Higgs cross-block Φ lives entirely in the fiber sub-bimodule $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$, structurally decoupled from any GeoVac-side bimodule data; and the Morita-triviality constraint ("could $\Phi = u^* v$ for GeoVac-derived u, v?") does not survive matter–antimatter doubling. The H1 POSITIVE-THIN verdict survives unchanged. The "dual" in dual modular propinquity is the bridge $\leftrightarrow$ tunnel construction duality [57], NOT Morita equivalence (a common misreading); the Morita route was structurally barred independently. *Net:* the framework gains substantive new content on the chemistry / bound-state QED side (Sub-sprints M-X, M-Y, M-Z) but not on the SM-unification side (M-H1). Source: `debug/sprint_modular_propinquity_mH1_higgs_memo.md`, `debug/sprint_modular_propinquity_synthesis_memo.md`.

Sprint M-Z partition tested on a second concrete case (W1c chemistry wall, Sprint β-neg 2026-05-23). The Sub-sprint M-Z cross-shift / endomorphism partition was tested on a second concrete case beyond the bound-state QED context where it was originally derived: the W1c-residual chemistry wall at NaH alkali hydrides (Paper 19, Sec. `sec:w1c_residual`). The bridge sprint (`debug/sprint_w1c_mz_partition_analysis_memo.md`, 2026-05-23) proposed a *three-bucket refinement* — cross-shift (handled) / basis-closable cross-shift (handled at sufficient basis size) / endomorphism (external input) — and tested it via three prespecified falsifiable predictions for NaH max_n=3 (`debug/data/sprint_w1c_mz_partition_predictions.json`). The F1 max_n=3 sprint (`debug/sprint_f1_maxn3_predictions_test_memo.md`) returned **CLEAN NEGATIVE**: 1 of 3 predictions pass; P1 (internal PES minimum) fails on the exact falsification criterion, the proposed basis-closable cross-shift sub-class evaporates. *The two-bucket M-Z partition (cross-shift / endomorphism) stands as the canonical framing.*

The substantive structural finding the test produced is a sharper localization of chemistry-side endomorphisms: at max_n=3 under combined W1c + multi-zeta architecture, the framework DOES construct the right bonding orbital (dominant natural orbital $-0.698 \cdot \phi_{\text{Na, 3s}} - 0.687 \cdot \phi_{\text{H, 1s}}$ with 50/50 amplitude split), but the constructed bonding combination has higher energy than the separated configuration at every R. The framework can BUILD the right orbital but cannot energetically PREFER it. Sub-layer 3 of the W1c hierarchy reclassifies from "basis-closable cross-shift" to "module endomorphism (basis-irreducible at tested scales)" and joins sub-layer 2 in the endomorphism bucket.

Same-day same-arc follow-on sprints (F2 + F3) sharpen the structural reading further into an architectural-extension ladder. The F1 max_n=3 verdict "module endomorphism, basis-irreducible at tested scales" was further refined by two same-day same-arc follow-on sprints (F2 + F3) executing the diagnostic-before-engineering rule at the sprint-cycle level. **F2** (cross-V_{ne} kernel-shape substitution diagnostic, KERNEL-NOT-IT verdict): the multipole expansion at default $L_{\max} = 4$ is bit-faithful to converged 3D numerical quadrature within $\sim 2 \times 10^{-5}$ Ha at NaH $R_{\text{eq}} = 3.566$ bohr, six to ten orders of magnitude below wall depth. Ruled out kernel-shape substitution as a closure mechanism. Substantive new content: the framework's h1 in `composed_qubit.build_composed_hamiltonian` is strictly block-diagonal — the cross-block off-diagonal h1 slot is *architecturally absent*, not just kernel-error-flawed. New sub-wall name **W1d** (cross-block h1 architectural extension), structurally distinct from a missing calibration input.

F3 (cross-block h1 architectural extension): implemented the missing matrix slot via new production module `geovac/cross_block_h1.py`; FCI naturals at NaH max_n=2 R=3.5 transform $[1.0000, 1.0000] \rightarrow [\mathbf{1.9991}, \mathbf{0.0007}]$ — four-orders-of-magnitude advance in dominant natural-orbital occupation signature, with the bonding combination as a doubly-occupied FCI ground state. **W1d CLOSED at the FCI level.** But F3 mini-PES remains monotonically descending with $D_e^{\mathrm{F3}} = +4.37$ Ha $= 58\times$ experimental: cross-block h1 enables bonding without supplying opposing Pauli repulsion against the [Ne] frozen core. **New sub-layer W1e (inner-region overattraction)**, F4 target via a bonding-orbital Phillips-Kleinman extension.

Net refined reading of the chemistry sub-walls under M-Z: chemistry sub-walls split into THREE structurally distinct sub-categories within the two-bucket M-Z partition: cross-shifts (handled by existing architecture), module endomorphisms (split chemistry-side mitigable vs QED-side strictly-external), and **architectural-absence** (missing matrix slots in the existing architecture, conditionally closable via architectural extension). The architectural-absence category is a structurally distinct refinement WITHIN the two-bucket partition, not a contradiction of it. The architectural-extension ladder (W1c-cross-screening $\rightarrow$ W1c-multi-zeta-basis $\rightarrow$ W1d-cross-block-h1 $\rightarrow$ W1e?) is the chemistry-side analog of refinements that the QED side does not natively support: QED endomorphisms (LS-8a Z_2-1, δm; Sprint H1 Yukawa) are strictly externally specified via UV-completion physics, while chemistry endomorphisms are intrinsically external but framework-internally computable (FrozenCore + atomic-physics fits) AND admit architectural extensions (cross-block h1, bonding-orbital PK). This explains why the W1c arc produced a five-sub-layer hierarchy (W1c, W1c-multi-zeta, W1d, W1e, plus the FALSIFIED three-bucket M-Z refinement candidate) while the QED-side LS-8a wall has remained at a single-counterterm-input characterization.

Sources: `debug/sprint_beta_neg_synthesis_memo.md` (Sprint β-neg synthesis, F1-maturity; superseded for framing by the full-arc memo); `debug/sprint_f2_cross_vne_kernel_memo.md` (F2 KERNEL-NOT-IT + architectural-absence finding, W1d named); `debug/sprint_f3_cross_block_h1_memo.md` (F3 W1d CLOSED at FCI level + W1e newly named); `debug/sprint_w1c_full_arc_synthesis_memo.md` (comprehensive full-arc synthesis at F3 maturity, supersedes Sprint β-neg synthesis at the framing level). Cross-references Paper 19 §`sec:w1c_residual` (W1c-residual wall + F1-F2-F3 arc extension with revised five-sub-layer hierarchy), Paper 17 §6.10 (composed-architecture cross-reference, updated to F3 maturity), Paper 34 §`sec:conv_three_bucket_falsified` (running-catalogue entry, parent), and Paper 34 §`sec:conv_w1c_f2_f3` (running-catalogue entry, F2+F3 follow-on).

F4–F6 refinement of W1e characterization (β-update, 2026-05-23). Three diagnostic sprints in immediate succession (F4, F5, F6) further refined the W1e (inner-region overattraction) sub-layer characterization beyond F3's working hypothesis of "missing Pauli repulsion against the [Ne] frozen core," tested all three candidate closure mechanisms F3 §6 had named, and sharpened the chemistry-side sub-mechanism dissection of W1e in the M-Z partition. **F4** (bonding-orbital Phillips–Kleinman extension): RULED OUT at Step 1 diagnostic — rank-1 PK on the F3-constructed bonding orbital saturates at ~43% closure ceiling in the FCI sensitivity test, far below the 96% required for the $2\times$ experimental closure window. **W1e is therefore NOT a single-particle Pauli- orthogonality wall** (which would have been the cross-shift-class mechanism within M-Z); it is a multi-determinant FCI correlation effect. **F5** (explicit-core Hartree treatment of [Ne] electrons via cross-block $\sum_c (J-K)$): RULED OUT at Step 1 diagnostic — right SIGN (repulsive, +1.12 Ha) but only 25.7% of wall depth; predicted residual $D_e^{\mathrm{F5}} \approx +3.25$ Ha $=$ $43\times$ experimental. **W1e is NOT addressable by mean-field Hartree-class core-bonding interaction** (which would have been closable by extended-bimodule machinery of standard-Hartree class operators within M-Z). **F6** (valence basis enlargement to NaH max_n=4 with full F3 stack): PARTIAL_IMPROVEMENT — 10.2% PES-level closure (the 2-point gate suggests 26.1% but is artifactual; the inner-region collapse continues from $R = 3.5$ down to $R = 2.0$ with another 0.696 Ha descent the 2-pt gate misses). **W1e is NOT addressable by basis enlargement alone at max_n=4** (which would have been a basis-truncation cross-shift candidate within M-Z).

Net structural reading of the chemistry-side sub-mechanism dissection of W1e under M-Z: the chemistry-side dissection of W1e refines the framework's understanding of where "correlation" sits in the M-Z partition. W1e correlation is neither single-particle Pauli (which is the cross-shift-class mechanism within M-Z, ruled out by F4) nor mean-field Hartree-level (which would be closable by extended-bimodule machinery for standard-Hartree-class operators, ruled out by F5), but **genuine multi-determinant content that requires beyond-sprint-scale architectural lift** (Schmidt orthogonalization at the basis-construction level, DMRG-class beyond-FCI methods, or explicit correlation factors — all are beyond-sprint-scale architectural investments). W1e characterizes a **three-sub-sub-mechanism decomposition with structurally-independent closure ceilings**: basis-truncation (∼10% PES, F6 measured) + Hartree-pressure (∼25% 2-pt, F5 predicted) + deep-correlation-residual (∼65–90% of wall, untested by sprint-scale compute). The chemistry-side architectural-extension ladder (W1c-cross-screening → W1c-multi-zeta-basis → W1d-cross-block-h1) has plausibly hit a natural pause point at W1e: the next architectural extension candidates are beyond-sprint-scale commitments rather than sprint-scale diagnostic-then-implementation cycles, and the chemistry arc's forward direction shifts from "find the right closure mechanism" to "characterize the framework's structural-skeleton scope limits empirically for second-row hydride binding" — consistent with the broader CLAUDE.md §1.7 multi-focal-composition wall taxonomy finding (the framework's structural-skeleton scope does not autonomously deliver binding energetics at this precision).

Methodological side finding for future M-Z sub-mechanism sprints. The F6 2-point-vs-PES discrepancy reveals that 2-point gates at fixed reference geometry ($R = 3.5$ bohr vs $R = 10$ bohr) overestimate PES-derived closure by ∼2.5× on monotonically-descending PES in the W1c-residual regime. Future sprint-cycle diagnostics in this regime should use PES well depth at $R_{\min}$ as the load-bearing closure metric rather than 2-point energy differentials. The F3-reported $D_e^{\mathrm{F3}} = +4.37$ Ha is itself a 2-pt value; the corresponding F3 PES well depth at $R_{\min} = 2.0$ is likely larger (untested but expected by analogy to F6). This is a methodological refinement of the diagnostic-before-engineering discipline (CLAUDE.md §1.7) at the sprint-cycle level: not just "run a diagnostic before production wiring," but also "choose the diagnostic geometry that captures the load-bearing physical content rather than a convenient reference."

Sources (F4–F6 refinement): `debug/sprint_f4_bonding_pk_memo.md` (F4 STOP-at-Step-1, rank-1 PK insufficient); `debug/sprint_f5_explicit_core_memo.md` (F5 STOP-at-Step-1, mean-field Hartree insufficient); `debug/sprint_f6_maxn4_nah_memo.md` (F6 PARTIAL_IMPROVEMENT, basis enlargement to max_n=4); `debug/sprint_beta_update_f4f6_memo.md` (β-update F4-F6-APPENDED-TO-F3-BASELINE synthesis). Cross-references Paper 19 §`sec:w1e_refinement_f4_f6` (refined six-sub-layer hierarchy with three W1e sub-sub-mechanisms), Paper 17 §6.10 (composed-architecture cross-reference, updated to β-update F4–F6 maturity), Paper 34 §`sec:conv_w1e_f4_f6_refinement` (running-catalogue entry, F4–F6 follow-on to F3 entry).

W1e sprint-scale candidate space closed at Schmidt + core correlation Day-1 diagnostics (2026-05-23). The two beyond-sprint-scale candidates F4–F6 named (Schmidt orthogonalization at basis-construction level; fully correlated [Ne] cores) were tested by Day-1 diagnostics. **Both ruled out.** Schmidt-orthogonalizing H 1s against the Na [Ne] core at $R = 3.5$ bohr produces $\sum_c |S_c|^2 = 1.00 \times 10^{-3}$ (only 0.1% projection mass; the H atom at $R = 3.5$ is structurally outside the [Ne] core's significant amplitude) and a total diagonal-element differential $\Delta_{\mathrm{total}} = +8.6$ mHa — $\sim 12\times$ below the prespecified GO threshold and $\sim 500\times$ below the wall depth. [Ne] core-correlation differential at NaH R_{eq} converges across three independent literature anchors (Iron-Oren-Martin 2003, Sylvetsky-Martin 2018, atomic Na correlation × activefraction) to $\sim$ 0.001–0.003 Ha, $10^3\times$ below the wall depth. *Five independent sprint-scale W1e closure mechanisms are now ruled out* (F4, F5, F6, Schmidt, core correlation). The chemistry-side architectural-extension ladder (W1c-cross-screening $\to$ W1c-multi-zeta-basis $\to$ W1d-cross-block-h1) has reached the sprint-scale boundary; further closure candidates are substantial architectural commitments. The W1e wall is the **sixth structurally-independent instance of the multi-focal-composition wall pattern** across the framework, joining five physics observables (Sprint H1 Yukawa, LS-8a $Z_2/\delta m$, HF-3 recoil, HF-4 Zemach, HF-5 multi-loop a_e). *The pattern's cross-domain transferability — five physics + one chemistry, two structurally distinct calibration-data classes hitting the same structural-skeleton-scope boundary — is the substantive new content of this closeout, extending the structural-skeleton scope statement from physics-only to physics+chemistry. Sources:* `debug/sprint_w1e_schmidt_diagnostic_memo.md` (Schmidt Day-1 STOP), `debug/sprint_w1e_core_correlation_estimate_memo.md` (core correlation DEFER). Cross-references Paper 19 §`sec:w1e_schmidt_core_correlation` and Paper 34 §`sec:conv_w1e_cross_domain_wall`

8.3 Sprint G3: Electroweak chirality co-location on S^3

Sprint H1 (Sec. 8.2) closed G2 up to the Yukawa-undetermined verdict and left G3 (electroweak unification, requiring identification of γ_{GV} with the SM weak-isospin chirality on $\mathcal{H}_F$) as the next physics question. The G3 sprint (May 2026) is a constructive test: build γ_{GV} on the GeoVac side and γ_F on the AC side as separate $\mathbb{Z}_2$ gradings each satisfying the standard NCG axioms, lift both to the tensor-product Hilbert space $\mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$, and ask whether the residual

$$\Delta := \gamma_{\mathrm{GV}} \otimes \mathbb{1}_F - \mathbb{1}_{\mathrm{GV}} \otimes \gamma_F \tag{5}$$

vanishes (or vanishes on the physical sector under standard left/right convention swap). $\Delta = 0$ would close G3 in the positive; $\Delta \neq 0$ robustly would close G3 in the negative on S^3 alone and push electroweak unification onto G4 cross-manifold territory.

Constructions. The sprint executed three parallel tracks (memos `debug/g3a_chirality_memo.md`, `debug/g3b_chirality_F_audit_memo.md`, `debug/g3c_tensor_chirality_memo.md`). Track G3-A constructed $\gamma_{\mathrm{GV}} = \sigma_x \otimes \mathbb{1}_w$ on the R3.5 full Dirac operator system at $n_{\max} \in \{1, 2, 3\}$, with $w = \mathrm{spinor\_dim}(n_{\max})$ and the basis ordering $[\chi = +1, \chi = -1]$ from `full_dirac_basis`. All three $\mathbb{Z}_2$-grading axioms are bit-exact at every tested $n_{\max}$: $\gamma_{\mathrm{GV}}^2 = \mathbb{1}$, $\{\gamma_{\mathrm{GV}}, D_{\mathrm{truthful}}\} = 0$, and $\gamma_{\mathrm{GV}} M \gamma_{\mathrm{GV}} = M$ as a matrix equality for every scalar multiplier $M \in \mathcal{O}_{n_{\max}}$. The natural σ_z candidate (the D-eigenvalue sign) commutes with D_{truthful} rather than anticommuting and is exposed only as a diagnostic. Module `geovac/chirality_grading.py` (50/50 tests passing).

Track G3-B added `ElectroweakFiniteTriple.chirality_F()` to `geovac/almost_commutative.py` with the standard Connes–Marcolli SM convention at KO-dim 6 of $\mathcal{T}_F$:

$$\gamma_F^{\mathrm{matter}} = \mathrm{diag}(+1, +1, -1, -1), \qquad \gamma_F^{\mathrm{anti}} = -\gamma_F^{\mathrm{matter}}. \tag{6}$$

The antimatter sign-flip is forced by $\{J_F, \gamma_F\} = 0$: the naive "identical on both copies" convention would put $\mathcal{T}_F$ at KO-dim 0 and silently break the H1 module's $\{J_F, D_F\}$ axiom suite. All grading axioms ($\gamma_F^2 = \mathbb{1}$, $\{\gamma_F, D_F\} = 0$ at any imposed Yukawa, $\gamma_F \pi_F(a) \gamma_F = \pi_F(a)$, $\gamma_F = \gamma_F^*$, $\{J_F, \gamma_F\} = 0$) hold to bit-exact zero residual; 15 new tests in `TestChiralityF`, all passing. The combined KO-dimension of $\mathcal{T}_{\mathrm{AC}} = \mathcal{T}_{\mathrm{GV}} \otimes \mathcal{T}_F$ is therefore $3 + 6 = 9 \equiv 1 \pmod 8$, consistent with H1's combined-Dirac construction.

Tensor-product diagnostic. Track G3-C computed Δ at $n_{\max} \in \{1, 2, 3\}$ using the production $\gamma_{\rm GV}$ and γ_F from G3-A and G3-B. Sanity invariants are bit-exact: $\gamma_{\rm GV}^2 = \mathbb{1}$, $\gamma_F^2 = \mathbb{1}$, and $[\gamma_{\rm GV} \otimes \mathbb{1}_F, \mathbb{1}_{\rm GV} \otimes \gamma_F] = 0$ (the two operators commute because they act on disjoint tensor factors). The residual itself is the structurally cleanest possible non-zero object:

$n_{\max}$	$\dim \mathcal{H}_{\rm GV}$	$\dim \mathcal{H}_{\rm total}$	$\|\Delta\|_{\rm op}$	$\|\Delta\|_F$	spec(Δ)
1	4	32	2.000	8.000	$\{-2{:}8,\ 0{:}16,\ +2{:}8\}$
2	16	128	2.000	16.000	$\{-2{:}32,\ 0{:}64,\ +2{:}32\}$
3	40	320	2.000	25.298	$\{-2{:}80,\ 0{:}160,\ +2{:}80\}$

Table 6: Tensor-product chirality residual $\Delta = \gamma_{\rm GV} \otimes \mathbb{1}_F - \mathbb{1}_{\rm GV} \otimes \gamma_F$ at $n_{\max} \in \{1, 2, 3\}$, computed in `debug/g3c_tensor_chirality.py` from the production $\gamma_{\rm GV}$ (G3-A, $\sigma_x \otimes \mathbb{1}_w$) and γ_F (G3-B, KO-dim 6). Operator norm is exactly 2 at every $n_{\max}$; eigenvalue spectrum is $\{-2, 0, +2\}$ with multiplicities in ratio 1:2:1, $n_{\max}$-independent. Frobenius norm scales as $\sqrt{2 \cdot \dim \mathcal{H}_{\rm total}}$ exactly. Data `debug/data/g3c_tensor_chirality.json`.

Convention swaps do not absorb the residual. Three standard convention swaps were tested at every $n_{\max}$: (a) global sign-flip $\gamma_F \to -\gamma_F$ (left/right convention), (b) restriction to the matter sector of $\mathcal{H}_F$, and (c) restriction to the Weyl chirality of $\mathcal{H}_{\rm GV}$. Swap (a) leaves the baseline invariant. Swaps (b) and (c) reduce $\|\Delta\|_F$ by dropping basis vectors, but $\|\Delta\|_{\rm op}$ remains 2 (or 1 under swap (c) with σ_x-$\gamma_{\rm GV}$, where the Weyl restriction collapses $\gamma_{\rm GV}|_w$ to 0_w and $\Delta_c = -\mathbb{1}_w \otimes \gamma_F$, still non-zero). Combined swaps yield the same conclusion. No standard NCG sign convention absorbs Δ.

Robustness. A σ_x-vs-σ_z cross-check at $n_{\max} = 2$ confirms that the spectrum of Δ is identical under both $\gamma_{\rm GV}$ conventions (both produce $\{-2{:}32,\ 0{:}64,\ +2{:}32\}$, since σ_x and σ_z share eigenvalues ± 1 with identical multiplicities and both commute with γ_F). The structural verdict is therefore convention-independent.

Sanity check on $\gamma_5 := \gamma_{\rm GV} \otimes \gamma_F$. The product operator $\gamma_5 := \gamma_{\rm GV} \otimes \gamma_F$ is the canonical Connes–Chamseddine combined chirality on $\mathcal{T}_{\rm AC}$ and satisfies $\gamma_5^2 = \mathbb{1}$ to bit-exact precision at every $n_{\max}$, with spectrum $\{-1, +1\}$ in equal multiplicities. The product works; the *difference* does not. γ_5 exists and is well-defined as the combined grading; $\gamma_{\rm GV}$ and γ_F do not identify across the tensor product.

Verdict: NEGATIVE on S^3 alone. The two chirality gradings $\gamma_{\rm GV}$ (Camporesi–Higuchi Dirac sign on S^3) and γ_F (weak-isospin grading on the AC factor) are *independent commuting* $\mathbb{Z}_2$ gradings on $\mathcal{H}_{\rm GV} \otimes \mathcal{H}_F$. They cannot be identified by any standard NCG sign convention because, by construction, $\gamma_{\rm GV}$ acts as identity on $\mathcal{H}_F$ and γ_F acts as identity on $\mathcal{H}_{\rm GV}$; no basis change on the tensor product turns one into the other. Electroweak chirality co-location does not close on S^3 alone.

The G2–G3 corollary. Sprint H1 left G2 (Higgs as inner fluctuation) at the verdict "construction admits a Higgs given an imposed Yukawa Y, but no GeoVac mechanism selects Y." G3's NEGATIVE supplies the structural reason. The Yukawa sits in the off-diagonal $\mathbb{C} \leftrightarrow \mathbb{H}$ block of D_F that anticommutes with γ_F; the candidate GeoVac sources for that block (offdiag CH chirality bridging, Sturmian inter-shell coupling) have no kinematic relation to γ_F, because $\gamma_{\rm GV}$ is structurally independent of γ_F. Equivalently: no GeoVac-side data fixes Y because the GeoVac chirality lives on a different $\mathbb{Z}_2$ grading from the one the Yukawa flips. G2 and G3 are therefore one structural fact, not two. This is a *sharpening* of H1's G2 verdict (the Yukawa-undetermined gap is named structurally), not a new gap.

Files. Modules `geovac/chirality_grading.py` (G3-A, 50/50 tests) and the `chirality_F` additions to `geovac/almost_commutative.py` (G3-B); tests `tests/test_chirality_grading.py` and `tests/test_almost_commutative.py` (53 passing, `TestChiralityF` block); diagnostic `debug/g3c_tensor_chirali` data `debug/data/g3a_chirality.json`, `debug/data/g3c_tensor_chirality.json`; memos `debug/g3a_chiralit` `debug/g3b_chirality_F_audit_memo.md`, `debug/g3c_tensor_chirality_memo.md`.

8.4 Sprint G4: Cross-manifold scoping (G4a reachable, G4b NCG-framework-blocked)

With G2 and G3 collapsed into one structural fact and electroweak unification on S^3 alone closed in the negative, G4 (cross-manifold) becomes *the* remaining route to a GeoVac-internal SM construction rather than the dominant gap among several. A scoping pass (May 2026, memo `debug/g4_cross_manifold_sco` sharpens G4 into two sub-gaps with sharply different reachability.

G4a (reachable): Connes SM on $\mathcal{T}_{S^3}$ alone with $\mathcal{A}_F = \mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$. The minimal extension of Sprint H1's electroweak slice $\mathcal{A}_F = \mathbb{C}\oplus\mathbb{H}$ to the full Connes SM finite algebra is reachable as a 1–2 month sprint extending `geovac/almost_commutative.py`. The $M_3(\mathbb{C})$ summand supplies SU(3) at the inner-fluctuation level on the existing GeoVac S^3 triple, without recourse to the S^5 Bargmann construction. Predicted outcome: *positive-thin*, in the exact sense of Sprints H1 and G3. Inner fluctuations decompose into a gauge-1-form sector (recovering $U(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3)$ at the operator level, with the abelian and SU(2) parts already verified by H1) plus a Higgs sector that is nontrivial whenever the Yukawa-Dirac off-diagonal blocks are imposed. The G2–G3 corollary applies verbatim: G4a defines the construction, but no GeoVac mechanism selects the Yukawa, hypercharge assignments, or generations. G4a therefore does not close the SM-derivation question; it relocates the open data-selection items (Y, hypercharge, generation count) from "electroweak slice" (H1, G3) to "full Connes finite algebra," without changing their status. G4a remains worth pursuing because it makes Sprint ST-SU3's Bargmann SU(3) construction structurally redundant for SM-derivation purposes (one S^3 construction with $M_3(\mathbb{C})$ delivers the same gauge content) and because it is the most published-precedent-compatible target available.

G4b (NCG-framework-blocked): genuine cross-manifold $\mathcal{T}_{S^3}\otimes\mathcal{T}_{S^5}$. The structurally deeper version of G4 asks whether the GeoVac S^3 Coulomb triple and the Paper 24 S^5 Bargmann lattice can be combined into a single spectral triple with both as commuting factors, so that the SU(3) of Sprint ST-SU3 is honestly cross-manifold-located rather than recapitulated on S^3. This sub-gap is blocked at the *NCG-framework level*, not at the GeoVac level. Connes' tensor product of spectral triples [2] requires both factors to be Riemannian spectral triples in the same sense. $\mathcal{T}_{S^3}$ carries the Camporesi–Higuchi Riemannian spinor Dirac (KO-dim 3). Paper 24's S^5 Bargmann construction is built from the holomorphic Hardy sector $H^2(S^5)$ restricted to symmetric SU(3) irreps $(N, 0)$; its natural diagonal operator is the first-order complex Euler operator $\hat{N} = \sum_i z_i\partial_i$, not a Riemannian Dirac. Two candidate $\mathcal{T}_{S^5}$ exist, and neither preserves both standard NCG axioms and the GeoVac-specific content of Sprints 25–30 / ST-SU3:

- *Option A (Riemannian round-S^5).* Take D_{S^5} to be the standard Riemannian spinor Dirac on round S^5 (KO-dim 5). The tensor product is a textbook Connes object with combined KO-dim $3+5=8 \equiv 0 \pmod 8$, but it discards the $(N,0)$-tower structure that made Paper 24's S^5 "natural," and the bit-exact π-free certificate of Paper 24 §III is irrelevant to round-S^5 (calibration π reappears via standard Weyl asymptotics).

- *Option B (Bargmann–Euler-based).* Promote the Euler operator $\hat{N}$ on the Bargmann graph to a Dirac-like object via chiral doubling. This preserves Sprint ST-SU3 SU(3) Wilson content and the π-free certificate, but the resulting object is *not* a Riemannian spectral triple in the published sense: KO-dimension is not classically defined for a Hardy-sector first-order complex operator, and there is no published prescription in the NCG literature for tensoring a Riemannian spectral triple with a Hardy-sector / Berezin–Toeplitz first-order complex object. The closest analogs (Hawkins 2000 on the unit disk; Hekkelman 2022, 2024; Hekkelman–McDonald 2024 on flat structures) stay within Berezin–Toeplitz on a single manifold or within Riemannian on a single manifold, never crossing.

The structural reading is that G4b is the Coulomb/HO asymmetry of Paper 24 §V resurfacing at the spectral-triple level. Paper 24 §5.3 now records four layers of this asymmetry, all Coulomb-specific: (i) spectrum-computing role of L_0; (ii) calibration π; (iii) Wilson gauge with natural matter coupling, Sec. 8.1; and (iv) modular-Hamiltonian Pythagorean orthogonality on the hemispheric wedge, the structural mechanism behind the L2-E signature-independence finding (Paper 43 §10.2 [52]; Remark 46 above). G4b is a *sibling* of these four layers, not the lone fourth layer: the Coulomb S^3 side carries a Riemannian spectral triple in the standard NCG sense and supports the BW-aligned wedge-modular construction at signatures $(3,0)$ and $(3,1)$, while the Bargmann S^5 side has none of the four ingredients (no spinor bundle on the $(N,0)$ Hardy tower; no second-order-vs-first-order distinction with a separate Dirac; no half-integer wedge; no calibration π or natural Wilson matter coupling). The Connes tensor-product framework cannot bridge them as currently published. Closing G4b is therefore not a sprint-scale GeoVac task; it requires either an extension of NCG to handle Riemannian–Hardy mixed tensor products, or the identification of a unifying ambient manifold whose GeoVac sub-structure contains both S^3 Coulomb and S^5 Bargmann as natural sub-objects. Both directions are recorded as honest open structural questions, not as sprint targets.

Net. After Sprints H1 + G3 + G4 scoping, the four-gap analysis of Sec. 8.1 compresses to:

- G2 and G3 are one structural fact: the Yukawa is undetermined because γ_{GV} and γ_F are independent commuting $\mathbb{Z}_2$ gradings on the tensor product (G3 NEGATIVE).
- G4 splits cleanly: G4a (Connes SM on $\mathcal{T}_{S^3}$ alone with $M_3(\mathbb{C})$) is **closed at POSITIVE-THIN** (2026-05-31): $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ recovered from inner fluctuations, all Connes axioms bit-exact, Yukawa/hypercharge/generations remain imposed; G4b (genuine cross-manifold) is blocked at the NCG-framework level by the Coulomb/HO structural asymmetry.
- G1 (three generations) is unchanged from H1: formally definable, not selected by GeoVac structure.

GeoVac does not autonomously generate the Standard Model; the construction admits SM gauge content given imposed Yukawa, hypercharge, and generation choices, and the GeoVac data is silent on each of those choices. This is a clean structural verdict, in the sense of CLAUDE.md §1.5: the framework is consistent with a GeoVac SM construction, but the framework does not select one.

G4a closure (Sprint G4a, 2026-05-31). The full Connes SM finite algebra $\mathcal{A}_F = \mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ has been implemented and verified on $\mathcal{T}_{S^3}$ at $n_{\max} \in \{1,2,3\}$ with $\dim \mathcal{H}_F = 32$ (16 matter + 16 antimatter; $\dim \mathcal{H}_{\text{total}} = 1280$ at $n_{\max} = 3$). All six Connes axioms ($J^2 = -I$, $JD = +DJ$, $D = D^\dagger$, $\gamma^2 = I$, order-zero, order-one) pass at literal machine-zero residual at every tested $n_{\max}$, with no finite-resolution degradation. Inner fluctuations decompose into $\mathrm{U}(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3)$ gauge 1-forms plus a Higgs scalar that is non-trivial whenever the Yukawa-Dirac off-diagonal blocks are imposed and identically zero otherwise. Matter-antimatter and lepton-quark cross-sectors decouple exactly. The predicted POSITIVE-THIN verdict is confirmed: the construction admits the full SM gauge content with imposed Yukawa, hypercharge, and generation choices, and the GeoVac data is silent on each of those choices. Module `geovac/standard_model_triple.py` (~ 530 lines), 45 tests passing.

H1 Higgs inner fluctuation (Sprint H1, May 2026). The full fluctuated Dirac $D_A = D + A + JAJ^{-1}$ has been constructed at $n_{\max} \in \{2,3\}$ ($\dim \mathcal{H}_{\text{total}} = 512, 1280$). Inner fluctuations decompose cleanly into gauge ($\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$, matching G4a) plus Higgs (complex SU(2) doublet Φ in the L–R off-diagonal block, non-trivial iff Yukawa $Y \neq 0$). The spectral-action Mexican-hat potential is confirmed: $\mathrm{Tr}(D_F^2) > 0$ and $\mathrm{Tr}(D_F^4) > 0$ give the standard CCM sign structure $V(\Phi) = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$.

Theorem 6 (H1 Yukawa non-selection)**.** *Within the almost-commutative spectral triple* $(\mathcal{A}_{\text{GV}} \otimes \mathcal{A}_F, \mathcal{H}_{\text{total}}, D)$ *with* $\mathcal{A}_F = \mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$*, the space of admissible finite Dirac operators* D_F *satisfying all Connes axioms (Hermiticity, chirality grading, J-reality, order-one condition) carries, after the matter-sector reductions* $1024 \to 512$ *(Hermiticity)* $\to 128$ *(chirality grading)* $\to 128$ *(J-reality; order-one imposes no further constraint), exactly* 128 *real parameters in the matter sector. (This* 128 *is the matter-sector parameter count and mirrors the chain of Theorem 7; the full-axiom* D_F *moduli dimension at* $n_{\max} = 2$ *is* 260*, not* 128 *— see that theorem's proof. The Yukawa non-selection conclusion is independent of this count.) For one generation,* $Y = \mathrm{diag}(y_\nu, y_e, y_u, y_d)$ *accounts for* 8 *of these. The gauge sector* $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ *is **forced**; the Yukawa values are **not selected** by any morphism in* $\mathcal{A}_{\text{GV}}$*.*

Proof sketch. The parameter count follows from direct enumeration on the chirality-graded J-real subspace of Hermitian operators on $\mathcal{H}_F$. The non-selection of the Yukawa is the G3 structural finding (Sprint G3, §8.3): γ_{GV} and γ_F are independent commuting $\mathbb{Z}_2$'s on $\mathcal{H}_{\text{GV}} \otimes \mathcal{H}_F$, so no GeoVac-side data couples to the γ_F flip that carries the Yukawa. The full fluctuated Dirac $D_A = D + A + JAJ^{-1}$ was constructed at $n_{\max} \in \{2,3\}$ ($\dim \mathcal{H}_{\text{total}} = 512, 1280$) and the inner fluctuations decompose cleanly into the forced gauge piece (matching G4a) plus the Higgs piece (complex SU(2) doublet Φ in the L–R off-diagonal block, non-trivial iff $Y \neq 0$). The spectral-action Mexican-hat potential is confirmed: $\mathrm{Tr}(D_F^2) > 0$ and $\mathrm{Tr}(D_F^4) > 0$ give the standard CCM sign structure $V(\Phi) = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$. See `debug/h1_higgs_inner_fluctuation_memo.md`. □

The forced/free seam (forcing-catalogue Door 4, 2026-06-01). The Yukawa non-selection theorem admits a sharper structural reading. The almost-commutative factorization $D^2 = D_{\text{GV}}^2 \otimes 1 + 1 \otimes D_F^2$ (the cross term vanishes by chirality anticommutation) factorizes the master Mellin engine into an *outer* ring — the M1/M2/M3 transcendentals of the GeoVac sector — tensored with an *inner* Yukawa Dirichlet ring $\mathbb{Q}[y_i^{-2s}]$ generated by the Yukawa eigenvalues. The two rings share no generator: the η-trivialization ($\{\gamma_F, D_F\} = 0$ annihilates the odd-k heat traces that would couple them), together with the G3 commuting-$\mathbb{Z}_2$ result, makes the inner Yukawa ring *disjoint* from every forced outer ring. This is the precise content of the structural-skeleton scope (§1.5 of the project guidelines): the bone (the forced, integer/rational skeleton) and the calibration data (free) live in provably non-overlapping rings, and the Yukawa is free *because* it generates its own ring — a quantity cannot be derived from a ring it generates. The reading also explains, after the fact, why the H1, W3, and Koide attempts to select the Yukawa all returned clean negatives: each probed *inside* the almost-commutative category, where the seam already forbids selection.

The inner algebra structure is mostly forced (Door 4b, 2026-06-01). The seam places the Yukawa *values* on the free side; it does *not* place the inner *algebra structure* there. A finite-algebra enumeration shows that among all finite semisimple real $*$-algebras with at most three summands and factor size at most three, exactly *two* reproduce the gauge group $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ as their unitary/inner-automorphism content: the Standard Model finite algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ and $\mathbb{C} \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C})$. The same Bertrand $\times$ complex-Hopf-tower truncation at $n \leq 3$ that forces the gauge content (Sec. 8.1) therefore also forces the factor count (3, one per Hopf rung $n = 1, 2, 3$), the $n = 1$ factor ($\mathbb{C}$), and the $n = 3$ factor ($M_3(\mathbb{C})$); the residual non-uniqueness collapses to a single binary fork — $\mathbb{H}$ versus $M_2(\mathbb{C})$ at the $n = 2$ rung. This fork was conjectured (Door 4b) to have a GeoVac-native handle: the outer triple's $J_{\mathrm{GV}}^2 = -1$ at KO-dimension 3 (the real-structure result of Sec. 3, the pseudoreal $\mathrm{Spin}(3) = \mathrm{SU}(2)$ two-spinor sign — the *same* $S^3 = \mathrm{SU}(2)$ fact that produces the gauge SU(2)) might select $\mathbb{H}$ over $M_2(\mathbb{C})$. A direct J sign-table audit (Door 4c, 2026-06-01) closed this conjecture *negative*: $J_{\mathrm{GV}}^2 = -1$ *admits* but does not *force* $\mathbb{H}$. The decisive fact is that over $\mathbb{C}$ the two candidates are the same complex algebra ($\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$); the $\mathbb{H}$-vs-$M_2(\mathbb{C})$ distinction is a choice of *real form*, fixed by the sign of an *internal* $\mathbb{C}^2$ antilinear involution ($J_{\mathrm{int}}^2 = -1 \Leftrightarrow \mathbb{H}$; $J_{\mathrm{int}}^2 = +1 \Leftrightarrow M_2(\mathbb{C})$), which lives on the algebra side and is invisible to the combined $J = J_{\mathrm{GV}} \otimes J_F$. The combined J^2, the combined $(\varepsilon, \varepsilon')$ signs, and the combined KO-dimension are *bit-identical* for both candidates at $n_{\max} \in \{1, 2, 3\}$, and the Chamseddine–Connes–Marcolli order-one-with-Yukawa selector is satisfied by both (matter/antimatter decoupling). The $\mathbb{H}$ choice is therefore a *literature import* (the CCM complex-fermion-representation / second-order condition on the finite factor), not a GeoVac forcing; the fork stays open. The honest ledger: *forced* are the factor count and the $\mathbb{C}$ and $M_3(\mathbb{C})$ summands; *admitted but not forced* is the $n = 2$ real form ($\mathbb{H}$ vs $M_2(\mathbb{C})$); *free* are the generation count (invisible to inner automorphisms), the inner KO-dimension, and the Yukawa values. The net sharpening is real but bounded: the geometry forces the Standard Model gauge group, the factor count, and two of the three matrix factors — more internal algebraic *shape* than the gauge group alone — but the third factor's real form, the generation multiplicity, and the Yukawa values remain outside the forcing. Sources: `debug/door4_gauge_yukawa_boundary_memo.md`, `debug/door4b_inner_algebra_forcing_memo.md`, `debug/door4c_j_signtable_audit_memo.md`.

A geometric handle on the $n = 2$ fork (Door 4d, 2026-06-02). A second handle, structurally orthogonal to the combined-J sign table of Door 4c, is the *Division-Algebra-of-the-Sphere* (DAS) criterion: *at each Hopf rung n, the inner algebra is the natural associative real normed division algebra whose unit-norm sphere IS the rung's Hopf bundle total space S^{2n-1}.* The identifications are bit-exact at all three rungs: S^1 is the unit-norm sphere of $\mathbb{C}$ under complex multiplication (closure 1.1×10^{-16}); S^3 is the unit-norm sphere of $\mathbb{H}$ under quaternion multiplication and is bit-exactly isomorphic to $\mathrm{Sp}(1) = \mathrm{SU}(2)$ via the standard $\mathbb{H} \to M_2(\mathbb{C})$ embedding (closure 2.2×10^{-16}, SU(2)-isomorphism error 3.3×10^{-16}); and by Hurwitz's classification of finite-dimensional real normed division algebras (only $\mathbb{R}, \mathbb{C}, \mathbb{H}$ are associative, with unit spheres S^0, S^1, S^3 respectively), S^5 has no associative-division-algebra realization, so the inner algebra at $n = 3$ falls back to a matrix algebra. DAS *agrees* with Door 4b's elimination forcings at $n = 1$ ($\mathbb{C}$) and $n = 3$ ($M_3(\mathbb{C})$), and *closes* the $n = 2$ fork by selecting $\mathbb{H}$ — the direct realization with $\mathrm{U}(\mathbb{H}) = \mathrm{Sp}(1) = \mathrm{SU}(2)$ on the 3-dimensional S^3, in contrast to the $M_2(\mathbb{C})$ realization whose unit-norm sphere is the 4-dimensional U(2) with SU(2) reached only via the U(2)/U(1) unimodularity quotient. The DAS argument never invokes any J or sign-table data; it is a purely geometric criterion identifying the rung-n sphere with the unit-norm subset of the rung's associative division algebra. The verdict is *partial-door*: the existing argument of Sec. 8.1 extracts only the gauge group from the Hopf-rung sphere; adopting DAS as a forcing requires the additional structural theorem that the AC tensor product $\mathcal{H} = \mathcal{H}_{\mathrm{GV}} \otimes \mathcal{H}_F$ transfers the rung-n sphere's division-algebra structure to a constraint on $\mathcal{H}_F$'s inner algebra. This theorem is sprint-reachable and paper-level; if it closes, the inner-algebra status upgrades from "mostly forced with the $\mathbb{H}$-vs-$M_2(\mathbb{C})$ fork closed by literature import (CCM second-order condition)" to "mostly forced with the fork closed by a GeoVac-coherent geometric criterion," leaving only the generation count, the inner KO-dimension, and the Yukawa values outside the forcing. Source: `debug/door4d_division_algebra_sphere_memo.md`.

Falsifier A′ on DAS (Door 4e, 2026-06-02). The DAS handle introduced above is partial-door because the existing construction of Sec. 8.1 extracts only the gauge group from the Hopf-rung sphere. The natural test was whether any standard AC tensor-product compatibility condition fills the gap, i.e., distinguishes $\mathbb{H}$ from $M_2(\mathbb{C})$ as the $n = 2$ inner algebra. Three candidates were tested literally: (C1) SU(2)-equivariance under the Ad action $g \cdot m \cdot g^{-1}$; (C2) Hopf U(1)-equivariance under scalar phase $e^{i\theta} I$ acting from the right; (C3) principal-bundle compatibility with the Hopf U(1) fiber. All three returned *negative*: (C1) both $\mathbb{H}$ and $M_2(\mathbb{C})$ are SU(2)-Ad-invariant real $*$-subalgebras of $M_2(\mathbb{C})$ (the former is the unique minimal non-abelian such, the latter is the maximal; the distinction requires a minimality criterion that is itself extra input); (C2) every 2×2 complex matrix commutes with the scalar phase $e^{i\theta} I$ trivially, so the condition is vacuous; (C3) both candidate actions are principal-bundle compatible at machine precision by associativity of matrix multiplication and centrality of scalar phases. *No standard AC tensor-product compatibility condition forces $\mathbb{H}$ over $M_2(\mathbb{C})$.* The substantive content of the test is not the negative outcome (which was the expected one) but the *import comparison* it enables: the standard Chamseddine–Connes–Marcolli derivation of $\mathbb{H}$ at the $n = 2$ factor invokes *three* axioms beyond the bare Connes spectral-triple data (second-order condition with nonzero finite Dirac, complex chiral fermion representation, and the $2N^2 = 32$ dimension count), whereas DAS invokes *one* (the rung-n sphere's division-algebra structure is inherited by the inner algebra). Both are imports relative to the bare axioms; DAS is leaner. The path to upgrading partial-door to door is therefore not a theorem about existing CCM machinery but a paper-level structural axiom: adopting "the inner algebra at rung n is the $*$-algebra whose unit group IS the rung-n sphere S^{2n-1} as a Lie group, when one exists; else fallback $M_n(\mathbb{C})$." This reads the Hopf-rung sphere fully (Lie-group structure plus topology) where the existing construction reads it partially (gauge group only), and gives $\mathbb{C}$ at $n = 1$ (via $S^1 = \mathrm{U}(1) = \mathrm{U}(\mathbb{C})$), $\mathbb{H}$ at $n = 2$ (via $S^3 = \mathrm{Sp}(1) = \mathrm{U}(\mathbb{H})$), and $M_3(\mathbb{C})$ at $n = 3$ (since S^5 is not a Lie group at all, by Adams 1958). Source: `debug/door4e_falsifier_A_prime_memo.md`.

Closure of the Door 4 series (Door 4f, 2026-06-02; net axiom savings audited). The natural next test was whether adopting Upgrade B introduces any new downstream constraint on existing GeoVac observables. Twelve sub-tests were run: inner KO-dimension (T1), chirality grading γ_F (T2), Yukawa eigenvalues (T3), generation count (T4), Connes order-zero (T5), order-one (T6), and second-order (T7) conditions, bosonic spectral-action gauge-sector coefficient (T8), Higgs vacuum manifold (T9), cross-rung Yukawa coupling (T10), extensibility to rung $n \geq 4$ via Adams 1958 (T11), and consistency with the Bertrand $\times$ Hopf-tower reading of Sec. 8.1 (T12). *All twelve* return either SILENT (Upgrade B unconstrained on the target), COMPATIBLE (the pre-existing constraint passes for $\mathbb{H}$, the mechanism unaffected), IDENTICAL to CCM (same observable from either $\mathbb{H}$ or $M_2(\mathbb{C})$ post-unimodularity), or a CLEAN extensibility/strengthening rule (T11, T12). Zero sub-tests introduce a new constraint; zero contradict any existing GeoVac result. Upgrade B is therefore a *clean reformulation* of the CCM $\mathbb{H}$-selection with no new predictive content.

Honest axiom-savings audit. The naive comparison in the preceding paragraph (CCM 3 axioms vs DAS 1 axiom) over-counts. Sub-test T7 (Connes second-order condition) shows that the second-order condition is *still* required in the Upgrade-B formulation, because it constrains the off-diagonal Yukawa structure of D_F (which $\mathbb{H}$-selection alone does not determine). Upgrade B therefore replaces *two* of CCM's three $\mathbb{H}$-selection axioms — the complex chiral fermion representation requirement and the $2N^2 = 32$ dimension count — with the single sphere-Lie-group axiom; the second-order condition persists in both formulations. Net axiom savings: *one*. The Door 4 series is structurally complete at **partial-door (final)**. The remaining open questions are paper-level (whether to adopt Upgrade B's axiom-minimality advantage as the working construction principle) and the deep wall (force N_{gen} and the inner KO-dim from a packing principle; no known handle; deferred). Source: `debug/door4f_falsifier_A_double_prime_memo.md`.

The deep wall is a relative necessity, not a gap (Direction 2 scoping, 2026-06-03). The "deep wall" named above — forcing N_{gen} and the inner KO-dimension from a packing principle — is sharper than an open gap: relative to the packing construction of Paper 0 as presently formulated, it is a *necessity.* Two established facts compose. First, the Door 4 seam: the inner Yukawa/real-structure data lives in a ring (and a set of $(\mathcal{H}_F, D_F, J_F, \gamma_F)$ data) provably disjoint from every forced outer ring. Second, Paper 0 §VII: the packing construction is *kinematic* — it emits the label set (n, ℓ, m, s) and the graph topology, and emits no operator, no real structure, no coupling, and no Hilbert-space multiplicity. But N_{gen} is exactly a Hilbert-space multiplicity (the factor in $\mathcal{H}_F = \mathbb{C}^{N_{\text{gen}}} \otimes \mathcal{H}_F^{(1)}$, invisible to every algebra- and automorphism-level mechanism), and the inner KO-dimension is exactly a (J_F, γ_F) datum — both of the type the packing construction does not produce. The packing axiom therefore cannot constrain either: the residue is packing-inaccessible by composition of these two facts. The single logically-open escape is a *packing-native multiplicity* — a datum the construction emits on the outer labels that would map to a factor-N_{gen} tripling of $\mathcal{H}_F$. It is empty: Paper 0's entire multiplicity inventory is the $2\ell+1$ angular slots (odd, and *growing* with ℓ, hence not a generation-universal multiplicity) together with a single $\mathbb{Z}_2$ already consumed by spin; no constant factor of three appears anywhere. This promotes the structural-skeleton scope from an observation to a *relative theorem of necessity* for the N_{gen}/inner-KO-dimension residue, and gives a single structural reason the H1, W3, and Koide selection attempts failed: all probed inside the almost-commutative category, on the side of the seam where the disjointness forbids selection and the packing axiom has no reach. The qualifier "relative" is load-bearing: the statement forecloses the residue for the *current* packing construction, not for every conceivable discrete one. Lifting it would require a *sibling axiom* that emits fiber multiplicities — a new primitive Paper 0 does not contain — recorded as the open (long-range) direction with no present handle. The inner *algebra shape* $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is the one inner datum the construction does reach (Door 4b/4d, via the Hopf tower and Hurwitz's classification); the residue it cannot reach is precisely the multiplicity- and real-structure-valued data. Source: `debug/seam_packing_scoping_memo.md`.

The Forced-Count Theorem (synthesis, 2026-06-03). The eight preceding paragraphs (Sprint H1 through Door 4f and the Direction 2 scoping) compose into a single Bertrand-shaped statement that crystallizes the structural content already proved across the sub-sprints. Stated as a theorem it makes explicit what the distributed prose shows in pieces: the framework forces the *dimension* of the D_F moduli space, but does not (and at the current packing primitive cannot) force the points within it.

Theorem 7 (Forced D_F moduli dimension)**.** *Let $\mathcal{T} = (\mathcal{A}_{\rm GV} \otimes \mathcal{A}_F,\ \mathcal{H}_{\rm GV} \otimes \mathcal{H}_F,\ D_{\rm GV} \otimes 1 + 1 \otimes D_F,\ J,\ \gamma)$ be the GeoVac almost-commutative spectral triple of Sec. 3, assuming:*

1. *The Bertrand-with-complex-Hopf-tower truncation $n \le 3$ of Sec. 8.1, with the Upgrade B sphere-Lie-group axiom of Door 4e, forcing $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$.*
2. *The canonical Chamseddine–Connes–Marcolli SM representation of $\mathcal{A}_F$ on $\mathcal{H}_F$, with generation multiplicity $N_{\rm gen}$ imported as an external input (Direction 2: not packing-derived).*
3. *The standard real-spectral-triple axioms: Hermiticity, KO-dim-6 chirality grading $\{\gamma_F, D_F\} = 0$, J-reality $[J_F, D_F] = 0$, and Connes order-one.*

Then the space of admissible finite Dirac operators D_F has real dimension forced by these inputs alone, with the reduction chain at $N_{\rm gen} = 1$:

$$\dim_{\mathbb{R}} \mathcal{M}(D_F) :\ 1024 \xrightarrow{\text{Herm.}} 512 \xrightarrow{\{\gamma_F, D_F\}=0} 128 \xrightarrow{[J_F, D_F]=0 + \textit{order-one}} 128. \tag{7}$$

The physical-slice dimension under restriction to diagonal Yukawa $Y = \operatorname{diag}(y_\nu, y_e, y_u, y_d)$ is 8 per generation. For $N_{\rm gen}$ generations with full generation-mixing Yukawa matrices the count is $128 \cdot N_{\rm gen}^2$.

Proof sketch. The chain in eq. (7) is the H1 sprint accounting at $n_{\max} = 2$ on the canonical $\mathcal{A}_F$-representation (32×32 at $N_{\rm gen} = 1$): Hermiticity halves the free parameters; $\{\gamma_F, D_F\} = 0$ kills the chirality-diagonal blocks; J-reality identifies the matter and antimatter sectors; the order-one condition is satisfied for the SM representation without further reduction (the lepton and quark off-chirality blocks each satisfy order-one because $\mathcal{A}_F$ acts block-diagonally in lepton/quark and $M_3(\mathbb{C})$ commutes with the Yukawa flavour structure). Each constructed-triple axiom-consistency check passes at $n_{\max} \in \{2, 3\}$ in module `geovac/standard_model_triple.py` (45 tests). *QA correction (2026-06-14):* those 45 tests verify the axiom consistency of one constructed D_F, *not* the moduli-dimension counts of the chain. A direct parameter-count of the constraint solution space (`tests/test_trunk_qa_forced_count_moduli.py`) gives, on the full 32-dimensional finite geometry, $2048 \to 1024\,(\text{Herm.}) \to 512\,(\{\gamma_F, D_F\}{=}0) \to 272\,(J\text{-real}) \to 260\,(\text{order-one})$—so the chain's "128" intermediates hold only under the restricted 16-dimensional *matter-sector* count ($512 \to 256 \to 128$), not the full-axiom count (260). The Forced-Count statement should be read as the matter-sector parameter count, with the $\{\gamma_F, D_F\}{=}0$ step relabelled accordingly. The $N_{\rm gen}^2$ scaling follows from the off-chirality block dimension scaling linearly with $N_{\rm gen}$ in each chirality. The inputs (1)–(3) are themselves theorems / proven structural forcings: (1) by Door 4b + Door 4d/4e/4f (the Hopf-tower truncation forces the factor count; the sphere-Lie-group axiom forces $\mathbb{H}$ at $n = 2$); (3) is the standard real-spectral-triple axiom set; (2) is the one external input, recorded by the Direction 2 paragraph as packing-unreachable in the current Paper 0 construction. □

Corollary 5 (Bertrand analogy)**.** *Theorem 7 is the inner-factor analogue of Bertrand's theorem on closed-orbit central forces. Bertrand forces the* form *of the central potential ($-k/r$ or kr^2) without forcing the coupling k; Theorem 7 forces the* dimension *of the D_F moduli space without forcing the D_F-entries themselves. The two seams are structurally identical: the framework determines what the moduli space* is*, not where in it the realised system sits.*

Remark 48 (Paired with the Door 4 negative half)**.** *The positive content of Theorem 7 sits beside two negative companions already recorded above:*

- ***Door 4 seam theorem (Yukawa-value wall):*** *the η-trivialization plus tensor factorization plus the G3 commuting-$\mathbb{Z}_2$ result prove that the D_F-entries inside $\mathcal{M}(D_F)$ are in a Dirichlet ring disjoint from every forced outer ring, so the framework cannot derive them. Sprint Yukawa-PSLQ (2026-06-03) added empirical confirmation: a 162-cell sweep of the nine measured Yukawa values against the forced inner-factor pure-Tate basis $\mathbb{Q}[\pi, \pi^{-1}]$ (basis sharpened by the η-trivialization audit ruling out M_3/Dirichlet L on the inner factor) returned zero hits at $M \leq 10^3$, at both MS-bar M_Z and MS-bar $\mu = 2 \times 10^{16}$ GeV, across three transforms (y_f, y_f^2, $\log y_f$). Source:* `debug/sprint_yukawa_pslq_memo.md`.
- ***Direction 2 packing-reach theorem:*** *N_{gen} and the inner KO-dimension are packing-unreachable in the current Paper 0 construction (preceding paragraph, source* `debug/seam_packing_scoping_m`

Theorem 7 thus crystallises the full *forced/free seam at the level of D_F: the forced data is the moduli space and its dimension; the free data is (i) the point in the moduli space (Yukawa values, Door 4 wall), (ii) the multiplicity factor (N_{gen}, Direction 2), and (iii) the real-structure signature (inner KO-dimension, Direction 2). All three free data are calibration inputs in the structural-skeleton-scope sense (§1.5 of the project guidelines).*

Theorem 8 (N_{gen} non-selection under canonical CCM representation)**.** *Let $\mathcal{A}$ denote the framework's current axiom set {Paper 0 packing axiom; standard real-spectral-triple axioms (Hermiticity, J-reality, KO-dim chirality, order-one); Hopf-tower truncation (Hurwitz); Bertrand's classical theorem; Upgrade B (sphere-Lie-group axiom)}, and fix the canonical Chamseddine–Connes SM representation of $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ on the inner Hilbert space.*

For every integer $N \geq 1$, let $\mathcal{T}_N$ denote the GeoVac spectral triple with inner Hilbert-space multiplicity N (i.e., $\dim \mathcal{H}_F = 32N$ under the canonical rep). Then $\mathcal{T}_N$ satisfies every axiom in $\mathcal{A}$. Consequently, $\mathcal{A}$ contains no morphism that selects a specific value of N.

Proof sketch. Each axiom in $\mathcal{A}$ is independent of N:

- The Paper 0 packing axiom outputs (n, l, m, m_s) labels for a single S^3 outer graph; N is not in this output.
- The standard real-spectral-triple axioms constrain the relations (D, J, γ) on $\mathcal{H} = \mathcal{H}_{\text{GV}} \otimes \mathcal{H}_F$. $\dim \mathcal{H}_F$ scales linearly with N, but Hermiticity, J-reality, KO-dim chirality, and order-one are structural relations independent of multiplicity.
- Hopf-tower truncation (Hurwitz, Doors 4b/4d/4e) operates on $\mathcal{A}_F$; the algebra $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is fixed independent of N.
- Bertrand's classical theorem selects the outer manifold S^3; N-independent.
- Upgrade B constrains the inner-algebra structure via the sphere-Lie-group identification of Door 4e; N-independent.
- The canonical CCM SM representation distributes 32 fermion degrees of freedom per generation across the three algebra factors; the multiplicity N enters as an external Hilbert-space multiplicity rather than as a constraint emergent from the rep structure.

The Read 2 scoping (`debug/sprint_read2_n_gen_scoping_memo.md`, 2026-06-03) provides three independent obstructions to any candidate Hopf-tower-to-representation shortcut that would correlate N with the algebra factor count under canonical representation: the SM fermion DOF count per generation distributes across all three algebra factors; the Yukawa-Higgs cross-generation coupling structure required for CKM/PMNS phenomenology prevents algebra-factor reorganisation; and the alternative "3 factors = 3 generations" assignment gives 12 DOF per gen versus the 32 DOF of the SM. Each obstruction independently closes the candidate shortcut.

Therefore $\mathcal{T}_N$ is a valid GeoVac spectral triple for every $N \geq 1$ under canonical representation; no axiom in $\mathcal{A}$ has been exhibited that selects a specific value. □ □

Remark 49 (Scope and the unconditional question)**.** *Theorem 8 is conditional on the canonical CCM SM representation choice. The unconditional question — whether any SM-phenomenologically-consistent representation of $\mathcal{A}_F$ on $\mathcal{H}_F$ could correlate N with the algebra factor count under an alternative rep — remains the open long-range structural- research target named by the Read 2 sharpest falsifier: exhibit a representation in which the* 32 *fermion DOF per generation are distributed across the three algebra factors in a way that makes the three generations the three rep blocks of the algebra, and the cross-generation Yukawa-Higgs coupling (CKM/PMNS) is naturally encoded as the algebra cross-factor coupling. Absent such a construction, Theorem 8 stands as the conditional impossibility result corresponding to the Read 2 NO-GO scoping verdict.*

The Theorem identifies N_{gen} as Class 1 calibration data (external input on which the framework's outer-factor mechanism is structurally silent), formalising the Direction 2 packing-reach and Read 2 representation-level NO-GO scopings into a single structural statement. It is the multiplicity-side analogue of the H1 Yukawa non-selection theorem: both are non-selection results for inner-factor content, both conditional on the canonical rep, and together with the Forced-Count Theorem (7) they characterise the full free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level.

Paper 57 §6.1 (the Forced/Free Seam observations paper) records this theorem as the C3 closure of the empirical Direction 2 / Read 2 NO-GO scopings, and as one of three theorem-grade non-selection results in the corpus (alongside H1 Yukawa non-selection and the Forced-Count Theorem).

Theorem 9 (Inner KO-dim is external calibration data under canonical-rep convention)**.** *Let $\mathcal{A}$ denote the framework's current axiom set (as defined in* Theorem 8*). The Paper 0 packing axiom is kinematic (Paper 0 §VII.B "What the construction does not provide"): it outputs the labels (n, l, m, m_s) and the S^3 graph topology, but neither a Dirac operator, nor a real structure, nor a chirality grading. The inner KO-dimensional signature is a property of the real-structure data (D_F, J_F, γ_F), none of which is in the packing axiom's output.*

The remaining axioms in $\mathcal{A}$ (standard real-spectral-triple axioms, Hopf-tower truncation, Bertrand's classical theorem, Upgrade B) constrain the consistency of (D_F, J_F, γ_F) with the outer GeoVac triple's KO-dimension 3*, but do not uniquely select an inner KO-dim signature. Different choices of (D_F, J_F, γ_F) that are consistent with these constraints yield different inner KO-dimensional signatures, each producing a valid GeoVac spectral triple. The canonical Chamseddine–Connes SM representation (inner KO-dim* 6*) is one such choice, supplied as external input to the framework.*

Proof sketch. The argument is a direct composition of two established facts:

Fact 1 (Paper 0 §VII.B, explicit scope statement). The packing construction is *kinematic*: it produces labels and graph topology, and explicitly does *not* produce operators, real structures, couplings, or fiber-Hilbert-space multiplicities.

Fact 2 (Door 4f T1, Door 4b Q3). The inner KO-dimensional signature is defined as a property of the real-structure data (D_F, J_F, γ_F). The chirality-grading axiom $\{\gamma_F, D_F\} = 0$ and the J_F-reality conditions $J_F^2 = \pm\mathbf{1}$, $J_F D_F = \pm D_F J_F$, $J_F\gamma_F = \pm\gamma_F J_F$ specify the residue classes mod 8 that are consistent, but do not uniquely select one in the absence of additional constraints (such as the phenomenological constraints imposed by SM fermion doubling and Majorana neutrino-mass structure).

Composition. $\mathcal{A}$ produces no (D_F, J_F, γ_F) data autonomously (Fact 1), and the inner KO-dim is a function of that data (Fact 2). Therefore $\mathcal{A}$ does not autonomously select an inner KO-dim signature. The canonical CCM SM rep supplies a specific (D_F, J_F, γ_F) triple (KO-dim 6); alternative specifications consistent with the outer KO-dim 3 and the standard real-spectral-triple axioms produce alternative KO-dim signatures, each consistent with $\mathcal{A}$. □ □

Remark 50 (The full inner-factor structural-skeleton boundary)**.** *Theorems 7, the H1 Yukawa non-selection (§10), 8, and 9 together characterise the full free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level:*

- ***Forced data:*** *the moduli space $\mathcal{M}(D_F)$ and its dimension (128 per generation).*
- ***Free data:*** *(i) the point in $\mathcal{M}(D_F)$ (Yukawa values — H1 non-selection), (ii) the Hilbert-space multiplicity ($N_{\rm gen}$ — Theorem 8), (iii) the real-structure signature (inner KO-dim — Theorem 9).*

These four theorems together formalise the structural-skeleton scope pattern at the inner-factor level for the GeoVac framework's current axiom set. Each is conditional on the canonical CCM SM representation. The unconditional versions — whether alternative phenomenologically-consistent representations could correlate any of the three free data with framework-derivable structure — remain long-range structural-research targets; Read 2 sharpest falsifier names the candidate construction for $N_{\rm gen}$, and analogous targets exist for KO-dim (a non-canonical rep that admits a different KO-dim while retaining SM phenomenology) and for Yukawa values (a packing-derivable structure on $\mathcal{A}_F$ that selects within the diagonal slice).

Paper 57 §6.3 (Forced/Free Seam observations paper) records the trio plus Forced-Count Theorem as the C3 closure of the inner-factor non-selection arc.

Definition 5 (Multi-cutoff spectral action structure)**.** *A multi-cutoff spectral action on a real spectral triple $\mathcal{T} = (\mathcal{A}, \mathcal{H}, D)$ is a functional of the form*

$$S^{\rm multi}(D, \{\Lambda_i\}_{i=1}^k, \{f_i\}_{i=1}^k) \;=\; \sum_{i=1}^{k} c_i \,\mathrm{Tr}\, f_i(D/\Lambda_i)$$

or more generally any functional of D depending on $k \geq 2$ independent energy scales $\Lambda_1, \ldots, \Lambda_k$ in a manner not reducible to a function of a single scale by reparametrization.

Theorem 10 (Single-cutoff spectral action)**.** *The framework's current axiom set $\mathcal{A}$ (as defined in* Theorem 8*) specifies the Chamseddine–Connes spectral action $S(D,\Lambda) = \mathrm{Tr}\, f(D/\Lambda)$ as a single-cutoff functional in the sense of* Definition 5*. No morphism derivable from $\mathcal{A}$ produces a multi-cutoff spectral action structure.*

Proof sketch. The argument is a structural inspection of each axiom in $\mathcal{A}$:

1. **Spectral action axiom (CCM).** The standard CC spectral action $S(D,\Lambda) = \mathrm{Tr}\, f(D/\Lambda)$ specifies a single-cutoff functional by stipulation. The argument D/Λ is a dimensionless ratio against a single energy scale.
2. **Paper 0 packing axiom + standard real-spectral-triple axioms + Hopf-tower truncation + Bertrand + Upgrade B.** These axioms are algebraic, kinematic, and topological: they specify $(\mathcal{A}_F, \mathcal{H}_F, D_F, J_F, \gamma_F)$ and the outer manifold geometry, but introduce no energy-scale data.

3. **Inner fluctuations $D \to D + A$ for $A = a[D, b]$.** These preserve the single-Dirac structure of the underlying spectral triple. The fluctuated operator $D' = D + A$ enters the spectral action as $S(D', \Lambda)$, still a single-cutoff functional.

4. **Tensor products $\mathcal{T}_1 \otimes \mathcal{T}_2$ (Paper 39).** These produce a new spectral triple with combined Dirac $D_1 \otimes 1 + 1 \otimes D_2$, on which the spectral action $S(D_1 \otimes 1 + 1 \otimes D_2, \Lambda)$ remains single-cutoff with the new combined D. No second cutoff is introduced by the tensor-product construction.

No axiom in $\mathcal{A}$ introduces a second independent energy scale, and no morphism derivable from $\mathcal{A}$ produces a functional of the form $S^{\text{multi}}(D, \Lambda_1, \Lambda_2, \ldots, \Lambda_k, \ldots)$ with $k \geq 2$ independent scales. Therefore the framework's spectral action machinery is single-cutoff. □ □

Corollary 6 (Multi-loop renormalization counterterms are external calibration data)**.** *Multi-loop QED renormalization counterterms (wavefunction renormalization Z_2, mass counterterm δm, vertex counterterm Z_1, etc.) require a multi-cutoff structure (Λ_{UV}, μ_R) where Λ_{UV} is the ultraviolet cutoff for the bare integrand and μ_R is the renormalization scale at which finite subtraction is performed. By* Theorem 10, *$\mathcal{A}$ contains no morphism producing such a structure. Therefore the multi-loop QED counterterms are external calibration data, supplied to the framework rather than derived from it.*

Consequently the following catalogue entries from Paper 57 [55] are classified as multi-cutoff renormalization calibration data:

- ***E7:*** *two-loop self-energy counterterms $Z_2, \delta m$.*
- ***E8:*** *multi-loop QED corrections beyond the one-loop LS-7 closure.*
- ***G3:*** *HF-5 multi-loop QED on hyperfine.*
- ***G4:*** *LS-8a two-loop SE renormalization (identical to E7 under the cross-domain unity of Paper 57 §4).*

The LS-8a sprint (May 2026) empirically confirmed this structural verdict: the framework's iterated Chamseddine–Connes spectral action reproduces the two-loop UV-divergent integrand at the right prefactor, sign, and divergence rate $\sim N^{3.79}$ (Paper 36 LS-8a), but cannot autonomously generate the $Z_2, \delta m$ counterterms required for finite multi-loop closure. Theorem 10 formalises the structural reason for the LS-8a wall: single-cutoff machinery cannot produce multi-cutoff observables.

Remark 51 (Honest scope)**.** *Theorem 10 is theorem-grade within the framework's current axiom set $\mathcal{A}$. Two scope clarifications:*

(i) The theorem does not claim "no multi-cutoff structure exists." *It claims that no morphism derivable from $\mathcal{A}$* as currently stated *produces a multi-cutoff structure. A sibling axiom adding renormalization-group flow machinery would extend the axiom set and could in principle produce multi-cutoff observables. Such a sibling axiom is the analog of the N_{gen} sibling-axiom direction (*Theorem 8 *unconditional version): a logically-possible escape, named here, that requires new primitive content beyond $\mathcal{A}$.*

(ii) The theorem does not address the spatial-composition sub-sector of the multi-focal-composition wall. *The catalogue entries G1 (HF-3 recoil cross-register $V_{eN}(\hat{r}_e, \hat{R}_N)$), G2 (HF-4 Zemach magnetization density), and G5 (W1e chemistry inner-region correlation) lie outside this theorem's scope: they are spatial $\times$ spatial composition walls, structurally distinct from the multi-cutoff renormalization sub-sector. A general multi-focal-composition wall theorem unifying both sub-sectors remains the named open structural-research target.*

Paper 57 §4 (cross-domain unity) and Paper 57 §6.3 (Sprint E7/E8 closure) record this theorem as the renormalization sub-sector closure of the multi-focal-composition wall.

Theorem 11 (Tensor-product radial coupling does not autonomously match physical two-body profiles)**.** *Let $\mathcal{T}_a = (\mathcal{A}_a, \mathcal{H}_a, D_a^{\mathrm{CH}})$ and $\mathcal{T}_b = (\mathcal{A}_b, \mathcal{H}_b, D_b^{\mathrm{CH}})$ be two GeoVac spectral triples on the same outer manifold S^3 at distinct focal lengths $\lambda_a \neq \lambda_b$, equipped with the Camporesi–Higuchi spinor Dirac operators. The framework's natural tensor-product spectral triple $\mathcal{T}_a \otimes \mathcal{T}_b$ [41] produces composed two-body operators whose:*

1. ***Angular structure** is forced: Gaunt selection rules, m-conservation, and monopole structure are all autonomously reproduced by the composed algebra (Paper 54 [54] Thm 3).*
2. ***Radial coupling profile** is determined autonomously by the combined Dirac structure $D = D_a^{\mathrm{CH}} \otimes 1 + \gamma_a \otimes D_b^{\mathrm{CH}}$, but does not autonomously coincide with the physical 1/r Coulomb form, the dipole-magnetization form, or other specific two-body radial profiles required by empirical observables.*

The framework's axiom set $\mathcal{A}$ contains no morphism enforcing that the composed radial coupling profile matches a specific physical two-body observable; matching a given observable requires external specification.

Proof sketch. The two assertions of the theorem stand on independent structural content:

(1) Angular structure forcing. Paper 54 Thm 3 [54] establishes that the composed algebra structure on $\mathcal{A}_a \otimes \mathcal{A}_b$ acts on the two-body Hilbert space through Wigner 3j coefficients in a way that exactly reproduces the angular selection rules (Gaunt, m-conservation, monopole hierarchy) of physical two-body Coulomb-style interactions. This is a direct consequence of the SU(2) representation-theoretic structure of the angular momentum operators on S^3, which the framework inherits via the Fock projection (Paper 7) and the Hopf-tower truncation.

(2) Radial coupling non-reproduction. The structural reason is the non-commutativity of the Fock-projection conformal factor with the tensor-product construction:

1. Paper 7's chordal-distance identity maps continuum $-Z/r$ on $\mathbb{R}^3$ to the unit-S^3 graph Laplacian via a focal-length- dependent stereographic conformal factor $c(\lambda)$. Two different focal lengths λ_a, λ_b give two different conformal factors c_a, c_b.
2. The tensor-product spectral triple $\mathcal{T}_a \otimes \mathcal{T}_b$ has continuum limit on the product manifold $S_a^3 \times S_b^3$. Under Fock projection, this corresponds to a six-dimensional space $\mathbb{R}_a^3 \times \mathbb{R}_b^3$ parametrised by two independent position vectors, with combined conformal factor $c_{ab} \neq c_a \cdot c_b$ in general.
3. Physical two-body observables ($\mathrm{V}_{eN}(\hat{r}_e, \hat{R}_N)$, Zemach magnetization, chemistry inner-region $V_{ee}(r_{12})$) are functions of the *relative* coordinate $\vec{r}_a - \vec{r}_b$ on $\mathbb{R}^3$ (three dimensions, not six). The reduction from the six-dimensional product manifold to the three-dimensional relative manifold is not in the framework's axiom set.
4. Empirical verification: Sprint resolvent_two_body (2026-06-01, `debug/sprint_resolvent_two_body_memo.md`) tested four weightings of the resolvent $(D_a^{\mathrm{CH}})^{-2}$ against the physical 1/r Coulomb profile and found Pearson correlations ≤ 0.81 decreasing with $n_{\max}$. Paper 54 verified the angular selection rules but reported only partial radial matching (75% connected fraction). Together these establish that the framework's natural composition mechanisms (tensor product, resolvent, gauged spectral action) do not autonomously match physical two-body radial profiles.

No axiom in $\mathcal{A}$ supplies the relative-coordinate reduction from the six-dimensional product manifold to the three-dimensional relative manifold, nor enforces a specific composed radial profile on the tensor-product spectral triple. Therefore the framework specifies the angular structure of two-body operators autonomously (via the SU(2) representation theory of S^3) but does not autonomously match the radial structure of any specific physical two-body observable. □ □

Corollary 7 (Spatial-composition catalogue entries are calibration data)**.** *The following catalogue entries from Paper 57 [55] are classified as spatial-composition calibration data:*

- ***G1:*** *HF-3 recoil cross-register $V_{eN}(\hat{r}_e, \hat{R}_N)$ (Track NI cross-register, Sprint HF 2026-05-07);*
- ***G2:*** *HF-4 Zemach magnetization density (Sprint HF 2026-05-07);*
- ***G5:*** *W1e chemistry inner-region overattraction (Sprint W1c arc 2026-05-23).*

Each requires a specific physical radial coupling profile that Theorem 11 establishes is not autonomously produced by the framework's tensor-product composition mechanism. The angular content is forced; the radial content is external calibration.

Remark 52 (Multi-focal-composition wall fully theorem-bound)**.** *Together with Theorem 10 + Corollary 6 (E7, E8, G3, G4), Theorem 11 + Corollary 7 (G1, G2, G5) close all six catalogue instances of the multi-focal-composition wall family (Paper 57 §4 cross-domain unity) at theorem-grade level under two structurally distinct theorems:*

- ***Renormalization sub-sector*** *(E7, E8, G3, G4): closed by Theorem 10 — spectral action is single-cutoff; no multi-cutoff structure in $\mathcal{A}$.*
- ***Spatial-composition sub-sector*** *(G1, G2, G5): closed by Theorem 11 — tensor-product composition reproduces angular structure but not radial structure; no relative-coordinate-reduction morphism in $\mathcal{A}$.*

The seventh wall instance (G6) is the H1 Yukawa non-selection, closed by Theorem 8's companion result (§ 10). All six unique multi-focal-composition wall instances (G6 = F1 is not independent) are now characterised at theorem-grade level by one of the three theorems.

The multi-focal-composition wall is theorem-bound at the canonical- representation level. Together with the four inner-factor non- selection theorems (Theorems 7, 8, 9, plus H1), the corpus now carries six theorem-grade non-selection results covering the inner-factor structural-skeleton boundary, the multi-loop QED renormalization sub-sector, and the spatial- composition sub-sector.

Paper 57 §6.3 (Sprint G1/G2/G5 closure) records this completion.

Theorem 12 (No single-mechanism generation of the combination rule $K = \pi(B+F-\Delta)$)**.** *Under the framework's axiom set $\mathcal{A}$ and the master Mellin engine case-exhaustion classification, the three spectral homes that appear in the empirical combination rule $K = \pi(B + F - \Delta)$ for α^{-1} (Paper 2) live in two distinct Mellin sub-rings:*

- *$B = 42$ — Casimir trace on the scalar/representation sector at $m = 3$; M_2 sub-ring (heat-kernel period).*
- *$F = \pi^2/6$ — Fock-degeneracy Dirichlet at the packing exponent $d = 3$ (the $\zeta(2)$ value); M_2 sub-ring.*
- *$\Delta = 1/40 = c^2(3, 2)$ — Dirac mode degeneracy g_3^{Dirac}; M_1 sub-ring (Hopf-base counting).*

No morphism in $\mathcal{A}$ produces the combination $K = \pi(B + F - \Delta)$ as an output of one Mellin mechanism, nor as a composition of two or three Mellin mechanisms with the specific weights $(+1, +1, -1)$ and the external π prefactor. Twelve candidate mechanisms have been empirically eliminated for single-principle generation of K (Paper 2 §IV.G; Phases 4B–4I, April 2026; Sprint A α-SP, 2026-04-18; Sprint K-CC, 2026-05-03).

Proof sketch. The non-selection statement follows from a *structural* argument (Facts 1+2); Fact 3 is independent empirical corroboration that does not enter the proof but is recorded as a robustness witness.

Structural argument.

Fact 1 (Master Mellin engine case-exhaustion, Theorem 4). Every transcendental appearing in a GeoVac observable is classified into the M_1, M_2, or M_3 Mellin sub-ring of $\mathcal{M}[\mathrm{Tr}(D^k \cdot e^{-tD^2})]$ at $k \in \{0, 1, 2\}$. The three sub-rings correspond to distinct operator-order $\times$ bundle-type combinations. Under $\mathcal{A}$, the master Mellin engine is the unique mechanism producing transcendental output, and a morphism producing output simultaneously in two distinct sub-rings would require simultaneously evaluating $\mathcal{M}[\mathrm{Tr}(D^k \cdot e^{-tD^2})]$ at two distinct values of k, which is excluded.

Fact 2 (Sub-ring partition of B, F, Δ, Paper 2 §IV + Phases 4B–4I). $B = 42$ is a finite Casimir trace in the representation sector (M_2); $F = \pi^2/6$ is a Fock-degeneracy Dirichlet sum at the packing exponent $d = 3$ (M_2); $\Delta = 1/40$ is the Dirac mode-degeneracy count g_3^{Dirac} (M_1). The combination $B + F - \Delta$ therefore mixes outputs of M_2 ($k = 2$) and M_1 ($k = 0$).

Composition. Under $\mathcal{A}$, by Fact 1 no single morphism produces output in both M_1 and M_2 simultaneously. By Fact 2 the combination $K = \pi(B+F-\Delta)$ mixes M_1 and M_2 output. Therefore no morphism in $\mathcal{A}$ generates K as a single-mechanism output. The non-selection conclusion follows structurally; the specific weights $(+1, +1, -1)$ and the external π prefactor enter the empirical observation but not the non-selection argument.

Empirical corroboration (Fact 3, not load-bearing). Twelve candidate single-principle generators of K have been independently tested and eliminated: Phases 4B–4I (April 2026, nine mechanisms); Sprint A α-SP (2026-04-18, Connes–Chamseddine direct spectral-action derivation); Sprint K-CC (2026-05-03, three further mechanisms via PSLQ at 100 dps against a 244-element basis, a natural Λ search from S^3 spectral data, and a unified CC heat-kernel route blocked by the algebraic obstruction Theorem T9 of Paper 28). These eliminations confirm the structural result empirically but are not used to establish it. □ □

Remark 53 (E6 scope and the observation-label discipline). *The Theorem identifies the structural reason for the empirical 12-mechanism elimination: the three spectral homes live in distinct Mellin sub-rings with no shared generator in $\mathcal{A}$. It does not derive K (which would contradict the empirical eliminations). It does not claim K is mathematically impossible to generate from some extended axiom set; sibling axioms producing K from a single mechanism remain logically possible but none is currently exhibited.*

Per CLAUDE.md §13.5, the Paper 2 combination rule is labeled an Observation *(not a conjecture or derivation), a status protected at the rule level and preserved by this Theorem: the structural impossibility result strengthens the empirical non-derivability, but does not promote K from observation to theorem. Paper 2's status as Observations-folder paper (post 2026-05-02 curve-fit-audit migration) is unchanged.*

Catalogue entry E6 (the combination rule for α^{-1}) is now classified as no-single-mechanism calibration data: the framework provides three structural homes for the ingredients but no single principle for their combination.

Theorem 13 (Cutoff function f and its Mellin moments are external calibration data)**.** *The framework's axiom set $\mathcal{A}$ specifies the Chamseddine–Connes spectral action functional $S(D, \Lambda, f) = \mathrm{Tr}\, f(D/\Lambda)$ on a real spectral triple. The cutoff function f enters as a Schwartz-class test function on $[0, \infty)$ (Paper 51) but is not constrained by any axiom in $\mathcal{A}$. Consequently the Mellin moments*

$$\varphi(s) \;=\; \int_0^\infty x^{s-1} f(x)\, dx$$

are external calibration data: no morphism in $\mathcal{A}$ selects the moment values $\varphi(0), \varphi(1), \varphi(2), \ldots$.

The Wald two-term-exactness $\Rightarrow$ pure Einstein $\Rightarrow$ action-G $\equiv$ entropy-G argument (Paper 51 §G7) forces relations *among the moments via the sector-wise Mellin moment map (tip $\leftrightarrow$ $\varphi(0)$, Einstein–Hilbert $\leftrightarrow$ $\varphi(1)$, $\Lambda_{cc} \leftrightarrow \varphi(2)$); these relations are theorem-grade within $\mathcal{A}$. The moment* values *are not constrained by $\mathcal{A}$.*

Corollary 8 (CC fine-tuning is external moment selection)**.** *The observed cosmological-constant problem in the framework's sector-wise Mellin moment formulation (Paper 51 §G4-5) requires*

$$\Lambda_{cc} \cdot G_{\mathrm{eff}} \;\approx\; 36\pi \cdot \frac{\varphi(2)}{\varphi(1)^2} \;\approx\; 10^{-122}$$

(dimensionless), which is consistent with the framework's structural relations between G_N, Λ_{cc}, and S_{BH} (Wald, theorem-grade) provided $\varphi(2)/\varphi(1)^2 \approx 10^{-124}$ with $\varphi(0), \varphi(1) = O(1)$. No natural cutoff function f produces this fine-tuned moment ratio, and no axiom in $\mathcal{A}$ constrains the moment values to it.

Catalogue entries D5 (cutoff function moments $\varphi(0), \varphi(1), \varphi(2)$) and D6 (cosmological constant value Λ_{cc}) are external calibration data inheriting from the cutoff function selection.

Remark 54 (D5/D6 scope and the sibling-axiom direction)**.** *Theorem 13 formalises the precise location of the cosmological-constant problem in the GeoVac framework: the gravitational* relations *are forced (Wald, Paper 51 §G7); the cutoff-function* moments *are external. The CC problem in this framing is a moment-selection problem on the cutoff function f, structurally similar to the Higgs–Yukawa selection problem (*Theorem *H1) but on a different calibration object.*

A sibling axiom that constrains the cutoff function or its moments (e.g., a renormalisation-group flow axiom, a regulator-independence principle, an asymptotic-safety constraint) would extend $\mathcal{A}$ and could in principle select the moment values. None is currently exhibited; the long-range structural-research direction is named here as the natural analog of the inner-factor sibling-axiom directions.

Paper 57 §6.3 records this theorem as the gravity sector closure of the C-arc.

Remark 55 (Chemistry-side η-trivialization analog: open structural-research target)**.** *The SM-side η-trivialization theorem (Paper 18 `thm:eta_trivialization`) forces the inner-factor period ring to $M_1 \cup M_2$ on any Krajewski-class even finite spectral triple satisfying the chirality grading axiom $\{\gamma_F, D_F\} = 0$: for all odd k, $\mathrm{Tr}(D_F^k \cdot e^{-tD_F^2}) \equiv 0$, so M_3 (vertex-parity Hurwitz / Catalan G / Dirichlet L) vanishes on the inner factor. This is the structural reason F1 (Yukawa values) and F4–F7 (Higgs VEV, CKM, PMNS, neutrino masses) live outside the master Mellin engine's structural skeleton: the inner-factor period ring is forced to a sub-ring categorically disjoint from where the framework's outer-factor mechanism could generate values.*

The chemistry-side analog would be a structural property of the FrozenCore $Z_{\mathrm{eff}}(r)$ pipeline that mirrors $\{\gamma_F, D_F\} = 0$ on the chemistry inner-factor side: a chirality-style grading on the Clementi–Raimondi $Z_{\mathrm{eff}}(r)$ structure that forces a similar period-ring restriction on chemistry inner-factor *data. Such a property is not currently exhibited. The most concrete candidate direction would identify a discrete grading on the FrozenCore radial profile pipeline (e.g., parity of multi-zeta exponents, partition of core orbital basis into chirality-graded blocks) that produces an η-trivialization-style theorem for chemistry inner-factor data.*

Sprint-scale path is unclear: unlike the SM-side η-trivialization, which follows directly from a single axiom ($\{\gamma_F, D_F\} = 0$, established in Krajewski 1998 and canonically adopted in CCM 2010), the chemistry-side analog would need to identify the analog grading axiom from scratch. This is a substantial NCG-research target.

Catalogue entries H6 (Clementi–Raimondi Z_{eff} exponents) and H7 (multi-zeta coefficients in second-row valence basis) remain empirical-only pending this structural-research target's completion. The empirical-grade content (W1e period-class diagnostic, `debug/sprint_w1e_period_class_memo.md`*, 2026-06-04) establishes that H6, H7 are not in $M_1 \cup M_2 \cup M_3$ at audit ceiling 100; the open question is whether they sit in a structurally distinct chemistry-side ring forced by a hypothetical analog axiom.*

Paper 57 §6.3 records this remark as the named substantial structural-research target for the chemistry sector; the C-arc's sprint-scale run closes at v3.103.0 with this target named but not attempted.

Remark 56 (The C-arc terminal state)**.** *After Sprints C2 + C3 + F3 + E7/E8 + G1/G2/G5 + E6 + D5/D6 + Chemistry-Analog (all 2026-06-08, v3.97.0–v3.103.0), the corpus carries* eight *theorem-grade non-selection results across the following sectors:*

Inner-factor sector *(closed at canonical-rep level):*

1. *Forced-Count Theorem (7).*
2. *H1 Yukawa non-selection (§10).*
3. *N_{gen} non-selection (8).*
4. *Inner KO-dimension non-selection (9).*

Multi-focal-composition wall *(fully closed under two sub-sector theorems):*

5. *Single-cutoff spectral action (10).*
6. *Spatial-composition radial wall (11).*

α ***combination rule sector*** *(single-mechanism generation closed):*

7. *No-single-mechanism $K = \pi(B + F - \Delta)$ (12).*

Gravity sector *(cutoff function external-selection closed):*

8. *Cutoff function f and its moments are external (13).*

The chemistry-side η-trivialization analog (Remark 55) is the named substantial structural-research target; catalogue entries H6, H7 remain empirical-only pending its completion.

The remaining empirical-only catalogue entries (F4–F7 Higgs VEV / CKM / PMNS / neutrino mass values, structurally inheriting from F1 and F4; I-entries foundational calibration) are subsidiary residuals inheriting from theorems above (Higgs admission B7 + H1 Yukawa non-selection) or sitting outside the structural-skeleton scope (Born rule via Gleason).

The full forced/free seam (Paper 57) is therefore in a stable terminal state for the path-#1 vein C arc: eight theorem-grade non-selection results across four sectors, one named substantial follow-on (chemistry η-trivialization analog), all subsidiary residuals accounted for.

Files. Scoping memo `debug/g4_cross_manifold_scoping_memo.md` ($\sim 5\,100$ words). G4a sprint memo `debug/sprint_g4a_connes_sm_memo.md`. H1 sprint memo `debug/h1_higgs_inner_fluctuation_memo.md`. Sprint C3 N_{gen} non-selection theorem memo `debug/sprint_c3_n_gen_non_selection_memo.md` (2026-06-08). G4b is recorded as an open structural question.

8.5 Frontier-of-field framing: GeoVac-internal walls vs. spectral-action open problems

The five walls catalogued in the multi-focal-composition audit (`debug/multifocal_phase_a_synthesis_memo.md`, May 2026) are not all GeoVac-specific. Reading them through the spectral-triple construction of Sec. 3 and the Marcolli–van Suijlekom lineage of Sec. 9, three are GeoVac-internal architectural gaps that are tooling-addressable, and two sit at the broader frontier of the spectral-action / NCG programs. We name them here for honesty and to fix the scope of the open questions of Sec. 10.

GeoVac-internal walls (W1a, W1b, W1c).

- **W1a (cross-register coordinate operator):** no two-body spatial operator $V(\hat{\mathbf{r}}_e, \hat{\mathbf{R}}_n)$ has been implemented across the nuclear and electronic registers of the composed nuclear–electronic Hamiltonian (Paper 23 [32] §VI; `geovac/nuclear/nuclear_electronic.py`). The framework couples the registers through spin-spin hyperfine and a classical finite-size shift but evaluates spatial operators at the classical proton position. This is an architectural gap, not a no-go theorem; the calibrated atomic-physics target is the Pachucki–Patkoš–Yerokhin recoil Hamiltonian (PRL 130, 023004, 2023), which is exact in the mass ratio at $(Z\alpha)^6$.
- **W1b (magnetization-distribution operator):** no operator on the proton register represents the Zemach radius r_Z as a Layer-2 magnetization focal length distinct from the charge radius r_p (Paper 34 [38] § III; Sprint HF-4, May 2026). Calibrated against Eides et al. *Phys. Lett. B* (2024). Architectural gap; QCD-internal magnetization density is input data, not GeoVac-internal.
- **W1c (cross-center frozen-core screening):** the `FrozenCore` $Z_{\text{eff}}(r)$ provides same-center screening only; cross-center screening of bare V_{ne} from the opposite nucleus is not implemented (Sprint 7 NaH/MgH_2 balanced FCI; CLAUDE.md §2). Sprint 7b made partial progress for the spin-orbit splitting; the PES extension is mechanical once cross-center screening is wired through the multipole expansion. Architectural gap.

These three walls are tooling-addressable inside the present framework: the missing operators exist in principle, are calibrated against published atomic-physics targets, and require no new mathematics in the broader NCG literature. Each can be closed by a sprint at bounded scale. Their existence does not undermine the spectral-triple framing of Sec. 3; they catalogue what has not yet been built rather than what cannot be.

Frontier-of-field walls (W2a, W2b).

- **W2a (multi-loop UV/IR composition):** the bare iterated Connes–Chamseddine spectral action on the Camporesi–Higuchi spectrum reproduces UV-divergent integrands of multi-loop QED (Sprint LS-8a, May 2026; Paper 36 [40] §VII) but cannot autonomously generate Z_2, δm, Z_3 counterterms for finite extraction at two loops. This is not GeoVac-specific. The Marcolli–van Suijlekom 2014 rationality theorem for Robertson–Walker spectral actions [5] partially addresses spectral-action coefficient structure on homogeneous compact backgrounds, and the subsequent Hekkelman–McDonald 2024 noncommutative integral approximation [13] provides a single-cutoff Szegő-style asymptotic theory in the Connes–van Suijlekom paradigm; neither closes the multi-cutoff matching question that renormalization counterterm generation requires. The published QFT archetype is the matched effective-field-theory chain (e.g. HQET $\to$ NRQED $\to$ pNRQED $\to$ SCET, JHEP 05 (2025) 171), where each matching step is at a single physical scale and imports $\overline{MS}$ counterterms by hand. No published spectral-action framework reproduces this autonomously. W2a is therefore an open question of the spectral-action program, of which GeoVac is one specific instance; it is not a defect of the present construction relative to the lineage.

- **W2b (cross-manifold spectral-triple composition):** the tensor product of two infinite metric spectral triples $(\mathcal{T}_{S^3}, d_{D_{S^3}}) \otimes (\mathcal{T}_{S^5}, d_{D_{S^5}})$, needed to compose the GeoVac Coulomb sector (Sec. 3) with the Bargmann–Segal HO sector of Paper 24 [33], has no published Latrémolière-propinquity convergence theorem. The almost-commutative case (one infinite Riemannian factor + one finite factor) is closed by Latrémolière 2026 [16] (*Spectral continuity of almost commutative manifolds for the C^1 topology on Riemannian metrics*); the two-infinite case is explicitly not handled in that paper, nor in the closely related Farsi–Latrémolière 2024 inductive-sequence work (Adv. Math. 437, 109442). This is the structural content of the G4b cross-manifold blocker (Sec. 8.4): not a GeoVac-specific limitation but a genuinely open NCG question. Closing it would itself be a substantial mathematical contribution, parallel to but separate from Paper 38's SU(2) state-space GH convergence theorem.

Forecast (without claiming closure). We do not claim W1a/W1c can be closed without surprises; the multi-λ Shibuya–Wulfman algebraic check that gates the W1a attack (multipole termination across mismatched exponents) is not in the published Avery-school literature and could fail to preserve Gaunt-style closure (Track 3 literature memo, May 2026, §1). W1b is bounded by external QCD input. W2a and W2b are open in the broader literature, not just in the GeoVac record. The honest forecast is that closing W1a/b/c is internal-effort plausible as a substantial program per wall; closing W2a or W2b would be a contribution to the spectral-action / NCG programs rather than to GeoVac alone, and we do not commit to it as a deliverable here.

Implication for the Marcolli–vS lineage placement. Sec. 9 read the present construction as a graph spectral triple in the lineage of [5, 9, 10]. The frontier-of-field framing above sharpens this: the lineage placement is correct, and the W2a/W2b walls are the lineage's own open problems, not GeoVac's separate concerns. This is consistent with Observation 3 (each structural ingredient has published precedent; what is not duplicated is the α prediction at 8.8×10^{-8}). The W1 walls remain GeoVac-side because they are about the architectural extension of the present triple to multi-particle real-space coupling, which the SM-style almost-commutative construction does not address.

8.6 Lorentzian transfer audit and the master Mellin engine under signature extension (Sprint L0, May 2026)

The case-exhaustion theorem (Theorem 4) classifies every π in a finite chain of Paper 34 projections as engaging one of three sub-mechanisms of a master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k\, e^{-tD^2})]$ with $k \in \{0, 1, 2\}$: M1 (Hopf-base measure), M2 (Seeley–DeWitt heat-kernel coefficients), M3 (vertex-parity Hurwitz / Dirichlet-L content). The GH-convergence theorem (Theorem 5) and the Bisognano–Wichmann reading (Remark 42, with Unruh pendant) close WH1 on the Riemannian side. We close out Sec. 8 with a brief structural appendix that records what Sprint L0 (May 2026) found when the audit was extended to the Riemannian $(3,0) \to$ Lorentzian $(3,1)$ extension of the framework.

Sprint L0 verdict on the master Mellin engine. Sprint L0 (`debug/lorentzian_l0_audit_memo.md`, May 2026; machine-readable table at `debug/data/lorentzian_l0_audit.json`) classifies all twenty-eight entries of Paper 34's projection dictionary [38] by their transfer behavior under signature extension; the full per-projection table sits at Paper 34 § V.E (`sec:lorentzian_transfer_audit`), the structural framing of the Sig/Op partition is in Paper 31 § 8 (`sec:signature_partition`) [37]. Two sharpenings of the master Mellin engine fall out of the L0 audit.

(i) M1 sub-mechanism is the load-bearing free-transfer engine. The M1 sub-mechanism ($k = 0$) on a temporal S^1_β or proper-time Rindler η_E-circle inherits a Lorentzian reading via the four-witness Wick-rotation theorem (Hawking + Sewell + BW + Unruh, codified Sprint Unruh-pendant 2026-05-10; Remark 42 and its Unruh-pendant extension). Four entries of Paper 34's dictionary sit in the WICK-MAP FREE bucket via this mechanism: observation / temporal-window (§ III.15), vector-photon promotion (§ III.11) at the $1/(4\pi)$ per-loop level, gauge choice (§ III.26) at the Coulomb-gauge per-loop level, and the Wick-rotation projection itself (§ III.27). None requires Krein-space machinery for its metric-functional-level reading. Equivalently: the $2\pi = \mathrm{Vol}(S^1)$ signature of M1 in the master Mellin engine is also the modular-flow / Lorentz-boost orbit period via published QFT, with the same generator on both sides of the Wick rotation.

(ii) M3 sub-mechanism predicted to trivialize at $(3,1)$. A non-trivial structural prediction emerges from the Bizi–Brouder–Besnard [48] signature classification: at $(m,n) = (3,1)$, the BBB sign table flips the $\{J, \gamma_5\}$ anticommutation relation that drives M3's vertex-parity content on the Riemannian side (Paper 28's $\zeta(s, 3/4)$ and $\zeta(s, 5/4)$ Hurwitz reductions producing Catalan's G and $\beta(4)$). On chirality-symmetric spectrum at $(3,1)$, the vertex-parity asymmetry vanishes, M3's half-integer-Hurwitz content becomes structurally empty, and the case-exhaustion theorem (Theorem 4) trivializes to a *two-mechanism partition* M1 $\cup$ M2 on the Lorentzian side. This is not a contradiction with the Riemannian theorem: the theorem is stated at signature $(3,0)$ and the case-exhaustion is over mechanisms producing π in finite chains *at that signature*. The $(3,1)$ trivialization is a structural sharpening that becomes a Sprint L2 consistency check (a substantial program, gated on blocker B2; see Paper 31 § 8.5).

What this does and does not change. The Riemannian case-exhaustion theorem stands; WH1 PROVEN (Theorem 5) stands. The L0 audit is a documentation-level extension that classifies the framework's transfer behavior under signature change without re-deriving any of the Riemannian results. Sprint L1 (beyond sprint scale, MEDIUM-HIGH) would deepen the Wick-rotation reading from structural correspondence to literal operator-system identification by computing the modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$ for a Rindler-wedge-restricted Camporesi–Higuchi state, verifying $\sigma_{i\cdot 2\pi} =$ identity at finite $n_{\max}$, and taking the GH limit using Theorem 5's lemmas. That closure would lift the correspondence on all four Wick-rotation faces simultaneously via the shared KMS-Tomita-Takesaki algebra. The L0 audit names this as the principal Sprint L1 falsifier without itself closing it.

The Nieuviarts 2025 [50] twisted-spectral-triple morphism is the named trigger that motivated reopening the May 2026 NO-GO scoping; a parallel L2-Nieuviarts-scoping pass tests whether the construction applies cleanly to $S^3 = \mathrm{SU}(2)$ (odd-dimension caveat) and would shorten blocker B1 (Lorentzian propinquity, long-range) substantially if applicable. The current audit does *not* assume the morphism applies.

8.7 Krein space construction at signature $(3,1)$ (Sprint L2-B, 2026-05-16)

The first concrete deliverable of the Sprint L2 arc lifts the Hilbert space and the fundamental symmetry of the truncated triple from signature $(3,0)$ to signature $(3,1)$, following the van den Dungen 2016 prescription [49]. The construction is at the *Hilbert-space level only*; the Lorentzian Dirac and the Connes axiom audit are deferred to L2-C and L2-D below.

Krein space. For $n_{\max} \geq 1$ and $N_t \geq 1$ define

$$\mathcal{K}_{n_{\max},N_t} \;:=\; \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes L^2(\mathbb{R}_t)_{\mathrm{cutoff},N_t}, \tag{8}$$

where $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ is the Riemannian Hilbert space of Definition 2 and $L^2(\mathbb{R}_t)_{\mathrm{cutoff},N_t}$ is the truncated temporal slot: a uniform real grid $t_k = -T_{\max} + k \cdot 2T_{\max}/(N_t - 1)$ for $N_t \geq 2$, or the singleton $\{0\}$ for $N_t = 1$. The temporal slot is NOT compactified to S^1_β (which would create closed timelike curves per Geroch's theorem and is structurally incompatible with the bounded-wedge reading of Paper 42).

Conventions. We use the West-coast metric $\eta = \mathrm{diag}(+1,-1,-1,-1)$ and the Peskin–Schroeder chiral (Weyl) basis for the four $\mathrm{Cl}(3,1)$ gamma matrices. In this basis, $\gamma^5 = \mathrm{diag}(-I_2, +I_2)$ is diagonal and `FullDiracLabel.chirality` is identified with the γ^5 eigenvalue, so the Riemannian limit identification with $\mathcal{H}_{\mathrm{GV}}$ is a transparent basis-label match (no similarity transform required). The Dirac basis alternative (where γ^0 is diagonal instead) is unitarily equivalent at the operator level but would have required a unitary similarity transform for the Riemannian-limit check; the chiral basis was chosen on Riemannian-limit-clarity grounds.

Fundamental symmetry J. The fundamental symmetry is

$$J \;=\; J_{\mathrm{spatial}} \otimes I_{N_t}, \tag{9}$$

where J_{spatial} is the lift of γ^0 to the chirality-doubled $\mathcal{H}_{\mathrm{GV}}$: on $|n_D, \kappa, m_j, \chi\rangle$ the action is the chirality swap $|n_D, \kappa, m_j, -\chi\rangle$. In the chiral basis, $\gamma^0 = \left(\begin{smallmatrix} 0 & I_2 \\ I_2 & 0 \end{smallmatrix}\right)$ is the off-diagonal block-swap, so J_{spatial} is a real $\{0,1\}$-valued permutation matrix. The Krein inner product is $\langle \psi, \phi \rangle_{\mathcal{K}} := \langle \psi, J\phi \rangle$.

Krein axiom verification. At every panel cell $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$, the construction satisfies

1. $J^2 = +I$ (bit-exact Frobenius residual $= 0.0$),
2. $J^* J = I$ (J Hilbert-unitary, bit-exact),
3. $J^* = J$ (J Hilbert-Hermitian, bit-exact),
4. $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$ with $\dim \mathcal{K}^\pm = \dim \mathcal{K}/2$ and $J|_{\mathcal{K}^\pm} = \pm I$ on the eigenspaces (rank $\dim \mathcal{K}/2$ each, bit-exact).

The bit-exact-zero residuals follow from the structural fact that $J = J_{\mathrm{spatial}} \otimes I_{N_t}$ is the Kronecker product of a $\{0,1\}$-valued integer permutation matrix and an identity matrix, both involving only operations exact in IEEE 754 arithmetic on real integers.

Load-bearing Riemannian-limit check. At $N_t = 1$ the temporal slot collapses to the singleton $\{0\}$ and $\mathcal{K}_{n_{\max},N_t=1}$ recovers $\mathcal{H}_{\rm GV}^{n_{\max}}$ *bit-identically* at every $n_{\max} \in \{1,2,3\}$: the dimension formula $\dim \mathcal{K}_{n_{\max},1} = N_{\rm Dirac}(n_{\max}) = \frac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2) = 4, 16, 40$ matches the Paper 32 § III formula exactly, the basis labels are bit-identical under `FullDiracLabel.__eq__`, and the J matrix at $N_t = 1$ equals $J_{\rm spatial}$ as numpy arrays (Frobenius difference = 0.0). This confirms that the Camporesi–Higuchi spatial spinor bundle is structurally compatible with the $\mathrm{Cl}(3,1)$ gamma-matrix embedding in the West-coast chiral basis; WH1 PROVEN (Theorem 5) is not re-opened by this result. Source: `debug/l2_b_krein_construction_memo.md`, raw data `debug/data/l2_b_krein_construction.json`, implementation `geovac/krein_space_construction.py`, regression tests `tests/test_krein_space_construction.py` (109 cases, all pass; zero regression in 132 upstream tests).

Honest scope. The Krein-space construction lifts the Hilbert space and the fundamental symmetry only. The Lorentzian Dirac, the algebra $\mathcal{A}_L$ on $\mathcal{K}$, the operator-algebra Krein-adjoint structure, and the modular Hamiltonian on the Krein space are deferred to L2-C, L2-D, and L2-E respectively.

8.8 Lorentzian Dirac operator D_L on $S^3 \times \mathbb{R}$ (Sprint L2-C, 2026-05-16)

Sprint L2-C constructs the Lorentzian Dirac operator on the Krein space (8) via the van den Dungen Proposition 4.1 prescription [49]: Wick-rotate the Lorentzian metric g on $M = S^3 \times \mathbb{R}$ to its Riemannian counterpart g_r, construct the standard Riemannian Dirac $\not{D}_{g_r}$ on (M, g_r), and define the Krein-self-adjoint Lorentzian Dirac as $D_L = i^t\, \not{D}_{g_r}$ with $t = 1$ for signature $(s,t) = (3,1)$.

Construction. The Wick-rotated Riemannian Dirac decomposes via spatial-temporal splitting:

$$\not{D}_{g_r} \;=\; \gamma^0 \otimes \partial_t + D_{\rm GV} \otimes I_{N_t}, \tag{10}$$

with γ^0 from the Sprint L2-B chiral basis, $D_{\rm GV}$ the Paper 32 § III Camporesi–Higuchi truthful full-Dirac matrix on $\mathcal{H}_{\rm GV}^{n_{\max}}$, and ∂_t the temporal-derivative matrix on the $L^2(\mathbb{R}_t)_{{\rm cutoff},N_t}$ grid. The Lorentzian Dirac is then

$$D_L \;=\; i \cdot \left[\gamma^0 \otimes \partial_t + D_{\rm GV} \otimes I_{N_t}\right] \;\in\; \mathbb{C}^{\dim \mathcal{K} \times \dim \mathcal{K}}. \tag{11}$$

Temporal derivative. The temporal-derivative ∂_t uses centered finite differences with Dirichlet (zero) boundary conditions:

$$(\partial_t)_{k,l} \;=\; \frac{1}{2\Delta_t}\bigl(\delta_{l,k+1} - \delta_{l,k-1}\bigr), \qquad \Delta_t = \frac{2T_{\max}}{N_t - 1}. \tag{12}$$

This choice is *structurally forced*, not a free parameter: the centered-FD plus Dirichlet-zero combination is the unique discretization on a finite grid that gives anti-Hermitian ∂_t ($\partial_t^\dagger = -\partial_t$), which is the property required for Krein-self-adjointness $D_L^\times = D_L$ to close via the vdD recipe. Forward, backward, and upwind schemes give $\partial_t^\dagger \neq -\partial_t$ (breaking Krein-self-adjointness); periodic boundary conditions would re-introduce closed timelike curves (Sprint L2-A audit § 3.8).

Sign of i^t. With $t = 1$, the prescription gives $i^t = i$ (not $-i$). The sign is derived from the structural requirement $D_L^\times = D_L$ with $D_L^\times := \gamma^0 D_L^\dagger \gamma^0$, using $\{\gamma^0, D_{\rm GV}\} = 0$ (the chirality swap γ^0 anticommutes with the chirality-diagonal $D_{\rm GV}$) and $\partial_t^\dagger = -\partial_t$. Explicit symbolic calculation (sympy verification) gives $D_L^\times = +D_L$ for $i^t = +i$ and $D_L^\times = -D_L$ for $i^t = -i$.

Verification at finite cutoff. The construction verifies at every panel cell $(n_{\max}, N_t) \in \{1,2,3\}\times\{1,11,21\}$:

1. *Krein-self-adjointness.* $\|D_L^{\times} - D_L\|_F = 0$ bit-exact at every cell.
2. *Real spectrum on Krein-positive cone.* Restricting D_L to $\mathcal{K}^+$ (the $+1$ eigenspace of J) gives real eigenvalues to machine precision ($\max|\mathrm{Im}(\lambda)| \leq 2.32\times 10^{-17}$).
3. *Riemannian-limit recovery (LOAD-BEARING).* At $N_t = 1$, $D_L|_{N_t=1} = i\cdot D_{\mathrm{GV}}$ bit-identically. The absolute-value spectrum $\{|\lambda_n| = n_{\mathrm{Fock}} + 1/2\}$ with multiplicities $g_n^{\mathrm{Dirac}} = 2n(n+1)$ matches D_{GV} bit-exactly at $n_{\max} = 1,2,3$.

Chirality finding (non-zero $\{\gamma^5, D_L\}$). A non-trivial structural finding emerges: the BBB universal relation $\chi D = -D\chi$ (Sec 5(v) of [48]) does *not* hold on truthful D_{GV}. The anticommutator $\{\gamma^5, D_L\}$ has Frobenius norm $2\|\gamma^5 D_{\mathrm{GV}}\|_F = 2\|D_{\mathrm{GV}}\|_F$ at $N_t = 1$ (values $6.00, 18.33, 38.88$ at $n_{\max} = 1,2,3$), because GeoVac's truthful D_{GV} is chirality-diagonal in the Peskin–Schroeder chiral basis and therefore commutes with γ^5. This finding is structural, not a bug: it is a consequence of two locked Riemannian-side conventions (chirality-as-γ^5 from L2-B and chirality-diagonal D_{GV} from Paper 32 § III). The full diagnostic and the three resolutions are documented in Sec. 8.9 below.

Sources. Implementation: `geovac/lorentzian_dirac.py`. Regression tests: `tests/test_lorentzian_dirac.py` (108 cases, all pass; zero regression in 205 upstream tests). Sprint memo: `debug/l2_c_lorentzian_dirac_memo.md`; raw data `debug/data/l2_c_lorentzian_dirac.json`.

8.9 Connes axiom audit at $(m,n) = (4,6)$ (Sprint L2-D, 2026-05-16)

Sprint L2-D performs the Connes axiom audit on the pair (D_L, γ^5, J_L) built in Sprints L2-B/C, against the Bizi–Brouder–Besnard 2018 [48] classification at $(m,n) = (4,6)$.

BBB sign determination. Reading the BBB Table 1 entries at $(m,n) = (4,6)$ directly from arXiv:1611.07062 v2: $\varepsilon = +1$, $\varepsilon'' = -1$, $\kappa = -1$, $\kappa'' = +1$. These four primitive signs, combined with the four BBB defining relations $J^2 = \varepsilon$, $J\chi = \varepsilon''\chi J$, $J\eta = \varepsilon\kappa\,\eta J$, $\eta\chi = \varepsilon''\kappa''\,\chi\eta$, predict at $(4,6)$: $J^2 = +I$, $\{J,\chi\} = 0$, $\{J,\eta\} = 0$, $\{\eta,\chi\} = 0$. At the bare $\mathrm{Cl}(3,1)$ 4-spinor level with charge conjugation $U_4 = i\gamma^2$, all four relations hold bit-exact in the chiral basis (Frobenius residual $= 0.0$). The translation $(s,t) = (3,1) \leftrightarrow (m,n) = (4,6)$ uses BBB Table 3: $m \equiv t+s$, $n \equiv t-s \pmod 8$, giving $m = 4$, $n = -2 \equiv 6 \pmod 8$.

Lift to the Krein space. The 4-spinor charge conjugation $i\gamma^2 = (i\sigma_y)_{\mathrm{chir}} \otimes (i\sigma^2)_{\mathrm{spin}}$ decomposes into a chirality-swap-with-sign factor times the standard spin-1/2 charge conjugation. The lift to $\mathcal{K}_{n_{\max},N_t}$ is

$$J_L \;=\; J_{L,\mathrm{spatial}} \otimes K_t, \tag{13}$$

where $J_{L,\mathrm{spatial}}$ acts on $|n,l,m_j,\chi\rangle$ as $\sigma_{\mathrm{chir}}(\chi)\cdot\sigma_{\mathrm{spin}}(l,-m_j)\cdot|n,l,-m_j,-\chi\rangle$ with $\sigma_{\mathrm{chir}}(\pm 1) = \mp 1$ and $\sigma_{\mathrm{spin}}(l,m_j) = (-1)^{(2l+1-2m_j)/2}$, and K_t is complex conjugation on $\mathbb{C}^{N_t}$. The matrix $U_L := J_L \cdot K^{-1}$ is a real $\{0,\pm 1\}$-valued unitary; the antilinear $J_L = U_L K$ satisfies $J_L^2 = +I$ bit-exact via $U_L\overline{U_L} = U_L^2$ in IEEE 754.

Six-axiom panel. At every panel cell $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$ on truthful D_{GV}:

Axiom	Predicted at $(4,6)$	Residual
(i) $J_L^2 = +I$ (BBB $\varepsilon = +1$)	$+I$	0.0 bit-exact
(ii) $\{J_L, \chi\} = 0$ (BBB $\varepsilon'' = -1$)	0	0.0 bit-exact
(iii) $\{J_L, \eta\} = 0$ (BBB $\varepsilon\kappa = -1$)	0	0.0 bit-exact
(iv) $J_L D_L = +D_L J_L$ (BBB universal Sec 5(v))	$JD = +DJ$	0.0 bit-exact
(v) $\{\chi, D_L\} = 0$ (BBB universal Sec 5(v))	$\chi D = -D\chi$	STRUCTURAL FINDING (non-zero)
(vi) order-zero $[a, J_L b J_L^{-1}] = 0$	0	≤ 0.0675 finite-resolution (sample-of-3)
(vii) order-one $[[D_L, a], J_L b J_L^{-1}] = 0$	0	≤ 0.101 finite-resolution (sample-of-3)

Axioms (i)–(iv) hold bit-exactly at every cell. Axioms (vi)–(vii) exhibit the same 5–10% finite-resolution residuals as the Riemannian-side audit (Table 1, "Order zero $[a, JbJ^{-1}] = 0$"); the residuals are sample-of-3 estimates on the canonical $\mathcal{A}_{\mathrm{GV}}$ multiplier basis and match the structural-artifact reading of Connes–van Suijlekom operator-system multiplicative-closure failure. Axiom (v) is the structural finding documented in Table 2 and discussed below.

Structural finding on $\{\chi, D_L\}$. The BBB universal axiom $\chi D = -D\chi$ fails on truthful D_{GV}: $\|\{\gamma^5_K, D_L\}\|_F = 2\|D_{\mathrm{GV}}\|_F$ at $N_t = 1$, scaling as $\|D_{\mathrm{GV}}\|_F\sqrt{N_t}\cdot(2+\text{small})$ at $N_t > 1$. The mechanism is the chirality-diagonal structure of D_{GV} in the chiral basis (§ 8.8). This is the load-bearing scope finding of Sprint L2-D and is the same structural fact captured in Table 2. The three resolutions (R1 accept, R2 use offdiag CH, R3 redefine γ^5) are documented in the discussion immediately following Table 2, with R1 + R2 in parallel as the recommended path for Sprint L2-E.

M3 trivialization is convention-dependent. Sprint L0 (§ 8.6 above) predicted that the M3 sub-mechanism of the master Mellin engine would *trivialize at* $(3,1)$ *on chirality-symmetric spectrum*, via the BBB sign-flip $\{J, \gamma^5\} = 0$ removing the asymmetry that drives M3's vertex-parity content. Sprint L2-D tests this prediction and finds: **the prediction is convention-dependent**.

Under the n_{Fock}*-parity convention* (the Paper 28 §QED-vertex convention that accesses Catalan G and Dirichlet $\beta(4)$ via quarter-integer Hurwitz shifts), the vertex-parity sum $D_{\mathrm{even}}(4) - D_{\mathrm{odd}}(4)$ on the chirality-symmetric $(3,1)$ truncation is bit-identical to the Riemannian value. Each full-Dirac level n_{Fock} contributes its full degeneracy $g_n = 2n(n+1)$ summed over both chirality blocks; the parity is read from the chirality-independent n_{Fock} label. The L0 prediction is therefore **FALSIFIED** under this reading at every $n_{\max} \in \{1,\ldots,5\}$ (bit-identical difference).

Under the *chirality-pairing convention* (pair "even" with $\chi = +1$ Weyl and "odd" with $\chi = -1$ anti-Weyl), the chirality-symmetric truncation gives $D_+ = D_-$ identically by chirality symmetry, so the difference vanishes bit-exact. The L0 prediction is **trivially confirmed** under this reading, but tautologically (any chirality-symmetric truncation gives the result by symmetry alone, independent of signature).

The structural refinement is therefore: **the BBB sign-flip $\{J, \gamma^5\} = 0$ at $(m,n) = (4,6)$ is a spectral-triple-axiom statement, not a vertex-parity-sum statement.** It does not directly imply M3 trivialization at any natural parity convention. M3's mechanism is sectional in n_{Fock}-parity, NOT in spacetime signature. Paper 18 § III.7 records the corresponding master-Mellin-engine sharpening.

Sources. Implementation: `geovac/connes_axiom_audit_31.py` (700 lines incl. docstrings). Regression tests: `tests/test_connes_axiom_audit_31.py` (75 cases, all pass). Sprint memo: `debug/l2_d_connes_ax` raw data `debug/data/l2_d_connes_axiom_audit_31.json` (BBB-Table-1 signs, 4-spinor verification, full axiom panel, M3 trivialization panel).

Sprint L2-E handoff. The four BBB-predicted-sign axioms hold bit-exactly, so the operator-algebra structure on the Krein space is fully BBB-compatible at the J-relation level. Sprint L2-E (the Lorentzian-level Paper 42 redo) should construct the Krein-side modular Hamiltonian $K_L = -\log \Delta_L$ on the Krein-positive cone $\mathcal{K}^+$ and verify the analog of Paper 42's $\sigma_{2\pi}(O) = O$ at the operator-system level on the Krein space. The resolution choice R1 (truthful D_{GV}, preserves Riemannian-limit recovery) versus R2 (offdiag CH, satisfies BBB axiom (v)) is recommended to be tested in parallel. WH1 PROVEN (Theorem 5) is not re-opened by these results.

8.10 Krein-level unified-strong four-witness theorem at finite cutoff (Sprint L2-E, 2026-05-17)

Sprint L2-E (`geovac/modular_hamiltonian_lorentzian.py`, $\sim$ 860 lines including docstrings; tests in `tests/test_modular_hamiltonian_lorentzian.py`, 74 fast plus 3 slow, all passing) builds the operator-system-level Krein-side modular Hamiltonian on the hemispheric wedge of $S^3 \times \mathbb{R}$ at signature $(3,1)$ and verifies the four LOAD-BEARING falsifiers of the Sprint L2-F catalogue (`debug/sprint_l2_falsifiers.md` § 3) plus the Riemannian-limit recovery LOAD-BEARING falsifier plus the headline H_{local} verdict and the six-witness collapse, all bit-exact at every $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$. The closure mirrors Paper 42 §§ 4–8 [51] verbatim with the temporal slot added via the Sprint L2-B Krein space.

Lorentzian wedge, KMS state, and BW choice. The hemispheric wedge on the Krein space is $W_L = P_W^{\mathrm{spatial}} \otimes P_{t\geq 0}$, where $P_W^{\mathrm{spatial}} = (1/2)(I + R_{\mathrm{polar}})$ is the Paper 42 hemispheric wedge on $\mathcal{H}_{\mathrm{GV}}$ (the m_j-reflection involution) and $P_{t\geq 0}$ is the $t \geq 0$ half-line projector on $\mathbb{C}^{N_t}$. The wedge KMS state under the BW choice $H_{\mathrm{local}} := K_L^{\alpha,W}/\beta$ at $\beta = 2\pi$ is

$$\rho_W^L = \frac{e^{-K_L^{\alpha,W}}}{Z},$$

β-independent at the algebra-action level, with $K_L^{\alpha,W}$ the wedge-restricted BW-α generator inheriting integer eigenvalues two_m_j from Paper 42 Definition 5.1.

The Krein-level four-witness theorem at finite cutoff. The Lorentzian-side BW-α (geometric) and BW-γ (Tomita–Takesaki on the Krein-GNS Hilbert–Schmidt space $M_{\dim W_L}(\mathbb{C})$) constructions both verify $\sigma_{2\pi}^L(O) = O$ bit-exactly at every panel cell:

$(n_{\max}, N_t)$	$\dim K$	$\dim W_L$	$r_W^{L,\alpha}$	$r_W^{L,\mathrm{TT}}$
$(1,1)$	4	2	0.0	$\leq 10^{-16}$
$(1,11)$	44	12	$\leq 10^{-16}$	$\leq 10^{-16}$
$(1,21)$	84	22	$\leq 2\times 10^{-16}$	$\leq 2\times 10^{-16}$
$(2,1)$	16	8	$\leq 10^{-16}$	$\leq 10^{-16}$
$(2,11)$	176	48	$\leq 3\times 10^{-16}$	$\leq 3\times 10^{-16}$
$(2,21)$	336	88	$\leq 4\times 10^{-16}$	$\leq 4\times 10^{-16}$
$(3,1)$	40	20	$\leq 5\times 10^{-16}$	$\leq 5\times 10^{-16}$
$(3,11)$	440	120	$\leq 4\times 10^{-15}$	$\leq 4\times 10^{-15}$
$(3,21)$	840	220	$\leq 7\times 10^{-15}$	$\leq 7\times 10^{-15}$

Verdict at every cell: STRONG_IDENTIFICATION_LORENTZIAN. Residuals scale as $O(\sqrt{\dim}\cdot\varepsilon_{\text{machine}})$ — pure round-off, no structural drift. Flow conjugacy $\sigma_t^{L,\mathrm{TT}}(O) = \sigma_{-t}^{L,\alpha}(O)$ at general $t \in \{1, \pi, 2\pi\}$ holds bit-exactly with the same residual bound, and the six-witness cross-consistency residual is exactly 0.0 at every cell.

Theorem (Krein-level four-witness Wick-rotation theorem at finite cutoff). *On the Lorentzian Krein space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$, with the wedge $W_L = P_W^{\text{spatial}} \otimes P_{t\geq 0}$, the wedge KMS state $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$, and the BW unit normalisation $\kappa_g = 1$, the four-witness Wick-rotation theorem (Hartle–Hawking + Sewell + Bisognano–Wichmann + Unruh) closes at the operator-system level via:*

- *BW-α period closure:* $\sigma_{2\pi}^{L,\alpha}(O) = O$ bit-exactly for all $O \in B(\mathcal{K}_{W_L})$.
- *BW-γ Tomita period closure:* $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$ bit-exactly for all $a \in B(\mathcal{K}_{W_L})$ on the Krein-GNS space.
- *Flow conjugacy:* $\sigma_t^{L,\mathrm{TT}}(a) = \sigma_{-t}^{L,\alpha}(a)$ bit-exactly at general t.
- *Six-witness collapse:* all six instantiations $\{\mathrm{BW}, \mathrm{HH}_{M=1}, \mathrm{HH}_{M=2}, \mathrm{Sew}_{M=1}, \mathrm{Unruh}_{a=1}, \mathrm{Unruh}_{a=2}\}$ give bit-identical Δ_L and K_L^{TT}.

All four statements verified bit-exact at every tested $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$. Proofs mirror Paper 42 Theorems 5.4, 6.3, 7.1, and Cor. 8.1 [51].

The structural reading is clean: the same M1 mechanism (Hopf-base measure $\mathrm{Vol}(S^1)/2\pi$, master Mellin engine $k=0$ sub-mechanism per the case-exhaustion theorem of § VIII) that produces the Riemannian-side period closure on the Wightman-BW vacuum analog ρ_W produces the Lorentzian-side period closure on ρ_W^L at $(3,1)$. The Wick-rotation theorem (Hawking + Sewell + BW + Unruh) **lifts from structural correspondence at the metric-functional level (Sprint TD Track 4, Unruh-pendant) to literal identification at the operator-system level (Lorentzian, finite cutoff)** at every Krein cutoff.

Riemannian-limit recovery (load-bearing falsifier). At $N_t = 1$, $P_{W_L} = P_W^{\text{spatial}}$, $K_L^{\alpha,W} = K_\alpha^W$, $\rho_W^L = \rho_W$, $\Delta_L = \Delta$, and $K_L^{\mathrm{TT}} = K_{\mathrm{TT}}$ bit-identically (Frobenius residual exactly 0.0, not merely machine-precision) at every $n_{\max} \in \{1,2,3\}$. This is the load-bearing falsifier that ensures Sprint L2-E is the Lorentzian *extension* of Paper 42 § 6 (the Tomita BW-γ construction) [51], not a parallel-but-different construction.

H_{local} **signature-independence finding (headline structural finding of Sprint L2-E).** Paper 42 § 7.2 / open question O3 names a load-bearing scope finding: the framework's intrinsic Dirac D_W is NOT the right local Hamiltonian for the BW vacuum at $\beta = 2\pi$; $H_{\mathrm{local}} := K_\alpha^W/\beta$ is. Sprint L2-E tests whether this finding holds at signature $(3,1)$. **Result:** at the Riemannian limit ($N_t = 1$) on the Lorentzian Krein space, $\|H_{\mathrm{local}}|_{(3,1),N_t=1} - D_L^W|_{N_t=1}\|_F$ EQUALS the Riemannian-side residual bit-exact at every tested $n_{\max} \in \{1,2,3\}$ (values 2.1332, 6.5275, 13.854). At $N_t > 1$ the Lorentzian-side residual is *refined upward* by the temporal-derivative content of $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I)$. The **Paper 42 § 7.2 finding is signature-INDEPENDENT at the Riemannian limit**. The spectral-action–vs–modular-Hamiltonian generator distinction is therefore NOT a peculiarity of the round S^3 Riemannian truncation; it is a deeper structural feature of the framework's modular content. Paper 42 § 10 O3 is refined from "open question about signature-dependence" to "signature-independent structural distinction at the Riemannian limit, refined upward by temporal-derivative content at $N_t > 1$" [51].

Honest scope. Closure is at finite $n_{\max}$ and finite N_t, NOT in the continuum (GH-limit / Lorentzian propinquity) limit. No published Lorentzian propinquity construction exists as of May 2026 (Latrémolière 2017/2026 and Hekkelman–McDonald 2024 are strictly Riemannian). Constructing a Lorentzian propinquity is the named Sprint L3 target. The Sprint L2-D structural finding (BBB universal $\chi D = -D\chi$ fails on truthful D_{GV}) is preserved in L2-E: the wedge construction is $K_L^{\alpha,W}$-driven and does not invoke this axiom. Cross-manifold W2b ($\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$) remains **open** as a separate frontier (see § VIII.D "frontier-of-field framing"), structurally distinct from the $(3,1)$ extension within $\mathcal{T}_{S^3}$ closed here.

Sources. Memo: `debug/l2_e_modular_hamiltonian_lorentzian_memo.md`. Computational driver: `debug/l2_e_modular_hamiltonian_lorentzian_compute.py`. Raw data: `debug/data/l2_e_modular_hamiltoni` Implementation: `geovac/modular_hamiltonian_lorentzian.py`. Tests: `tests/test_modular_hamiltonian_lor` (74 fast + 3 slow, all pass; zero regression in 355 upstream tests). The math.OA-style standalone writeup of the Lorentzian extension is outlined in Paper 43 (`papers/group1_operator_algebras/paper_43_loren` companion document); when drafted, Paper 43 sits alongside Papers 38, 39, 40, 42 as the fourth math.OA paper in the series and does not supersede Paper 42 — Paper 42 is the Riemannian closure, Paper 43 is the Lorentzian extension.

8.11 Modular Hamiltonian at finite $n_{\max}$: Sprint L1 closure on the Riemannian side (2026-05-16)

The Sprint L1 falsifier named in Remark 42 and § 8.6 is closed in this subsection at the strongest possible level: the modular Hamiltonian K on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$ satisfies the period closure $\sigma_{2\pi}(O) = O$ *bit-exactly* at finite $n_{\max}$ for all four physical witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh). The result lifts the four-witness Wick-rotation theorem from "structural correspondence" (Remark 42 prior verdict) to "literal identification at the operator-system level (Riemannian)" at every finite cutoff.

Architecture. Sprint L1 implements the locked architectural decisions of L1-A (`debug/l1_modular_hamiltonian` L1-B (`debug/l1_infrastructure_audit_memo.md`), and L1-C (`debug/l1_witness_spec_memo.md`):

- **Wedge (W1):** hemispheric P_W on S^3 aligned with the Hopf-base axis, defined as $P_W = (1/2)(I + R_{\mathrm{polar}})$ where $R_{\mathrm{polar}} : m_j \mapsto -m_j$ is the equatorial reflection involution on the spinor basis; $P_W^2 = P_W$ and $P_W^* = P_W$ are bit-exact (no smoothing needed; the reflection is an exact involution on the discrete basis).

- **Unit normalisation (U-1):** natural rapidity units $\kappa_g = 1$ for the canonical BW witness; the four witnesses parameterize as Hartle–Hawking $\kappa_g = 1/(4M)$, Unruh $\kappa_g = a$, Sewell $\kappa_g = 1/(4M)$.
- **Construction:** the BW-α geometric realization $K = J_{\text{polar}}$ (rotation generator around the wedge axis), with integer eigenvalues two_m$_j$ on the full-Dirac basis (odd integers in $\{-(2l+1), \ldots, +(2l+1)\}$). The Tomita BW-$\gamma$ realization is also instantiated as a sanity check on the trace cyclicity of the KMS state at expectation level; both realizations close consistently.
- **Dirac:** truthful Camporesi–Higuchi (`camporesi_higuchi_full_dirac_matrix`), the only Dirac choice satisfying $J_{GV}D = +DJ_{GV}$ on the truncated operator system (see § 4). R3.2's n-degeneracy obstruction for the Connes-distance SDP does *not* bite for modular flow, which requires only cyclic-separatingness of the cyclic vector.

Result. The maximum periodicity residual $\max_O \|\sigma_{2\pi}(O) - O\|_F$, taken over the wedge-restricted versions of the first five non-identity multipliers in the `FullDiracTruncatedOperatorSystem` at cutoff $n_{\max}$, is at machine precision $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\text{machine}})$ for every tested $n_{\max}$ and every witness:

$n_{\max}$	$\dim_H$	wedge dim	$\max \|\sigma_{2\pi}(O) - O\|_F$	verdict
2	16	8	1.80×10^{-16}	STRONG_IDENTIFICATION
3	40	20	4.76×10^{-16}	STRONG_IDENTIFICATION
4	80	40	9.33×10^{-16}	STRONG_IDENTIFICATION
5	140	70	1.59×10^{-15}	STRONG_IDENTIFICATION

The $n_{\max} = 3$ cutoff is structurally distinguished: $\dim_H = 40 = g_3^{\text{Dirac}} = \Delta^{-1}$ (Paper 2's structurally distinguished selection-principle cutoff). The L1 closure at this cutoff is the operator-system-level confirmation that the M1 mechanism's 2π closure holds on the Paper-2-relevant truncation.

Cross-witness collapse. The cross-witness consistency residual $\big|\|\sigma_{2\pi}^{(\text{witness})}(O) - O\| - \|\sigma_{2\pi}^{\text{BW}}(O) - O\|\big|$ is exactly 0 at every tested $n_{\max}$ between BW ($\kappa_g = 1$), HH at $M = 1$ ($\kappa_g = 1/4$), HH at $M = 2$ ($\kappa_g = 1/8$), Sewell at $M = 1$ ($\kappa_g = 1/4$), Unruh at $a = 1$ ($\kappa_g = 1$), and Unruh at $a = 2$ ($\kappa_g = 2$). The period closure is independent of κ_g because the geometric K_boost has integer spectrum: $e^{i \cdot 2\pi \cdot n} = 1$ for any integer n. The four-witness theorem collapses to a single operator-system test at the unit normalisation U-1.

Structural reading. The period closure $\sigma_{2\pi}(O) = O$ is finite-cutoff exact, not a limit-statement. It does not require the Gromov–Hausdorff machinery of Theorem 5 (although it is consistent with it via the propinquity rate cross-check: residual / $\gamma_{n_{\max}}$ ratio is $O(\varepsilon_{\text{machine}})$, several orders below the qualitative-rate prediction). The structural reason is the spectral-spectrum property: the BW-α geometric K has integer spectrum on the full-Dirac basis (odd integers two_m$_j$), inherited from the SO(4) Casimir grading of the Camporesi–Higuchi spinor harmonics. This is the framework-internal version of Bisognano–Wichmann's continuum statement.

The π-source case-exhaustion theorem (Theorem 4) is enriched, not weakened, by this result. The 2π in the modular period is M1 content (Hopf-base measure $\mathrm{Vol}(S^1)$) at the operator-system level, not just at the metric-functional level. The dual physical reading (Riemannian Hopf-base measure / Lorentzian boost-orbit period) is now unified at the operator level inside the spectral triple, not only by the published Wick-rotation chain.

Honest scope. The L1 closure is at signature $(3, 0)$ (Riemannian). Three distinctions remain open:

- The Lorentzian extension to signature $(3, 1)$ via the Bizi–Brouder–Besnard 2018 [48] Krein-space spectral triple is the named Sprint L2 follow-up (long-range scale per Sprint L0 audit `debug/lorentzian_l0_audit_memo.md`). Sprint L0 predicts M3 trivializes at $(3, 1)$ but M1 stays (so the four-witness theorem extends to Lorentzian); Sprint L2 would verify this directly.
- Non-tracial states would require explicit Tomita polar decomposition of the modular S. The tracial Gibbs state on $\mathcal{T}_{n_{\max}}$ gives the trivial reduction $K_{\mathrm{mod}} = \beta H_{\mathrm{local}} + \log Z$ (BW-γ); non-tracial extension is an open structural question.
- Cross-manifold modular structure (the Paper 24 § V cross-manifold $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ blocker) remains unaddressed and is not the subject of Sprint L1.

The implementation is at `geovac/modular_hamiltonian.py` (production module, fresh) with `tests/test_modular_hamiltonian.py` (41 tests, 39 fast + 2 slow, all pass). Computational driver: `debug/l1_modular_hamiltonian_compute.py`. Machine-readable results JSON: `debug/data/l1_modular_hamil`
Closure memo: `debug/l1_modular_hamiltonian_results_memo.md`.

Sprint L1-tighten (same day): load-bearing Tomita–Takesaki closure. The L1 closure above uses the BW-α geometric realization $K = J_{\mathrm{polar}}$ as the load-bearing object; the Tomita BW-γ realization was instantiated only as an expectation-level KMS sanity check. Sprint L1-tighten (same day, 2026-05-16 evening) rebuilds the BW-γ construction at the full Tomita–Takesaki level on the GNS Hilbert–Schmidt space $H_{\mathrm{GNS}} = M_{\dim_W}(\mathbb{C}) \cong \mathbb{C}^{\dim_W^2}$, computes the modular operator Δ as a $(\dim_W^2 \times \dim_W^2)$ matrix via the polar decomposition $S = J_{\mathrm{TT}}\Delta^{1/2}$, extracts $K_{\mathrm{TT}} = -\log \Delta$, and verifies $\sigma_{2\pi}^{\mathrm{TT}}(O) = O$ at the operator-action level for the same five generators tested by BW-α.

The construction uses the natural local Hamiltonian $H_{\mathrm{local}} := K_\alpha^W/\beta$ on the wedge sub-algebra (so $\beta H_{\mathrm{local}} = K_\alpha^W$ is the wedge-restricted boost-class generator with integer spectrum two_m$_j$); this matches the operator-system analog of the Wightman BW vacuum $\rho_W \propto e^{-2\pi K_{\mathrm{boost}}}$ on the Rindler wedge.

Result (UNIFIED_STRONG): at every tested $n_{\max} \in \{2, 3, 4, 5\}$ for all six witness instantiations (BW, HH$_{M=1}$, HH$_{M=2}$, Sewell$_{M=1}$, Unruh$_{a=1}$, Unruh$_{a=2}$):

$n_{\max}$	$\dim_W$	$\dim_{\mathrm{GNS}}$	$\sigma_{2\pi}^{\alpha}$ residual	$\sigma_{2\pi}^{\mathrm{TT}}$ residual	$\sigma_t^{\mathrm{TT}} - \sigma_{-t}^{\alpha}$ at $t = 1$	$J_{\mathrm{TT}}^2 - I$
2	8	64	1.10×10^{-16}	1.09×10^{-16}	4.18×10^{-17}	0
3	20	400	2.22×10^{-16}	1.07×10^{-15}	9.79×10^{-17}	5.00×10^{-17}
4	40	1600	3.51×10^{-16}	3.03×10^{-15}	3.18×10^{-16}	7.22×10^{-17}
5	70	4900	4.97×10^{-16}	4.79×10^{-15}	4.03×10^{-16}	2.18×10^{-17}

All residuals at machine precision. The four-witness cross-collapse is preserved at the Tomita–Takesaki level (in fact, the density matrix $\rho_W = e^{-K_\alpha^W}/Z$ is itself β-independent, so Δ and K_{TT}

are identical across the six witnesses); all six instantiations give bit-identical $\sigma_{2\pi}^{\mathrm{TT}}$ residuals at every $n_{\max}$.

Structural relation between BW-α and BW-γ. The two constructions generate the same modular automorphism on the algebra, up to the sign of t:

$$\sigma_t^{\mathrm{TT}}(a) \;=\; \rho^{it}\, a\, \rho^{-it} \;=\; e^{-itK_\alpha^W}\, a\, e^{+itK_\alpha^W} \;=\; \sigma_{-t}^{\alpha}(a). \tag{14}$$

The cross-flow difference $\|\sigma_1^{\mathrm{TT}}(O) - \sigma_1^{\alpha}(O)\|_F \sim 0.16$ at non-period $t = 1$ is non-zero in general (the conjugate flows are not equal at arbitrary t); but $\|\sigma_1^{\mathrm{TT}}(O) - \sigma_{-1}^{\alpha}(O)\|_F < 10^{-16}$ verifies equation (14) bit-exactly. At $t = 2\pi$ the sign is invisible because $e^{\pm i2\pi n} = 1$ for any integer n, so both flows close to the identity bit-exactly.

$J_{\mathrm{TT}}^2 = +I$ at the FULL Tomita level. The polar-decomposition antilinear conjugation $J_{\mathrm{TT}}(X) = \rho^{1/2} X^* \rho^{-1/2}$ satisfies $J_{\mathrm{TT}}^2 = +I$ to machine precision at every $n_{\max}$ (residuals in the table above). This is structurally distinct from the L1 verification, which only checked the canonical sanity case $U = I$; L1-tighten verifies the $\rho^{1/2}$-weighted Tomita conjugation itself, which is the genuine state-dependent antilinear operator defined by the polar decomposition. This separates J_{TT} from the Connes KO-dim 3 charge conjugation J_{GV} (which has $J_{\mathrm{GV}}^2 = -I$); the two antilinear operators coexist on $H_{n_{\max}}$ without conflation, as required by L1-A § 10e.

Verdict graduation. The L1 closure is at the operator-system level via the geometric BW-α realization *and* via the Tomita–Takesaki BW-γ polar decomposition. The two constructions agree at the operator-action level (up to sign of t; identical at the period). The four-witness Wick-rotation theorem is operator-system-level closed at signature $(3, 0)$ via both readings, not just via the geometric one. The L1 § 7.3 "BW-α-only" caveat is removed.

Honest scope. The unified closure rests on the choice $H_{\mathrm{local}} = K_\alpha^W/\beta$, which makes the wedge KMS state the operator-system analog of the Wightman BW vacuum $\rho_W \propto e^{-2\pi K_{\mathrm{boost}}}$. An alternative choice $H_{\mathrm{local}} = D_W$ (wedge-restricted Camporesi–Higuchi Dirac) has half-integer spectrum, so its operator-level period lift is $e^{i2\pi D_W} = -I$ (the spinor double cover) rather than the $+I$ of the odd-integer K_α^W; the conjugation flow still closes for both (the global $-I$ cancels), but only K_α^W lifts the period to the identity operator, so the framework's intrinsic Dirac is not the canonical local Hamiltonian for the wedge KMS state at $\beta = 2\pi$. The L1-tighten unified closure is for the BW-aligned wedge state, the load-bearing choice for the Bisognano–Wichmann identification.

The L1-tighten implementation extends `geovac/modular_hamiltonian.py` with the `TomitaModularStructure` class (load-bearing BW-γ) and accessor methods (`tomita_structure`, `k_alpha_geometric`, `k_tomita`, `compare_alpha_vs_tomita`) on the `ModularHamiltonian` class. Test coverage: `tests/test_modular_hamiltonia` extended with 26 fast + 2 slow new tests (total 67 tests, 0 regressions of the 41 L1 tests). Computational driver: `debug/l1_tighten_tomita_compute.py`. Machine-readable results: `debug/data/l1_tighten_tomit` Closure memo: `debug/l1_tighten_tomita_results_memo.md`.

9 Marcolli–van Suijlekom Placement: Instance + GeoVac-Original Extension

The construction of Sec. 3 is an *instance* of the gauge networks framework of Marcolli and van Suijlekom [5] (with the Perez-Sanchez correction [10] that the continuum limit is Yang–Mills without Higgs) at the data level, together with three GeoVac-original structural extensions: the forced-graph claim (Bertrand's theorem $\times$ Fock projection forces S^3), the master Mellin engine $\mathcal{M}[\mathrm{Tr}(D^k e^{-tD^2})]$ at $k \in \{0, 1, 2\}$ classifying the framework's transcendentals, and the cosmic-Galois layer U^* constructed in Paper 56. MvS 2014 does not predict a Tannakian dual; the U^* sits on top of the gauge-network scaffolding as additional structure, not as an MvS corollary. The key parallels at the data level:

- *Vertex algebra.* Marcolli–vS take $\mathcal{A}_V = C(V)$ on a vertex set V; we take $\mathcal{A}_{\mathrm{GV}} = \mathbb{C}^{V_{\mathrm{Fock}}}$ on the Fock-projected S^3 vertex set (Sec. 3).
- *Edge connection.* Marcolli–vS take a connection on edges valued in a finite group G; Paper 25 takes $G = U(1)$ and Paper 30 takes $G = \mathrm{SU}(2)$.
- *Spectral action.* Marcolli–vS show that the spectral action of the resulting almost-commutative spectral triple equals a Wilson lattice gauge action on the connection; Paper 30 verifies this explicitly for SU(2) at $n_{\max} = 3, 4$.
- *Continuum limit.* Per Perez-Sanchez, the continuum limit of the gauge-network spectral action is Yang–Mills without a Higgs sector; Paper 28's heat-kernel Seeley–DeWitt expansion is the spectral-action expansion in this limit.

Observation 3 (Lineage placement)**.** *Each structural ingredient of the GeoVac spectral triple has published precedent: finite almost-commutative spectral triples ([17, 18, 19]), graph spectral triples ([5, 9]), spectral truncations ([11, 12, 13]), and the Standard Model spectral action ([4]). What is not duplicated in the literature is the* α *prediction at* 8.8×10^{-8} *from a discrete spectral construction; that remains the original empirical claim of Paper 2.*

This is part of what the working hypothesis WH1 in the project register rests on (CLAUDE.md §1.7, WH1 "PROVEN" as of 2026-06-10): the structural framing is in the published lineage; the predictive claim is not duplicated.

10 Open Questions

The construction of Sec. 3 and the audit of Sec. 4 leave the following structural questions genuinely open.

Q1. Order one beyond rank 1. The order-one condition holds trivially for the abelian $\mathcal{A}_{\mathrm{GV}}$ and non-trivially constrains the rank-1 non-abelian extension $\mathcal{A}_{\mathrm{GV}} \otimes M_2(\mathbb{C})$ of Paper 30. At rank ≥ 2 (e.g., $M_3(\mathbb{C})$ or sums of matrix algebras as in the Standard Model triple), order-one becomes a strong constraint and typically requires modification of the Dirac operator (*first-order condition* $\Rightarrow$ Higgs sector in [4]). Whether GeoVac admits a natural rank-≥ 2 extension—and what its spectral action would predict—is an open question. The Bargmann–Segal S^5 HO triple of Paper 24 was probed for an SU(3) extension in Sprint 5 (CLAUDE.md §2) with negative outcome; that is one negative data point, not the full picture. The electroweak slice $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H}$ (rank up to 2 via $\mathbb{H} \cong M_2(\mathbb{C})$ over $\mathbb{C}$, a real $*$-subalgebra) is the most reachable test, and Sprint H1 addresses it; see Sec. 8.2 below for the verdict.

Q2. Real structure at finite $n_{\max}$ (resolved). Proposition 5 establishes a real structure on the continuum spinor bundle that restricts to the truncation. The axiom audit of Sec. 4 pins the finite-$n_{\max}$ behaviour down to machine precision: the restricted $J_{\mathrm{GV}}^{(n_{\max})}$ satisfies $J^2 = -\mathbb{1}$ (KO-dimension 3, *not* $+\mathbb{1}$) bit-exactly at every $n_{\max}$ examined in the audit (`test_real_structure`), so the finite-versus-limit question is closed in favour of finite-$n_{\max}$ exactness.

Q3. KO-dimension and the four-way coincidence. Working hypothesis WH4 asks: is the four-way S^3 coincidence (Sec. 1) forced by the KO-dimension 3 structure of the present triple, or is it independent geometric data? The relevant test is whether any other rank-1 spectral triple of KO-dimension 3 fails to produce all four roles. S^3 is unique among compact rank-1 non-abelian simply-connected groups, so the question reduces to: within the Marcolli–vS gauge-network setting, does KO-dimension 3 + non-abelian fiber + non-trivial Hopf bundle uniquely select S^3? The lineage literature does not appear to address this directly.

Q4. Spectral-action selection of Λ. Paper 28's spectral-action computation produced an exact two-term form $\mathrm{Tr}\, e^{-D^2/\Lambda^2} = (\sqrt{\pi}/2)\Lambda^3 - (\sqrt{\pi}/4)\Lambda + O(e^{-\pi^2/\Lambda^2})$ but did not autonomously select Λ. Setting the trace equal to K/π gave a closed-form Cardano solution $\Lambda_\infty = 3.7102\ldots$ (50 dps), which was tautological in the sense that it was determined by demanding the equality. Whether there is a structural principle that selects Λ is open; if so, K/π would become a derived quantity rather than a coincidence. This is the most direct route by which Paper 2 could be promoted from observation to derivation, and it is also the one most resistant to incremental progress.

Q5. Higher-form structure (Breit / rank-2 connections). Paper 25 §VII.B asked whether the Breit interaction (rank-2 in electron-electron coupling) lifts the $U(1)$ 1-form picture to a higher-form gauge theory. In spectral-triple language this is the question whether the GeoVac triple admits a natural extension to a rank-2 fiber that produces the Breit retarded integrals. Paper 30's SU(2) result is rank 1 (vector); rank 2 (tensor) is a distinct extension; the answer is unknown.

11 Conclusion

This paper has constructed an explicit finite spectral triple $(\mathcal{A}_{\mathrm{GV}}, \mathcal{H}_{\mathrm{GV}}, D_{\mathrm{GV}})$ on the Fock-projected S^3 graph, audited it against the standard Connes axioms (Table 1, Theorem 2), and identified Papers 25, 28, 30, 31 as projections of the resulting object (Table 3). The construction sits inside the published Marcolli–van Suijlekom gauge-network framework with the Perez-Sanchez correction. The combination rule $K = \pi(B + F - \Delta)$ is now phrased as a sum across three structurally distinct sectors of the same triple (Observation 1), without changing its status as an observation. The Coulomb/HO asymmetry of Paper 31 is the statement that the GeoVac triple is one specific real odd KO-dimension-3 triple, not a generic class (Observation 2).

Three things this paper did not do. It did not derive α; Paper 2 stays in Observations. It did not claim the Standard Model; GeoVac is rank 1, the SM is higher rank. It did not prove that the four-way S^3 coincidence is forced by KO-dimension 3; that is open question Q3.

Three things this paper did do. It made the spectral-triple framing explicit and auditable. It placed the framework in a published mathematical lineage. It made the open questions about α, the fiber rank, and the four-way coincidence into precise spectral-triple questions instead of free-floating conjectures.

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